# SOLUTIONS TO SEVERAL PROBLEMS ON INFINITE SUMSET CONFIGURATIONS

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Abstract. We resolve a number of the problems collected by Kra, Moreira, Richter and Robertson in their survey *Problems on infinite sumset configurations in the integers and beyond*. Our main result is that in every countably infinite abelian group, every set of positive upper Banach density contains a sumset $B_1 + \cdots + B_k$ of $k$ infinite sets, for every $k$; the proof avoids Host–Kra structure theory altogether. We also answer, positively or negatively, questions on monochromatic sets $B + mB$, on sumsets in sets of logarithmic density, on polynomial sumset configurations, on sums of the form $B_{i_1} + \cdots + B_{i_m}$ with $B_i$ drawn from a density family, on random sets, on ordered sumsets in shifted primes, on sums and products in $\mathbb{N}$ and $\mathbb{Q}$, and on sumsets in abelian groups. Finally, we determine the lower Banach density thresholds for $B + B + t$ and $B + B$, answering a question of Kousek and Radić.

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## 1. Introduction

A well-known theorem of Moreira, Richter and Robertson [26], answering a question of Erdős, says that every set $A \subset \mathbb{N}$ of positive upper Banach density contains a sumset $B + C = \{b + c : b \in B,\ c \in C\}$ with $B$ and $C$ infinite. In the survey [22], Kra, Moreira, Richter and Robertson collected a large number of open problems around this theorem: which other infinite configurations must occur in sets of positive density, which density hypotheses are needed, and what happens in the primes, in random sets, in multiplicative settings and in other groups.

In this paper we answer twenty of the questions and conjectures posed in [22] (nineteen questions and Conjecture 4.5). Our main result concerns sums of more than two infinite sets in abelian groups.

**Theorem** (Theorem 2.1)**.** *Let $\Gamma$ be a countably infinite abelian group and let $k \geq 1$. Every set $A \subset \Gamma$ of positive upper Banach density contains a sumset $B_1 + \cdots + B_k$, where $B_1, \dots, B_k \subset \Gamma$ are infinite.*

This answers Question 5.14 of [22]. The proof does not use the Host–Kra structure theory or the theory of nilspaces: it extends the short proof of the case $k = 2$ by Kra, Moreira, Richter and Robertson [24] to an arbitrary number of summands, building the required configuration one level at a time and keeping track only of the supports of suitable measures on the Kronecker factor.

The remaining results are summarised in Table 1. In addition, in Section 13 we answer a question of Kousek and Radić [19] on lower Banach density: the thresholds for $B + B + t$ and for $B + B$ are exactly $1/2$ and $3/4$. Several of them are short, and some rely in an essential way on recent work of other authors; we indicate this in each case.

**Conventions.** We use the numbering of the second arXiv version of [22]. Some questions in the survey are posed in the running text without a number. We refer to them by the subsection in which they appear followed by a letter, in order of appearance: for instance, Question §3.2(a) is the first unnumbered question in Subsection 3.2 of [22]. Before answering each question we restate it in our notation.

Throughout, $\mathbb{N} = \{1, 2, 3, \dots\}$, $\mathbb{P}$ denotes the set of primes, and for $A \subset \mathbb{N}$

$$\overline{\mathsf{d}}(A) = \limsup_{N\to\infty} \frac{|A \cap [N]|}{N}, \qquad \mathsf{d}^*(A) = \lim_{N\to\infty} \sup_{M\geq 0} \frac{|A \cap [M+1, M+N]|}{N},$$

$$\overline{\mathsf{d}}_{\log}(A) = \limsup_{N\to\infty} \frac{1}{\log N} \sum_{n \in A\cap[N]} \frac{1}{n},$$

where $[N] = \{1, \dots, N\}$. Lower densities $\underline{\mathsf{d}}$ and natural and logarithmic densities $\mathsf{d}$, $\mathsf{d}_{\log}$ are defined in the same way. In a countable abelian group, or more generally a countable amenable group, upper Banach density $\mathsf{d}^*$ is the supremum of upper densities along Følner sequences. A set is *thick* if it contains arbitrarily long intervals (in a group: translates of every finite set). For sets $B, C$ we write $B + C = \{b + c : b \in B, c \in C\}$, $B + t = \{b + t : b \in B\}$ and $mB = \{mb : b \in B\}$.

**Questions whose answer depends on the formulation.** For several questions the answer we give concerns the question exactly as it is stated in the survey, and the literal statement differs from what the authors of the survey probably intended. We include these answers because the difference is not always obvious and the literal statements have been repeated in the literature, but in each case we point out the issue explicitly, give the natural corrected formulation, and say whether it remains open. This applies to Question 3.20 (where $m = 1$ is coprime to

every $q$), Question 3.23 (where $B+mB$ contains the diagonal terms $(m+1)b$), Question §3.2(a) (where the density condition can be read in two ways; we answer both readings), Question 5.6, Question 5.23 and Question §5.4(b) (where the fibre conditions constrain only one coordinate). The negative answer to Question §3.3(e) follows directly from known counterexamples (one from [22] itself and one due to Ackelsberg [2]); we record it for completeness. Similarly, our answer to Question 3.24 is preceded by an example of Kousek and Radić [19], which, as we check, already answers that question.

| Question | Answer | Section |
|---|---|---|
| 5.14 | Yes: $B_1 + \cdots + B_k$ in every countably infinite abelian group | 2 |
| 5.16, §5.3(a) | Exactly which sets of translates suffice when $[G : 2G] < \infty$ | 3 |
| 5.10 | No: fibrewise lower density does not suffice in $\mathbb{N}^2$ | 4 |
| 5.23, §5.4(b) | No, as stated; a corrected version is open | 5 |
| 3.20 | No (for $B + mB$ as in the survey), for two and for three colours | 8.1 |
| 3.23 | No, as stated | 8.2 |
| 3.24 | No (already implied by Kousek–Radić) | 8.3 |
| §3.2(a) | No, for both readings | 6 |
| §3.3(a) | Ordered polynomial sums: exactly the linear case | 7.1 |
| §3.3(e) | No (direct consequence of known examples) | 7.2 |
| 3.31, §3.6(a) | Yes, in the strongest $k$-fold form | 9 |
| 4.13 | Yes iff $\gamma \notin \mathbb{Z}$ (for almost every $x$) | 10 |
| §4.4(a) | Random sets: criterion for monotone $p_n$; no Borel criterion in general | 11 |
| 4.5 | Yes, unconditionally | 12 |
| 5.4 | No | 14.2 |
| 5.6 | No, as stated | 14.3 |
| §5.1(a) | Yes for both examples | 14.1 |

TABLE 1. Questions from [22] answered in this paper.

**Formalization.** The main theorem and most of the other results have been formalized in Lean 4 with Mathlib; the formalization is available at `https://github.com/daizisheng/kmrr25`, in the directory `lean/Kmrr25`. No file contains `sorry` or adds an axiom, and every main theorem depends only on the standard axioms of Lean. Where one of our proofs relies on a published theorem that is not formalized, that theorem is stated as a proposition and passed to the formal statement as an explicit hypothesis. Table 2 gives an overview, and after each formalized result a paragraph headed *Formalization* names the file and the main formal statements and says what is not covered.

**Organisation.** Section 2 contains the proof of the main theorem. The rest of the paper is organised by the setting: further results on groups and on $\mathbb{N}^2$, results on sumsets in the integers, and results on configurations mixing sums and products. The sections can be read independently.

| Question | File | Status |
|---|---|---|
| 5.14 | `ThmK*` (14 files) | conditional on four classical theorems (see the end of Section 2.9) |
| 5.16, §5.3(a) | `Q516` | conditional on [8, Corollary 1.13] |
| 5.10 | `Q510` | complete |
| 5.23, §5.4(b) | `Q523` | answer to the questions as stated: complete; repaired version (Theorem 5.5): not formalized |
| 3.20 | `Q320` | complete |
| 3.23 | `Q323` | complete |
| 3.24 | `Q324` | complete |
| §3.2(a) | `S32a` | complete |
| §3.3(a) | `S33a` | negative part: complete; positive part: conditional on [17, Theorem 1.2] |
| §3.3(e) | `S33e` | complete |
| 3.31, §3.6(a) | `Q331` | conditional on the dynamical input from [23] (Proposition 9.7) |
| 4.13 | `Q413`, `Q413a`, `Q413SecondMoment` | complete |
| §4.4(a) | `S44a`, `S44b` | complete, except that the non-Borel results use [16, Theorems 14.2 and 27.1] on ill-founded trees as hypotheses |
| 4.5 | `C45` | conditional on [30, Theorem 1.5] |
| 5.4 | `Q54` | complete |
| 5.6 | `Q56` | complete |
| §5.1(a) | `S51a` | complete |
| KR, Q. 5.2 | `KR52` | complete |

TABLE 2. Lean formalization. "Complete" means that the answer to the question proved in this paper is formalized without hypotheses beyond the standard axioms; the file names refer to `lean/Kmrr25/*.lean`.

# Part 1. Sumsets in groups

## 2. Sums of $k$ infinite sets in countable abelian groups

This section answers Question 5.14 of [22]. We show (Theorem 2.1) that in every countably infinite abelian group, every set of positive upper Banach density contains a sumset $B_1+\cdots+B_k$ of $k$ infinite sets, for every $k$. For finite groups the question has a trivially negative answer (Remark 2.2), so "countably infinite" is the right setting. The proof does not use Host–Kra theory, nilsystems or nilspaces, or any equidistribution theorem. It extends to $k$ summands, and to arbitrary countable abelian groups, the short proof of the case $k = 2$ for subsets of $\mathbb{N}$ given by Kra, Moreira, Richter and Robertson [24]. The cube is built one level at a time. At each level we re-base on the Kronecker factor of the current measure, and from one level to the next we keep only a condition on topological supports.

2.1. **The question and the result.** Let $\Gamma$ be a countable abelian group, written additively. A *Følner sequence* in $\Gamma$ is a sequence $\Phi = (\Phi_N)_{N\in\mathbb{N}}$ of nonempty finite subsets of $\Gamma$ such that

$|(\Phi_N+\gamma)\triangle\Phi_N|/|\Phi_N|\to 0$ for every $\gamma\in\Gamma$. The *upper Banach density* of $A\subset\Gamma$ is

$$\mathsf{d}^*(A)=\sup_{\Phi}\ \limsup_{N\to\infty}\frac{|A\cap\Phi_N|}{|\Phi_N|},$$

where the supremum is over all Følner sequences $\Phi$ in $\Gamma$.

**Question 5.14** ([22])**.** Is it true in every countable abelian group $(G,+)$ that every set $A$ of positive upper Banach density contains $B_1+\cdots+B_k$ for infinite sets $B_1,\dots,B_k\subset G$?

**Theorem 2.1.** *Let $\Gamma$ be a countably infinite abelian group and let $k\geq 1$. If $A\subset\Gamma$ satisfies $\mathsf{d}^*(A)>0$, then there are infinite sets $B_1,\dots,B_k\subset\Gamma$ with*

$$B_1+\cdots+B_k=\{b_1+\cdots+b_k:\ b_i\in B_i\}\subset A.$$

*Remark* 2.2 (Finite groups). Read literally, Question 5.14 includes finite groups, and for them the answer is no. A finite group has no infinite subsets, yet every nonempty $A\subset G$ has $\mathsf{d}^*(A)\geq|A|/|G|>0$. (For finite $G$, the Følner condition forces $\Phi_N=G$ for all large $N$.) So the meaningful version of the question concerns countably *infinite* groups, and Theorem 2.1 answers that version positively for every $k$.

*Remark* 2.3 (Previously known cases).

(a) For $k=2$ the conclusion holds in every countable amenable group, with products in place of sums, for sets of positive density along a two-sided Følner sequence. This is due to Moreira, Richter and Robertson [26, Theorem 1.3]. In an abelian group every Følner sequence is two-sided.

(b) For $\Gamma=\mathbb{Z}$, or rather for $A\subset\mathbb{N}$, the case of general $k$ is [20, Theorem 1.1] (Theorem 2.8 of [22]). Here and below, theorem numbers in [20] refer to arXiv:2206.01786v2. Its proof relies on pronilfactors: Kra, Moreira, Richter and Robertson construct Erdős cubes in an extension of the system with continuous factor maps to its pronilfactors [20, Section 5].

(c) For $k=3$ and $[\Gamma:6\Gamma]<\infty$, the conclusion follows from a theorem of Ackelsberg and Jamneshan [3, Theorem 1.9]. They find $t\in\Gamma$ and an infinite $B\subset\Gamma$ such that $A-t$ contains every sum of three distinct elements of $B$. Splitting $B$ into three disjoint infinite sets $B_1',B_2',B_3'$ then gives $(B_1'+t)+B_2'+B_3'\subset A$.

After stating Question 5.14, the survey [22] remarks that the techniques of [20], together with work of Griesmer [11], may suffice for $\Gamma=\mathbb{Z}^d$. In other groups, it says, "one cannot proceed in the same way, because the analogous ergodic structure theory is not currently available." The proof below uses no such structure theory.

*Remark* 2.4 (The integers). Our proof specialises to a proof of [20, Theorem 1.1] that uses no pronilfactors. To see this, let $A\subset\mathbb{N}$ have positive upper Banach density. Intervals form a Følner sequence in $\mathbb{Z}$, so $\mathsf{d}^*(A)>0$ also when $A$ is viewed as a subset of $\mathbb{Z}$. Theorem 2.1 then gives infinite $B_i\subset\mathbb{Z}$ with $B_1+\cdots+B_k\subset A\subset\mathbb{N}$. Fix $b_j\in B_j$ for $j\neq i$. Then every element of $B_i$ is at least $1-\sum_{j\neq i}b_j$, so each $B_i$ is bounded below. Hence $B_i\cap\mathbb{N}$ is still infinite, and $(B_1\cap\mathbb{N})+\cdots+(B_k\cap\mathbb{N})\subset A$.

### 2.2. **Strategy of the proof.**

*Erdős cubes.* Following [20], we look for a dynamical configuration called an Erdős cube. The correspondence principle (Theorem 2.9) gives the following data:

- an ergodic system $(X,\mu,T)$;
- an open set $E\subset X$ with $\mu(E)>0$;

- a point $a \in X$, generic for $\mu$, such that $A = \{\gamma : T^\gamma a \in E\}$.

An *Erdős cube* is a tuple $y = (y_\varepsilon)_{\varepsilon \in \{0,1\}^k}$ of points of $X$ with the following property. For every direction $i$, the lower $i$-face $(y_\varepsilon)_{\varepsilon_i=0}$ is moved arbitrarily close to the upper $i$-face $(y_{\varepsilon+e_i})_{\varepsilon_i=0}$ by infinitely many group elements $\beta$, acting diagonally; here $\varepsilon + e_i$ is $\varepsilon$ with its $i$-th digit changed to 1. If $y_{0\cdots0} = a$ and $y_{1\cdots1} \in E$, a greedy argument produces $B_1 + \cdots + B_k \subset A$ (Lemma 2.11). So the task is to construct such a cube.

*The case $k = 2$.* The argument of [24] can be read as constructing a 2-dimensional cube whose bottom face is $(a, y)$ and whose top face $(x, x')$ is random; we use our own notation. A point $y \in \operatorname{supp} \mu$ is chosen so that $(a, y)$ is generic for some joining $\lambda$ of $\mu$ with itself. The key claim (Claim 1 in the proof of [24, Theorem 3.1]) produces infinitely many $\beta$ with two properties:

- $T^\beta a$ is close to $y$;
- the top face $(x, x')$, which is distributed according to $\lambda$, still has positive measure after the direction-1 constraint imposed by $\beta$.

The proof decomposes $1_E = f_c + f_{wm}$ into its Kronecker and weak mixing parts.

- The weak mixing part averages out along every Følner sequence.
- The Kronecker part is handled by approximating $f_c$ by *continuous* eigenfunctions. This uses an enhanced correspondence principle in which every measurable eigenfunction is topological [24, Theorem 2.5(4)]. Continuity gives the equicontinuity estimate (4.2) of [24], which turns "$T^\beta a$ is close to $y$" into "$f_c \circ T^\beta$ is close to $f_c$ along the orbit".
- The positive frequency of the $\beta$ with $T^\beta a$ near $y$ comes from $y \in \operatorname{supp} \mu$.

*Where the difficulty lies for $k \geq 3$.* Suppose a $(k-1)$-dimensional bottom face $w = (w_\varepsilon)$ with $w_{0\cdots0} = a$ has been built, generic for an ergodic measure $\lambda$ on $X^{\{0,1\}^{k-1}}$. We try to add a random top face distributed according to $\lambda$.

- The direction $k$ (bottom face to top face) behaves exactly as in the case $k = 2$.
- In each of the other directions $i$, the relevant system is the face measure $\nu_i$, the distribution under $\lambda$ of the upper $i$-face. Its Kronecker factor carries information about $\mu$ beyond the Kronecker factor of $\mu$. For instance, for the skew product $(s, u) \mapsto (s + \theta, u + s)$ on $\mathbb{T}^2$, the measures on pairs that arise already have eigenfunctions of the form $e(u_1 - u_0)$, with eigenvalues that depend on the parameter of the pair measure. For a general system there is no reason for such eigenfunctions to be continuous in the topological model in which $a$ lives.

So the equicontinuity step of [24] is not available on faces. The proof of [20] for $\mathbb{Z}$ proceeds differently. It disintegrates the Host–Kra cubic measure $\mu^{[k]}$ over the first coordinate and needs the conditional measure at the prescribed point $a$. To define that conditional measure at every point, it passes to an extension in which the factor maps onto the pronilfactors are continuous, and it proves that the ergodic decomposition of $\mu^{[k]}$ is then continuous.

*The three ideas.*

(i) **Phase returns replace continuity** (Proposition 2.22). Let $\iota_i$ be the Kronecker map of $\nu_i$, with rotation $\alpha_i$.
   - By ergodicity, $\iota_i(\text{upper face}) - \iota_i(\text{lower face})$ is a $\lambda$-a.e. constant $r_i$.
   - By continuity of translation in $L^1$ of the Kronecker group, the Kronecker part is controlled for all $\beta$ whose "phase" $\alpha_i(\beta)$ is close to $r_i$.
   - So it suffices to have many $\beta$ for which $T^\beta(\text{lower face of } w)$ is close to the upper face of $w$ *and* $\alpha_i(\beta)$ is close to $r_i$. Here "many" means a set of positive upper Banach density.

(ii) **Phase returns come from supports** (Lemma 2.18). Such phase returns exist as soon as the upper face of $w$ lies in the topological support of $\nu_i$. The proof re-bases $w$: we change the Følner sequence so that $w$, decorated with a point of the Kronecker group, becomes generic for the graph measure. A stabiliser argument for the support of the graph measure then forces the decorated upper face into that support (Lemma 2.17).

(iii) **The support condition survives one more level** (Proposition 2.23). We build the cube level by level. At level $m$ we have a point $w \in X^{\{0,1\}^m}$ that is generic for an ergodic, face-symmetric measure $\lambda_m$ and whose upper faces lie in the supports of the face measures. To pass to level $m+1$:
- re-base $w$ on the Kronecker factor of $\lambda_m$;
- choose the new top face at random, so that the new cube is generic for an ergodic component of the relative product of $\lambda_m$ with itself over its Kronecker factor;
- check that the support condition persists. This is the content of an elementary convolution lemma (Lemma 2.19). The lemma holds at *every* support point, not only at typical ones.

From one level to the next, the induction carries only genericity, ergodicity, face symmetry, the vertex marginals and the support condition; the Kronecker data are rebuilt at each level. The only prescribed point is $a$. Every vertex added at an induction step is a typical point for the current measure, chosen after all the data it must avoid have been fixed, and the final top face is constructed by nested boxes. Apart from genericity, no statement about typical points is ever applied at a prescribed point, and this is why no equidistribution at a prescribed point is needed.

2.3. **Notation and standard facts.** Throughout, $\Gamma$ is a countably infinite abelian group. For a finite set $F$ and a function $u$ on $F$ we write $\mathbb{E}_{\gamma\in F}\, u(\gamma) = |F|^{-1}\sum_{\gamma\in F} u(\gamma)$.

**Lemma 2.5.** *If $\Phi$ is a Følner sequence in $\Gamma$, then $|\Phi_N| \to \infty$.*

*Proof.* Let $F \subset \Gamma$ be finite with $|F| = n$. The number of pairs $(x, g) \in \Phi_N \times F$ with $x+g \in \Phi_N$ is at most $|\Phi_N|^2$. It also equals $\sum_{g\in F} |\Phi_N \cap (\Phi_N - g)|$, which is $n|\Phi_N|(1-o(1))$ by the Følner property. Hence $|\Phi_N| \geq n(1-o(1))$. Since $\Gamma$ is infinite, $n$ is arbitrary. □

*Topological systems and generic points.* A *topological $\Gamma$-system* $(Y, T)$ consists of a compact metric space $Y$ and an action of $\Gamma$ by homeomorphisms $T^\gamma$, with $T^{\gamma+\delta} = T^\gamma T^\delta$. All measures are Borel probability measures. A measure $\nu$ is *invariant* if $T^\gamma_*\nu = \nu$ for all $\gamma$, and *ergodic* if moreover every invariant Borel set has measure 0 or 1. Since $\Gamma$ is countable, this is equivalent to asking that every Borel set $S$ with $\nu(S \triangle T^{-\gamma}S) = 0$ for all $\gamma$ has measure 0 or 1 (replace $S$ by $\bigcap_\gamma T^{-\gamma}S$); in particular a Borel map into a compact metric space which is $\nu$-a.e. invariant under each $T^\gamma$ is $\nu$-a.e. constant. A point $y \in Y$ is *generic for $\nu$ along* $\Phi$, written $y \in \mathrm{gen}(\nu, \Phi)$, if $\mathbb{E}_{\gamma\in\Phi_N} f(T^\gamma y) \to \int f\, d\nu$ for every $f \in C(Y)$. The (topological) support $\operatorname{supp}\nu$ is the smallest closed set of full measure. It is invariant whenever $\nu$ is. We use the following facts freely.

(G1) If $y \in \mathrm{gen}(\nu, \Phi)$ and $O \subset Y$ is open, then $\liminf_N |\{\gamma \in \Phi_N : T^\gamma y \in O\}|/|\Phi_N| \geq \nu(O)$. This follows by approximating $1_O$ from below by continuous functions. In particular, if $O$ meets $\operatorname{supp}\nu$, then the set of return times $\{\gamma : T^\gamma y \in O\}$ has positive upper Banach density. By Lemma 2.5 it is then infinite.

(G2) If $y \in \mathrm{gen}(\nu, \Phi)$, then $\operatorname{supp}\nu$ is contained in the orbit closure $\overline{\{T^\gamma y : \gamma \in \Gamma\}}$. This is immediate from (G1).

(G3) If $P\colon Y \to Y'$ is continuous and equivariant and $y \in \mathrm{gen}(\nu, \Phi)$, then $P(y) \in \mathrm{gen}(P_*\nu, \Phi)$.

(G4) If $\nu$ is ergodic, then for every Følner sequence $\Phi$ there is a subsequence $\Psi$ such that $\nu$-almost every point is generic for $\nu$ along $\Psi$ [8, Lemma 2.7]. This rests on Lindenstrauss' pointwise ergodic theorem.

*Cubes.* For $m \geq 0$ let $[m] = \{0,1\}^m$, with elements written $\varepsilon = \varepsilon_1 \cdots \varepsilon_m$. Let $\vec{0} = 0 \cdots 0$ and $\vec{1} = 1 \cdots 1$. $\Gamma$ acts on $X^{[m]}$ diagonally, $(T^\gamma x)_\varepsilon = T^\gamma x_\varepsilon$; we use the same letter $T$ for this action. We write $e_i \in [m]$ for the vertex with a single digit 1, in position $i$, and $\varepsilon + e_i$ for the vertex obtained from $\varepsilon$ (with $\varepsilon_i = 0$) by changing its $i$-th digit to 1. For $1 \leq i \leq m$, $\eta \in [m-1]$ and $j \in \{0,1\}$, let $\eta^{(i,j)} \in [m]$ be the vertex obtained by inserting the digit $j$ at position $i$. The *lower* and *upper $i$-faces* are the continuous equivariant maps $L_i, U_i \colon X^{[m]} \to X^{[m-1]}$ given by

$$(L_i x)_\eta = x_{\eta^{(i,0)}}, \qquad (U_i x)_\eta = x_{\eta^{(i,1)}}.$$

We identify $X^{[m+1]}$ with $X^{[m]} \times X^{[m]}$ by $y \mapsto (y', y'')$, where $y'_\varepsilon = y_{\varepsilon 0}$ and $y''_\varepsilon = y_{\varepsilon 1}$ (the last digit is split off). Under this identification:

- $L_i(y) = (L_i y', L_i y'')$ and $U_i(y) = (U_i y', U_i y'')$ for $i \leq m$;
- $L_{m+1}(y) = y'$ and $U_{m+1}(y) = y''$;
- $y_{\vec{0}} = y'_{\vec{0}}$ and $y_{\vec{1}} = y''_{\vec{1}}$.

For a measure $\lambda$ on $X^{[m]}$ and $\varepsilon \in [m]$, the *vertex marginal* at $\varepsilon$ is the image of $\lambda$ under $x \mapsto x_\varepsilon$.

*Spectral measures and Wiener's lemma.* Let $\widehat{\Gamma}$ be the (compact metrizable) dual group of $\Gamma$; its elements are homomorphisms $\psi \colon \Gamma \to S^1 = \{z \in \mathbb{C} : |z| = 1\}$. For an invariant measure $\nu$, write $\mathcal{U}_\gamma f = f \circ T^\gamma$ for the Koopman representation on $L^2(\nu)$. For $f \in L^2(\nu)$, the spectral measure $\sigma_f$ is the finite positive measure on $\widehat{\Gamma}$ with $\langle \mathcal{U}_\gamma f, f\rangle = \int \psi(\gamma)\, d\sigma_f(\psi)$ (Bochner–Herglotz); note $\sigma_f(\widehat{\Gamma}) = \|f\|_2^2$. An *eigenfunction with eigenvalue* $\psi \in \widehat{\Gamma}$ is a nonzero $f$ with $\mathcal{U}_\gamma f = \psi(\gamma) f$ for all $\gamma$. Write $H_\psi$ for the corresponding eigenspace (including 0).

**Lemma 2.6.** *Let $\nu$ be invariant and let $\Phi$, $\Theta$ be Følner sequences.*

*(a) For $\psi \in \widehat{\Gamma}$, $\mathbb{E}_{\gamma \in \Phi_N} \psi(\gamma) \to 1$ if $\psi = 1$, and $\to 0$ otherwise.*
*(b) For every unitary representation $V$ of $\Gamma$ on a Hilbert space and every vector $v$, $\mathbb{E}_{\gamma \in \Phi_N} V_\gamma v$ converges to the orthogonal projection of $v$ onto the $V$-invariant vectors.*
*(c) For $f \in L^2(\nu)$ and $\psi \in \widehat{\Gamma}$, $\sigma_f(\{\psi\}) = \|P_\psi f\|^2$, where $P_\psi$ is the orthogonal projection onto $H_\psi$. In particular, $\sigma_f$ has no atoms if and only if $f$ is orthogonal to every eigenfunction.*
*(d) If $\sigma_f$ has no atoms, then for every $g \in L^2(\nu)$, $\mathbb{E}_{\beta \in \Theta_M} |\langle \mathcal{U}_\beta f, g\rangle| \to 0$ as $M \to \infty$.*

*Proof.* (a): If $\psi(g) \neq 1$, then $|1 - \psi(g)| \cdot |\mathbb{E}_{\Phi_N} \psi| \leq |\Phi_N \triangle (\Phi_N + g)|/|\Phi_N| \to 0$.

(b): The space splits orthogonally into the invariant vectors and the closed span of $\{u - V_g u\}$. On a vector $u - V_g u$ the averages are bounded by $2\|u\|\, |\Phi_N \triangle (\Phi_N + g)|/|\Phi_N|$.

(c): Apply (b) to $V_\gamma = \overline{\psi(\gamma)}\mathcal{U}_\gamma$, whose invariant vectors form $H_\psi$. Then

$$\|P_\psi f\|^2 = \lim_N \mathop{\mathbb{E}}_{\gamma \in \Phi_N} \overline{\psi(\gamma)} \langle \mathcal{U}_\gamma f, f\rangle = \lim_N \int \mathop{\mathbb{E}}_{\gamma \in \Phi_N} (\psi' \overline{\psi})(\gamma)\, d\sigma_f(\psi') = \sigma_f(\{\psi\})$$

by (a) and dominated convergence.

(d): Let $Q$ be the orthogonal projection onto the cyclic subspace $Z(f)$, the closed span of $\{\mathcal{U}_\gamma f\}$. The map $\mathcal{U}_\gamma f \mapsto (\psi \mapsto \psi(\gamma))$ extends to a unitary isomorphism $Z(f) \to L^2(\sigma_f)$. Let $\phi$ be the image of $Qg$. Then $\langle \mathcal{U}_\beta f, g\rangle = \langle \mathcal{U}_\beta f, Qg\rangle = \int \psi(\beta)\, d\tau(\psi)$, where $\tau = \overline{\phi}\, \sigma_f$ is a complex

measure without atoms. By Fubini and (a),

$$\mathop{\mathbb{E}}_{\beta\in\Theta_M}\Big|\int\psi(\beta)\,d\tau\Big|^2=\iint\mathop{\mathbb{E}}_{\beta\in\Theta_M}(\psi\overline{\psi'})(\beta)\,d\tau(\psi)\,d\overline{\tau}(\psi')\longrightarrow\sum_{\psi}|\tau(\{\psi\})|^2=0.$$

The Cauchy–Schwarz inequality finishes the proof. □

*Kronecker maps.* Let $(Y,T)$ be a topological $\Gamma$-system and $\lambda$ an ergodic measure on $Y$. The *Kronecker subspace* $H_c(\lambda)\subset L^2(\lambda)$ is the closed linear span of all eigenfunctions. A *rotation* is a pair $(K,\alpha)$, where $K$ is a compact metrizable abelian group (written additively, with Haar measure $m_K$) and $\alpha\colon\Gamma\to K$ is a homomorphism with dense image; $\Gamma$ acts on $K$ by $z\mapsto z+\alpha(\gamma)$.

**Definition 2.7.** A *Kronecker map* for $(Y,\lambda)$ is a Borel map $\kappa\colon Y\to K$ to a rotation $(K,\alpha)$ with the following properties:

- $\kappa_*\lambda=m_K$;
- for every $\gamma$, $\kappa(T^\gamma y)=\kappa(y)+\alpha(\gamma)$ for $\lambda$-a.e. $y$;
- $L^2(Y,\sigma(\kappa),\lambda)=H_c(\lambda)$.

We write $\mathbb{E}_\lambda(f\mid\kappa)$ for the orthogonal projection onto $H_c(\lambda)$, that is, the conditional expectation onto $\sigma(\kappa)$.

**Lemma 2.8.** *Let $\lambda$ be ergodic.*

*(a) A Kronecker map for $(Y,\lambda)$ exists.*

*(b) Let $\kappa\colon Y\to K$ be any Kronecker map, with rotation $\alpha$. Every eigenfunction is a.e. equal to $c\cdot(\varphi\circ\kappa)$ for a constant $c$ and a character $\varphi\in\widehat{K}$. Here $\varphi$ is the unique character with $\varphi\circ\alpha$ equal to the eigenvalue.*

*(c) For $f\in L^2(\lambda)$: $\mathbb{E}_\lambda(f\mid\kappa)=0$ if and only if $\sigma_f$ has no atoms.*

*Proof.* (a): This is the discrete spectrum theorem (Halmos–von Neumann; for amenable groups see [8, Proposition 2.12]). For completeness we sketch the abelian case.

- The modulus of an eigenfunction is invariant, hence constant. Products of eigenfunctions are eigenfunctions. So the set $\mathcal{E}\subset\widehat{\Gamma}$ of eigenvalues is a subgroup.
- Eigenfunctions with distinct eigenvalues are orthogonal and $L^2(\lambda)$ is separable, so $\mathcal{E}$ is countable.
- Let $K$ be the dual of the discrete group $\mathcal{E}$, which is compact and metrizable, and let $\alpha(\gamma)=(\psi\mapsto\psi(\gamma))$. The image $\alpha(\Gamma)$ is dense, since its annihilator in $\mathcal{E}$ is trivial.
- Choose unimodular eigenfunctions $e_\psi$ for $\psi\in\mathcal{E}$, and set $\Theta(y)=(e_\psi(y))_\psi\in(S^1)^{\mathcal{E}}$. Then $\Theta(T^\gamma y)=\alpha(\gamma)\cdot\Theta(y)$ a.e., where $K$ is viewed as a closed subgroup of $(S^1)^{\mathcal{E}}$.
- Hence $\Theta_*\lambda$ is invariant under $\alpha(\Gamma)$, and so under $K$.
- The image of $\Theta$ in $(S^1)^{\mathcal{E}}/K$ is invariant, so by ergodicity it is a.e. a constant coset $Ku_0$.
- Put $\kappa=\Theta\cdot u_0^{-1}$, redefined on a null set to take values in $K$. Then $\kappa_*\lambda=m_K$ and $\kappa$ is a.e. equivariant.
- By ergodicity each eigenspace is one-dimensional, spanned by $e_\psi$. The characters of $K$ are the evaluations at elements of $\mathcal{E}$, so the functions $\chi\circ\kappa$, $\chi\in\widehat{K}$, are constant multiples of the $e_\psi$. Hence $L^2(\sigma(\kappa))$, the closed span of these functions, equals $H_c(\lambda)$.

(b): Let $h$ be an eigenfunction with eigenvalue $\psi$. Then $h\in H_c(\lambda)$, so $h=\tilde{h}\circ\kappa$ for some $\tilde{h}\in L^2(m_K)$, and $\tilde{h}(z+\alpha(\gamma))=\psi(\gamma)\tilde{h}(z)$ for $m_K$-a.e. $z$. Expanding $\tilde{h}=\sum_\varphi a_\varphi\varphi$ in characters gives $a_\varphi(\varphi(\alpha(\gamma))-\psi(\gamma))=0$ for all $\varphi$ and $\gamma$. Since $\alpha(\Gamma)$ is dense, at most one $\varphi$ satisfies $\varphi\circ\alpha=\psi$.

(c): $H_c(\lambda)$ is the closed span of the eigenspaces $H_\psi$, so this follows from Lemma 2.6(c). □

*Disintegrations.* Let $\kappa\colon Y \to K$ be a Kronecker map for $(Y,\lambda)$. There is a Borel family $z \mapsto \lambda_z$ of probability measures on $Y$ with the following properties [8, Theorem 2.2 and (2.1)]:

- $\lambda = \int_K \lambda_z \, dm_K(z)$;
- $\lambda_z(\kappa^{-1}(z)) = 1$ for $m_K$-a.e. $z$;
- for bounded Borel $f$ on $Y$, $\mathbb{E}_\lambda(f \mid \kappa) = \tilde{f} \circ \kappa$ $\lambda$-a.e., where

$$\tilde{f}(z) := \int f \, d\lambda_z.$$

The family is unique up to $m_K$-null sets. It is a priori defined only for $m_K$-a.e. $z$; we extend it to all $z \in K$ by setting $\lambda_z := \lambda$ on the exceptional null set, which affects none of the statements below. If $0 \le f \le 1$, then $0 \le \tilde{f} \le 1$ a.e. Since $\sigma(\kappa)$ is invariant modulo null sets, $\mathbb{E}_\lambda(\cdot \mid \kappa)$ commutes with $\mathcal{U}_\gamma$. Hence $\widetilde{f \circ T^\gamma}(z) = \tilde{f}(z + \alpha(\gamma))$ for $m_K$-a.e. $z$.

2.4. **The correspondence principle and Erdős cubes.** The following is Furstenberg's correspondence principle [10], in the form given by Charamaras and Mountakis.

**Theorem 2.9** ([8, Theorem 2.15])**.** *Let $A \subset \Gamma$, and let $\Phi$ be a Følner sequence such that $\mathsf{d}_\Phi(A) = \lim_N |A \cap \Phi_N|/|\Phi_N|$ exists. Then there exist an ergodic system $(X,\mu,T)$ on a compact metric space $X$, a clopen set $E \subset X$, a Følner sequence $\Psi$ and a point $a \in \mathrm{gen}(\mu,\Psi)$ such that $\mu(E) \ge \mathsf{d}_\Phi(A)$ and $A = \{\gamma \in \Gamma : T^\gamma a \in E\}$.*

In [8] this is stated for countable amenable groups. In the proof, $X = \{0,1\}^\Gamma$ with the shift action and $a = 1_A$. We do *not* need the strengthened form in which measurable eigenfunctions are topological (property (4) of [24, Theorem 2.5]). The re-basing lemma (Lemma 2.16) takes its place.

**Definition 2.10** (Erdős cubes; cf. [20, Definition 4.1])**.** Let $(X,T)$ be a topological $\Gamma$-system and $k \ge 1$. A point $y \in X^{[k]}$ is a *$k$-dimensional Erdős cube* if, for each $1 \le i \le k$, there are pairwise distinct $\beta_1, \beta_2, \ldots \in \Gamma$ with $T^{\beta_n} L_i(y) \to U_i(y)$ in $X^{[k-1]}$.

The next lemma is [20, Lemma 4.5 and Corollary 4.6], transferred to $\Gamma$. We formulate the acceptability condition so that it is meaningful while some of the sets are still empty.

**Lemma 2.11.** *Let $y \in X^{[k]}$ be an Erdős cube, and let $E \subset X$ be open with $y_{\vec{1}} \in E$. Then there are infinite sets $B_1, \ldots, B_k \subset \Gamma$ such that $T^{b_1+\cdots+b_k} y_{\vec{0}} \in E$ for all $b_i \in B_i$.*

*Proof.* For $\varepsilon \in [k]$ let $Z(\varepsilon) = \{j : \varepsilon_j = 0\}$. Call a tuple $(B_1, \ldots, B_k)$ of finite sets *acceptable* if, for every $\varepsilon \in [k]$ and every choice of $b_j \in B_j$ for $j \in Z(\varepsilon)$,

$$T^{\sum_{j \in Z(\varepsilon)} b_j} \, y_\varepsilon \in E. \tag{1}$$

For $\varepsilon = \vec{1}$ the sum is empty, and the condition reads $y_{\vec{1}} \in E$. Hence $(\emptyset, \ldots, \emptyset)$ is acceptable.

Let $(B_1, \ldots, B_k)$ be acceptable, and fix $i$. For $\varepsilon$ with $\varepsilon_i = 0$, let

$$H_\varepsilon = \bigcap T^{-\sum_{j \in Z(\varepsilon) \setminus \{i\}} b_j} E,$$

the intersection over all choices $b_j \in B_j$, $j \in Z(\varepsilon) \setminus \{i\}$ (an empty intersection is $X$). This is a finite intersection of open sets, so $H_\varepsilon$ is open. Since $Z(\varepsilon + e_i) = Z(\varepsilon) \setminus \{i\}$, acceptability at $\varepsilon + e_i$ says that $y_{\varepsilon+e_i} \in H_\varepsilon$.

Because $y$ is an Erdős cube, there are pairwise distinct $\beta_n$ with $T^{\beta_n} y_\varepsilon \to y_{\varepsilon+e_i}$ for all $\varepsilon$ with $\varepsilon_i = 0$. Hence for all large $n$ we have $T^{\beta_n} y_\varepsilon \in H_\varepsilon$ for every such $\varepsilon$. Choose such a $b^* = \beta_n$ with $b^* \notin B_1 \cup \cdots \cup B_k$, and put $B_i' = B_i \cup \{b^*\}$ and $B_j' = B_j$ for $j \ne i$.

We check (1) for the new tuple. Let $\varepsilon \in [k]$, and choose $b_j \in B_j'$ for $j \in Z(\varepsilon)$. There are two cases.

- If $i \notin Z(\varepsilon)$, or if $b_i \in B_i$, then (1) is an old condition.
- If $i \in Z(\varepsilon)$ and $b_i = b^*$, then $T^{\sum_{j\in Z(\varepsilon)} b_j} y_\varepsilon = T^{\sum_{j\in Z(\varepsilon)\setminus\{i\}} b_j}(T^{b^*} y_\varepsilon) \in E$, because $T^{b^*} y_\varepsilon \in H_\varepsilon$. This step uses commutativity of $\Gamma$.

Now add elements in the order $i = 1, 2, \dots, k, 1, 2, \dots$. This produces increasing acceptable tuples whose unions $B_1, \dots, B_k$ are infinite. Each finite subfamily is contained in an acceptable stage, so (1) holds for the unions. Taking $\varepsilon = \vec{0}$ gives the claim. (Since $b^*$ is always chosen outside $B_1 \cup \cdots \cup B_k$, the sets $B_i$ are in addition pairwise disjoint.) □

2.5. **Relative products over the Kronecker factor.** Throughout this subsection, $(Y, T)$ is a topological $\Gamma$-system, $\lambda$ is an ergodic measure on $Y$, and $\kappa\colon Y \to K$ is a Kronecker map with rotation $\alpha$ and disintegration $(\lambda_z)$.

**Definition 2.12.** For $t \in K$, the *relative product with parameter* $t$ is the measure on $Y \times Y$

$$J_t^\lambda := \int_K \lambda_z \otimes \lambda_{z+t}\, dm_K(z).$$

The *graph measure* is $\rho_\lambda := (\mathrm{id}, \kappa)_*\lambda$ on $Y \times K$. Here $\Gamma$ acts on $Y \times K$ by $T^\gamma \times (\cdot + \alpha(\gamma))$. We write $\Sigma_\lambda := \operatorname{supp} \rho_\lambda$.

For $\Gamma = \mathbb{Z}$ these measures appear in [20, (3.10) and Proposition 3.11], and for amenable groups in [8, Definition 4.7 and Theorem 4.8], where $(y, y') \mapsto J^\lambda_{\kappa(y')-\kappa(y)}$ is shown to be a continuous ergodic decomposition of $\lambda \times \lambda$ when $\kappa$ is continuous. Lemma 2.13 below is the version we need; we include its short proof.

Translations preserve $m_K$-null sets. Hence $J_t^\lambda$ does not depend on the version of the disintegration, and it is defined for *every* $t \in K$. The graph measure $\rho_\lambda$ is invariant and concentrated on the graph of $\kappa$. The projection onto $Y$ is therefore a measure-theoretic isomorphism $(Y \times K, \rho_\lambda) \to (Y, \lambda)$, and $\rho_\lambda$ is ergodic. For Borel sets $S \subset Y$ and $O \subset K$ we have

$$\rho_\lambda(S \times O) = \int_O \lambda_z(S)\, dm_K(z). \tag{2}$$

**Lemma 2.13.**

*(a) For bounded Borel $f, g$ on $Y$ and every $t \in K$, $\int f \otimes g\, dJ_t^\lambda = \int_K \tilde{f}(z)\, \tilde{g}(z+t)\, dm_K(z)$.*
*(b) Each $J_t^\lambda$ is invariant under the diagonal action, has both marginals equal to $\lambda$, and is concentrated on $\{(y, y') : \kappa(y') - \kappa(y) = t\}$. The map $t \mapsto J_t^\lambda$ is weak* continuous.*
*(c) $J_t^\lambda$ is ergodic for $m_K$-almost every $t$.*

*Proof.* (a) is the definition.

(b): The marginals are $\int \lambda_z\, dm_K = \lambda$ and $\int \lambda_{z+t}\, dm_K = \lambda$. Concentration follows from $\lambda_z(\kappa^{-1}(z)) = 1$ for a.e. $z$. For invariance, use (a) and $\widetilde{f \circ T^\gamma} = \tilde{f}(\cdot + \alpha(\gamma))$:

$$J_t^\lambda\big((f \circ T^\gamma) \otimes (g \circ T^\gamma)\big) = \int \tilde{f}(z + \alpha(\gamma))\, \tilde{g}(z + t + \alpha(\gamma))\, dm_K(z) = J_t^\lambda(f \otimes g).$$

For continuity, note that $t \mapsto \int \tilde{f}(z)\tilde{g}(z+t)\, dm_K$ is continuous, by continuity of translation in $L^2(m_K)$. Linear combinations of functions $f \otimes g$ with $f, g \in C(Y)$ are uniformly dense in $C(Y \times Y)$.

(c): Let $D(y, y') = \kappa(y') - \kappa(y)$ and $\Lambda = \lambda \times \lambda$.

*Step 1: $J$ disintegrates $\Lambda$ over $D$.* By (a) and Fubini, for bounded Borel $f, g$ on $Y$ and $h$ on $K$,

$$\int f \otimes g \cdot (h \circ D)\, d\Lambda = \iint \tilde{f}(z)\, \tilde{g}(z')\, h(z' - z)\, dm_K\, dm_K = \int_K h(t)\, J_t^\lambda(f \otimes g)\, dm_K(t). \tag{3}$$

By a monotone class argument, (3) holds with $f\otimes g$ replaced by any bounded Borel $F$ on $Y\times Y$, and $t\mapsto J_t^\lambda(F)$ is Borel. Together with (b), this shows that $t\mapsto J_t^\lambda$ is the disintegration of $\Lambda$ over $D$. In particular $\mathbb{E}_\Lambda(F\mid\sigma(D))(y,y') = J^\lambda_{D(y,y')}(F)$ for $\Lambda$-a.e. $(y,y')$ (recall $D_*\Lambda = m_K$).

*Step 2: invariant functions of $\Lambda$ are $\sigma(D)$-measurable.* Write $L^2(\lambda) = H_c\oplus H_w$, where $H_c = L^2(\sigma(\kappa))$ has orthonormal basis $\{\varphi\circ\kappa : \varphi\in\widehat{K}\}$. Let $P_1$ be the projection onto the invariant functions in $L^2(\Lambda)$.

- For $f\in H_w$ and $g\in L^2(\lambda)$, we have $\langle\mathcal{U}_\gamma(f\otimes g), f\otimes g\rangle = \widehat{\sigma_f}(\gamma)\widehat{\sigma_g}(\gamma)$, so $\sigma_{f\otimes g} = \sigma_f * \sigma_g$. This has no atoms because $\sigma_f$ has none (Lemma 2.8(c)). Hence $P_1(f\otimes g) = 0$ by Lemma 2.6(c). The same holds for $g\otimes f$.
- The function $\varphi\circ\kappa\otimes\varphi'\circ\kappa$ is an eigenfunction with eigenvalue $(\varphi\varphi')\circ\alpha$. Since $\alpha(\Gamma)$ is dense, this eigenvalue is trivial only if $\varphi' = \overline{\varphi}$.

Since $L^2(\Lambda)$ is the orthogonal sum of $H_c\otimes H_c$, $H_w\otimes L^2(\lambda)$ and $H_c\otimes H_w$, and $P_1$ maps an eigenfunction with nontrivial eigenvalue to 0, $P_1L^2(\Lambda)$ is the closed span of the functions $\varphi(\kappa(y))\overline{\varphi(\kappa(y'))} = \overline{\varphi}(D(y,y'))$, all of which are $\sigma(D)$-measurable. Conversely $D$ is invariant. So $P_1F = \mathbb{E}_\Lambda(F\mid\sigma(D))$, which is the function $(y,y')\mapsto J^\lambda_{D(y,y')}(F)$ by Step 1.

*Step 3: ergodicity of almost every $J_t^\lambda$.* Fix a Følner sequence $\Phi$, write $A_NF = \mathbb{E}_{\gamma\in\Phi_N} F\circ T^\gamma$, and let $F\in C(Y\times Y)$. The function $t\mapsto \|A_NF - J_t^\lambda(F)\|^2_{L^2(J_t^\lambda)}$ is continuous by (b). Since $J_t^\lambda$ is concentrated on $D^{-1}(t)$,

$$\int_K \left\|A_NF - J_t^\lambda(F)\right\|^2_{L^2(J_t^\lambda)}\,dm_K(t) = \left\|A_NF - J^\lambda_{D(\cdot)}(F)\right\|^2_{L^2(\Lambda)} \xrightarrow[N\to\infty]{} 0$$

by Lemma 2.6(b) and Step 2. For each $t$, the integrand converges to $\|P_1^{(t)}F - J_t^\lambda(F)\|^2_{L^2(J_t^\lambda)}$, where $P_1^{(t)}$ is the invariant projection in $L^2(J_t^\lambda)$. By Fatou's lemma this limit vanishes for a.e. $t$. Applying this to a countable dense subset of $C(Y\times Y)$ shows the following. For a.e. $t$, $P_1^{(t)}$ maps a dense subset of $L^2(J_t^\lambda)$, and hence all of it, to constants. That is, $J_t^\lambda$ is ergodic. □

**Lemma 2.14** (Kronecker factors of factors)**.** *Let $\lambda$ be ergodic on $Y$ with Kronecker map $\kappa\colon Y\to K$ (rotation $\alpha$). Let $P\colon Y\to Z$ be a Borel map into a topological $\Gamma$-system with $P_*\lambda = \nu$ and, for each $\gamma$, $P\circ T^\gamma = T^\gamma\circ P$ $\lambda$-a.e., and let $\iota\colon Z\to K'$ be a Kronecker map for $\nu$ (rotation $\alpha'$). Then:*

*(a) $\mathbb{E}_\lambda(f\circ P\mid\kappa) = \mathbb{E}_\nu(f\mid\iota)\circ P$ for $f\in L^2(\nu)$;*
*(b) there is a unique continuous homomorphism $\bar\chi\colon K\to K'$ with $\bar\chi\circ\alpha = \alpha'$. It is surjective, and there is $c\in K'$ with $\iota\circ P = \bar\chi\circ\kappa + c$ $\lambda$-a.e.;*
*(c) for* every *$s\in K$, $(P\times P)_*J_s^\lambda = J^\nu_{\bar\chi(s)}$.*

*Proof.* (a): Write $f = f_c + f_w$ with $f_c\in H_c(\nu)$ and $f_w\perp H_c(\nu)$. Eigenfunctions of $\nu$ pull back under $P$ to eigenfunctions of $\lambda$, so $f_c\circ P\in H_c(\lambda)$. Since $P$ is a factor map, $\sigma_{f_w\circ P} = \sigma_{f_w}$, which has no atoms. Hence $f_w\circ P\perp H_c(\lambda)$ by Lemma 2.8(c).

(b): Uniqueness holds because $\alpha(\Gamma)$ is dense. For existence, take $\psi\in\widehat{K'}$. Then $\psi\circ\iota\circ P$ is an eigenfunction of $\lambda$ with eigenvalue $\psi\circ\alpha'$. By Lemma 2.8(b) there is a unique $\varphi_\psi\in\widehat{K}$ with $\varphi_\psi\circ\alpha = \psi\circ\alpha'$. By uniqueness, $\psi\mapsto\varphi_\psi$ is a homomorphism $\widehat{K'}\to\widehat{K}$. By Pontryagin duality there is a continuous homomorphism $\bar\chi\colon K\to K'$ with $\psi\circ\bar\chi = \varphi_\psi$ for all $\psi$. Then $\bar\chi\circ\alpha = \alpha'$. The image $\bar\chi(K)$ is closed and contains the dense set $\alpha'(\Gamma)$, so $\bar\chi$ is onto. Finally, $\iota\circ P - \bar\chi\circ\kappa$ is $\lambda$-a.e. invariant, since both terms are shifted by $\alpha'(\gamma)$ under $T^\gamma$. By ergodicity it is a.e. equal to a constant $c$.

(c): Let $f, g$ be bounded Borel on $Z$. By (a) and (b), $\mathbb{E}_\lambda(f\circ P \mid \kappa) = \tilde f(\iota\circ P) = \tilde f(\bar\chi\circ\kappa + c)$, so $\widetilde{f\circ P}(z) = \tilde f(\bar\chi z + c)$ for $m_K$-a.e. $z$. Here $\tilde f$ is formed with respect to $(\nu,\iota)$. The surjective homomorphism $\bar\chi$ maps $m_K$ to $m_{K'}$. Hence, by Lemma 2.13(a),

$$\int f\otimes g\, d(P\times P)_* J_s^\lambda = \int_K \tilde f(\bar\chi z + c)\,\tilde g(\bar\chi z + \bar\chi s + c)\, dm_K(z) = \int_{K'} \tilde f(u)\,\tilde g(u+\bar\chi s)\, dm_{K'}(u),$$

and the right-hand side is $J^\nu_{\bar\chi(s)}(f\otimes g)$. Both sides are probability measures, so they agree. □

The next lemma is [20, Lemma 3.18] for $\Gamma=\mathbb{Z}$, and the Claim in the proof of [8, Theorem 4.10] for amenable groups. We include a short proof, based on the van der Corput inequality rather than on the continuity of the ergodic decomposition. In the argument of [24] its role is played by the choice of a generic $y$ such that $(x,y)$ is generic for some (not necessarily ergodic) joining.

**Lemma 2.15** (Generic pairs)**.** *Suppose in addition that $\pi\colon Y\to K$ is a* continuous *Kronecker map with $\pi(T^\gamma y) = \pi(y)+\alpha(\gamma)$ for* all *$y$ and $\gamma$. Let $(J_t)$ be the relative products formed with respect to $\pi$. Let $y_0\in \mathrm{gen}(\lambda,\Phi)$. Then there are a subsequence $\Phi'$ of $\Phi$ and a Borel set $Y_0\subset Y$ with $\lambda(Y_0)=1$ such that*

$$(y_0,y)\in \mathrm{gen}\big(J_{\pi(y)-\pi(y_0)},\Phi'\big) \qquad \text{for every } y\in Y_0.$$

*Proof.* Let $f,g\in C(Y)$. We show that, in $L^2(\lambda)$ (as functions of $y$),

(4)
$$\mathop{\mathbb{E}}_{\gamma\in\Phi_N} f(T^\gamma y_0)\, g(T^\gamma y) \;\longrightarrow\; \Lambda_{f,g}(y) := J_{\pi(y)-\pi(y_0)}(f\otimes g) = \int \tilde f(z)\,\tilde g(z+\pi(y)-\pi(y_0))\, dm_K(z).$$

Fix $\tau>0$. Choose a trigonometric polynomial $\tilde f' = \sum_j c_j\varphi_j$ on $K$ with $\|\tilde f-\tilde f'\|_{L^2(m_K)}<\tau$, and put $f'=\tilde f'\circ\pi$. The function $f'$ is continuous because $\pi$ is. Let $f''=f-f'=f^\perp+f^\sharp$, with $f^\perp = f-\mathbb{E}_\lambda(f\mid\pi)$ and $f^\sharp = (\tilde f-\tilde f')\circ\pi$. Then $\|f^\sharp\|_2<\tau$, and $\sigma_{f^\perp}$ has no atoms (Lemma 2.8(c)).

*The part $f'$.* We have $f'(T^\gamma y_0) = \sum_j c_j\varphi_j(\pi y_0)\,\varphi_j(\alpha(\gamma))$. By Lemma 2.6(b) applied to $V_\gamma = \varphi_j(\alpha(\gamma))\mathcal{U}_\gamma$, the average $\mathbb{E}_{\Phi_N}\varphi_j(\alpha(\gamma))\mathcal{U}_\gamma g$ converges to the projection of $g$ onto $H_{\overline{\varphi_j\circ\alpha}}$. That eigenspace is spanned by $\overline{\varphi_j}\circ\pi$, so the projection is $\big(\int \tilde g\,\varphi_j\,dm_K\big)\,\overline{\varphi_j}\circ\pi$. Hence the averages for $f'$ converge in $L^2(\lambda)$ to

$$\Lambda'_{f,g}(y) = \sum_j c_j\,\varphi_j(\pi y_0)\,\overline{\varphi_j(\pi y)}\int \tilde g\,\varphi_j\,dm_K.$$

A change of variables shows that $\Lambda'_{f,g}(y) = \int \tilde f'(z)\tilde g(z+\pi(y)-\pi(y_0))\,dm_K(z)$. It follows that $|\Lambda_{f,g}-\Lambda'_{f,g}|\le\tau\|g\|_\infty$ everywhere.

*The part $f''$.* Put $u(\gamma)=f''(T^\gamma y_0)$. By the spectral theorem,

$$\Big\|\mathop{\mathbb{E}}_{\gamma\in\Phi_N} u(\gamma)\mathcal{U}_\gamma g\Big\|_2^2 = \int_{\widehat\Gamma}\Big|\mathop{\mathbb{E}}_{\gamma\in\Phi_N} u(\gamma)\psi(\gamma)\Big|^2 d\sigma_g(\psi).$$

We bound the integrand uniformly in $\psi$ by a van der Corput argument. Fix a finite $H\subset\Gamma$. By the Følner property, $\mathbb{E}_{\Phi_N} u\psi = \mathbb{E}_{\gamma\in\Phi_N}\mathbb{E}_{h\in H}(u\psi)(\gamma+h)+o(1)$. By Cauchy–Schwarz,

$$\Big|\mathop{\mathbb{E}}_{\Phi_N} u\psi\Big|^2 \le \frac{1}{|H|^2}\sum_{h,h'\in H}\Big|\mathop{\mathbb{E}}_{\gamma\in\Phi_N} f''(T^{\gamma+h}y_0)\overline{f''(T^{\gamma+h'}y_0)}\Big| + o(1).$$

The function $f''$ is continuous and $y_0$ is generic, so the inner averages converge to $\langle \mathcal{U}_{h-h'} f'', f''\rangle$. Moreover, $|\langle \mathcal{U}_\delta f'', f''\rangle - \langle \mathcal{U}_\delta f^\perp, f^\perp\rangle| \le \tau(\|f^\perp\|_2 + \|f''\|_2) \le \tau(2\|f\|_\infty + \tau)$. Take $H = \Theta_M$ for a Følner sequence $\Theta$. By Cauchy–Schwarz and the computation in Lemma 2.6(d),

$$\frac{1}{|\Theta_M|^2} \sum_{h,h' \in \Theta_M} |\langle \mathcal{U}_{h-h'} f^\perp, f^\perp\rangle| \le \Big( \iint \Big| \mathop{\mathbb{E}}_{h\in\Theta_M} (\psi\overline{\psi'})(h)\Big|^2 d\sigma_{f^\perp}\, d\sigma_{f^\perp}\Big)^{1/2} \to 0.$$

Hence $\sup_\psi \limsup_N |\,\mathbb{E}_{\Phi_N} u\psi|^2 \le \tau(2\|f\|_\infty + \tau)$. The integrands are bounded and $\sigma_g$ is finite, so by reverse Fatou

$$\limsup_N \Big\| \mathop{\mathbb{E}}_{\Phi_N} u(\gamma)\mathcal{U}_\gamma g\Big\|_2^2 \le \tau(2\|f\|_\infty + \tau)\|g\|_2^2.$$

Combining the two parts,

$$\limsup_N \Big\| \mathop{\mathbb{E}}_{\gamma\in\Phi_N} f(T^\gamma y_0)\,\mathcal{U}_\gamma g - \Lambda_{f,g}\Big\|_2 \le \tau\|g\|_\infty + \big(\tau(2\|f\|_\infty + \tau)\big)^{1/2}\|g\|_2,$$

and letting $\tau \to 0$ proves (4).

Now let $\mathcal{D}$ be a countable dense subset of $C(Y)$. Convergence in $L^2$ implies a.e. convergence along a subsequence. A diagonal argument gives a subsequence $\Phi'$ and a conull Borel set $Y_0$ such that the convergence in (4) holds at every $y \in Y_0$ along $\Phi'$, for all $f, g \in \mathcal{D}$. Both sides of (4) are bounded bilinear in $(f, g)$ with respect to the uniform norm. The span of $\{f \otimes g\}$ is dense in $C(Y \times Y)$. Hence $(y_0, y) \in \mathrm{gen}(J_{\pi(y)-\pi(y_0)}, \Phi')$ for $y \in Y_0$. □

2.6. **Re-basing and two support lemmas.** The following lemma is the rotation case of [8, Lemma 3.5] (see the remark following that lemma, which allows any topologically distal factor in place of the Kronecker factor). Charamaras and Mountakis prove it for amenable groups, extending a result of Host and Kra for $\mathbb{Z}$-actions [14, Proposition 6.1] (see also [20, Proposition 3.19]); their proof is in [8, Appendix A]. For rotations the Ellis-semigroup step of that proof reduces to compactness of $K$, so we include the short argument.

**Lemma 2.16** (Re-basing)**.** *Let $(W, T)$ be a topological $\Gamma$-system and let $w \in W$ have dense orbit. Let $\lambda$ be an ergodic measure on $W$. Let $\kappa\colon W \to K$ be a Borel map into a compact metrizable abelian group, and $\alpha\colon \Gamma \to K$ a homomorphism, such that for each $\gamma$, $\kappa(T^\gamma x) = \kappa(x) + \alpha(\gamma)$ for $\lambda$-a.e. $x$. Let $\hat\lambda = (\mathrm{id}, \kappa)_*\lambda$ on $W \times K$, with the action $T^\gamma \times (\cdot + \alpha(\gamma))$. Then there are $d \in K$ and a Følner sequence $\Psi$ such that $(w, d) \in \mathrm{gen}(\hat\lambda, \Psi)$.*

*Proof.* The measure $\hat\lambda$ is invariant and ergodic, since it is isomorphic to $\lambda$ via the projection. By (G4) there are a point $(x_1, z_1)$ and a Følner sequence $\Phi$ with $(x_1, z_1) \in \mathrm{gen}(\hat\lambda, \Phi)$.

For each $N$, the finitely many maps $T^\gamma$, $\gamma \in \Phi_N$, are continuous at $x_1$, and the orbit of $w$ is dense. So there is $h_N \in \Gamma$ with $\sup_{\gamma\in\Phi_N} d_W(T^{\gamma+h_N} w, T^\gamma x_1) < 1/N$.

By compactness, $\alpha(h_{N_j}) \to u_*$ in $K$ along a subsequence. Put $d = z_1 - u_*$ and $\Psi_j = \Phi_{N_j} + h_{N_j}$; this is again a Følner sequence. For $F \in C(W \times K)$,

$$\mathop{\mathbb{E}}_{\gamma\in\Psi_j} F(T^\gamma w, d + \alpha(\gamma)) = \mathop{\mathbb{E}}_{\gamma\in\Phi_{N_j}} F\big(T^{\gamma+h_{N_j}} w,\ z_1 + \alpha(\gamma) + (\alpha(h_{N_j}) - u_*)\big).$$

The arguments are uniformly within $o(1)$ of $(T^\gamma x_1, z_1 + \alpha(\gamma))$: in the first coordinate by the choice of $h_N$, and in the second by uniform continuity of addition on $K$. Uniform continuity of $F$ and genericity of $(x_1, z_1)$ give the claim. □

**Lemma 2.17** (Support of the graph measure)**.** *Let $(Z, T)$ be a topological $\Gamma$-system, $\nu$ ergodic on $Z$, and $\iota\colon Z \to K$ a Kronecker map with rotation $\alpha$. Let $\rho = \rho_\nu$ and $\Sigma = \operatorname{supp}\rho \subset Z \times K$. Suppose $(p, \theta) \in Z \times K$ is generic for $\rho$ along some Følner sequence, and $p \in \operatorname{supp}\nu$. Then $(p, \theta) \in \Sigma$.*

*Proof.* The projection of $\Sigma$ to $Z$ is compact and has full $\nu$-measure, so it contains $\operatorname{supp}\nu$. Pick $\theta_0$ with $(p,\theta_0)\in\Sigma$ and put $\zeta=\theta-\theta_0$.

For $u\in K$ let $S_u(x,t)=(x,t+u)$. These maps commute with the action, so $S_\zeta\Sigma$ is closed and invariant. It contains $S_\zeta(p,\theta_0)=(p,\theta)$, and hence the orbit closure of $(p,\theta)$, which contains $\Sigma$ by (G2). So $\Sigma\subset S_\zeta\Sigma$, that is, $S_{-\zeta}\Sigma\subset\Sigma$, and hence $S_{-n\zeta}\Sigma\subset\Sigma$ for all $n\geq 1$.

By compactness there are $1\leq n_1<n_2<\cdots$ with $n_{l+1}-n_l\geq 2$ and $n_l\zeta\to\zeta_*$. Then $(n_{l+1}-n_l)\zeta\to 0$, so with $q_l=n_{l+1}-n_l-1\geq 1$ we get $-q_l\zeta\to\zeta$. Since $\Sigma$ is closed, $S_\zeta\Sigma\subset\Sigma$, and so $(p,\theta)=S_\zeta(p,\theta_0)\in\Sigma$. □

**Lemma 2.18** (Phase returns)**.** *Let $(W,T)$ and $(Z,T)$ be topological $\Gamma$-systems, $\lambda$ ergodic on $W$, and $w\in\operatorname{gen}(\lambda,\Psi)$ for some Følner sequence $\Psi$. Let $L,U\colon W\to Z$ be continuous equivariant maps with $L_*\lambda=U_*\lambda=:\nu$. Let $\iota\colon Z\to K$ be a Kronecker map for $\nu$ with rotation $\alpha$.*

*(a) There is $r\in K$ with $\iota\circ U=\iota\circ L+r$ $\lambda$-a.e.*

*(b) If $U(w)\in\operatorname{supp}\nu$, then for every open $V\ni U(w)$ and every open $O\ni 0$ in $K$, there is a Følner sequence $\Omega$ along which the set*

$$\{\gamma\in\Gamma:\ T^\gamma L(w)\in V,\ \alpha(\gamma)\in r+O\}$$

*has positive lower density; in particular this set has positive upper Banach density.*

*Proof.* (a): Both $\iota\circ U$ and $\iota\circ L$ are shifted by $\alpha(\gamma)$ under $T^\gamma$ (a.e.), so their difference is $\lambda$-a.e. invariant, hence a.e. constant.

(b): Let $W'$ be the orbit closure of $w$. Then $\lambda(W')=1$ by (G2), and $w$ has dense orbit in $W'$. The map $\kappa'=\iota\circ L$ satisfies the hypotheses of Lemma 2.16 on $W'$. That lemma gives $z\in K$ and a Følner sequence $\Omega$ with $(w,z)$ generic along $\Omega$ for $\hat\lambda=(\mathrm{id},\iota\circ L)_*\lambda$. Consider the continuous equivariant maps $(x,t)\mapsto(Lx,t)$ and $(x,t)\mapsto(Ux,t+r)$ from $W'\times K$ to $Z\times K$. They push $\hat\lambda$ forward to $(L,\iota\circ L)_*\lambda=\rho_\nu$ and to $(U,\iota\circ L+r)_*\lambda=(U,\iota\circ U)_*\lambda=\rho_\nu$, respectively. By (G3), both $(L(w),z)$ and $(U(w),z+r)$ are generic for $\rho_\nu$ along $\Omega$. By Lemma 2.17, $(U(w),z+r)\in\operatorname{supp}\rho_\nu$. So $V\times(z+r+O)$ is an open set meeting $\operatorname{supp}\rho_\nu$, and by (G1) the set

$$\{\gamma:(T^\gamma L(w),z+\alpha(\gamma))\in V\times(z+r+O)\}=\{\gamma:T^\gamma L(w)\in V,\ \alpha(\gamma)\in r+O\}$$

has positive lower density along $\Omega$. □

**Lemma 2.19** (Pointwise support)**.** *Let $\nu$, $\iota\colon Z\to K$, $\rho=\rho_\nu$, $\Sigma=\operatorname{supp}\rho$ and $J_t=J_t^\nu$ be as above. If $(p,\theta)\in\Sigma$, then for $\rho$-almost every $(q,\varphi)\in Z\times K$,*

$$(p,q)\in\operatorname{supp}J_{\varphi-\theta}.$$

*More precisely, the set of $(q,\varphi)$ with $(p,q)\notin\operatorname{supp}J_{\varphi-\theta}$ is contained in a $\rho$-null Borel set.*

*Proof.* Fix open sets $O_1\ni p$ and $O_2\subset Z$. Define Borel functions $f(\xi)=\nu_\xi(O_1)$ and $g(\xi)=\nu_\xi(O_2)$ on $K$, with values in $[0,1]$, and let $F(t)=J_t(O_1\times O_2)=\int f(\xi)g(\xi+t)\,dm_K(\xi)$. The function $F$ is continuous.

By (2), $\int_N f\,dm_K=\rho(O_1\times N)>0$ for every open $N\ni\theta$, because $O_1\times N$ is a neighbourhood of $(p,\theta)\in\Sigma$.

Let $C=\{\varphi\in K:F(\varphi-\theta)=0\}$, a closed set. We claim $\int_C g\,dm_K=0$. Suppose not. Put $R(k)=\int_C g(\varphi)\,g(k+\varphi-\theta)\,dm_K(\varphi)$. Then $R\geq 0$, and $R$ is continuous since $|R(k)-R(k')|\leq\|g(\cdot+k)-g(\cdot+k')\|_{L^1(m_K)}$. Moreover $R(\theta)=\int_C g^2\,dm_K>0$. So $R>R(\theta)/2$ on some open $N\ni\theta$, and

$$\int f\,R\,dm_K\ \geq\ \tfrac12 R(\theta)\int_N f\,dm_K>0.$$

On the other hand, by Fubini,

$$\int f(k)R(k)\,dm_K(k) = \int_C g(\varphi)\int f(k)\,g(k+\varphi-\theta)\,dm_K(k)\,dm_K(\varphi)$$
$$= \int_C g(\varphi)F(\varphi-\theta)\,dm_K(\varphi) = 0,$$

a contradiction.

By (2), $\int_C g\,dm_K = \rho\big(O_2\times C\big)$. So

$$\mathcal{N}_{O_1,O_2} := \{(q,\varphi):\ q\in O_2,\ J_{\varphi-\theta}(O_1\times O_2)=0\} = O_2\times C$$

is a $\rho$-null Borel set. Let $\mathcal{B}$ be a countable base of $Z$. If $(p,q)\notin \operatorname{supp} J_{\varphi-\theta}$, then some $O_1\times O_2$ with $O_1,O_2\in\mathcal{B}$, $p\in O_1$ and $q\in O_2$ satisfies $J_{\varphi-\theta}(O_1\times O_2)=0$, so $(q,\varphi)\in\mathcal{N}_{O_1,O_2}$. The union of these countably many $\rho$-null sets is $\rho$-null. □

*Remark* 2.20. For a fixed point $p$, the proof uses only that $\theta\in\operatorname{supp}(f_{O_1}m_K)$ for every open $O_1\ni p$, where $f_{O_1}(\xi)=\nu_\xi(O_1)$. By (2) this is equivalent to $(p,\theta)\in\Sigma$.

2.7. **Cube data and the cube criterion.** From now on, $(X,\mu,T)$ is an ergodic system on a compact metric space, $a\in X$, and $E\subset X$ is open with $\mu(E)>0$.

**Definition 2.21.** Let $m\ge 0$. An *$m$-dimensional cube datum* is a triple $(w,\lambda,\Psi)$ with the following properties:

- (D1) $\lambda$ is an ergodic measure on $X^{[m]}$ (for the diagonal action), $\Psi$ is a Følner sequence, $w\in\operatorname{gen}(\lambda,\Psi)$, and $w_{\vec 0}=a$;
- (D2) every vertex marginal of $\lambda$ equals $\mu$;
- (D3) (face symmetry) $(L_i)_*\lambda=(U_i)_*\lambda=:\nu_i$ for $1\le i\le m$;
- (D4) (upper faces in the support) $U_i(w)\in\operatorname{supp}\nu_i$ for $1\le i\le m$.

If $a\in\operatorname{gen}(\mu,\Psi)$, then $(a,\mu,\Psi)$ is a 0-dimensional cube datum, since (D3) and (D4) are empty. Each face measure $\nu_i$ is ergodic, being a factor of $\lambda$. In what follows, $\iota_i\colon X^{[m-1]}\to K_i$ denotes a Kronecker map for $\nu_i$, with rotation $\alpha_i$.

The next proposition is our version of Claim 1 in the proof of [24, Theorem 3.1], together with the iteration that follows it. The structure of the argument comes from there:

- the splitting into a Kronecker part and a weak mixing part;
- the positivity remark [24, Remark 2.2];
- the averaging of the weak mixing part along *every* Følner sequence.

What is new is the treatment of the Kronecker part. We use continuity of translation on the Kronecker group of the face measure and the phase returns of Lemma 2.18. No continuity of eigenfunctions is needed.

**Proposition 2.22** (Cube criterion)**.** *If an $m$-dimensional cube datum $(w,\lambda,\Psi)$ exists, then there is $x\in X^{[m]}$ such that $y=(w,x)\in X^{[m+1]}$ is an $(m+1)$-dimensional Erdős cube with $y_{\vec 0}=a$ and $y_{\vec 1}=x_{\vec 1}\in E$.*

*Proof.* Fix a metric on $X$ and use the maximum metric on products. An *open box* is a set $V=\prod_{\varepsilon\in[m]}V_\varepsilon\subset X^{[m]}$ with each $V_\varepsilon$ open. For $1\le i\le m$ and $\beta\in\Gamma$ put $U_i(V)=\prod_{\eta\in[m-1]}V_{\eta^{(i,1)}}$ and

$$W_i(V,\beta)=\{x\in V:\ T^\beta L_i(x)\in U_i(V)\}=\{x\in V:\ T^\beta x_{\eta^{(i,0)}}\in V_{\eta^{(i,1)}}\ \forall\eta\in[m-1]\}.$$

This set is open, and $V\subset U_i^{-1}(U_i(V))$.

*Claim A (direction $m+1$).* If $V$ is open with $\lambda(V) > 0$, there are infinitely many $\beta$ with $T^\beta w \in V$.

This is (G1) together with (D1).

*Claim B (directions $i \le m$).* If $V$ is an open box with $\lambda(V) > 0$, $1 \le i \le m$ and $\delta > 0$, then there are infinitely many $\beta$ such that

$$d\big(T^\beta L_i(w), U_i(w)\big) < \delta \qquad \text{and} \qquad \lambda\big(W_i(V,\beta)\big) > 0.$$

*Proof of Claim B.* Let $F = 1_{U_i(V)}$ on $X^{[m-1]}$. Since $L_i$ is equivariant,

$$\lambda(W_i(V,\beta)) = \int 1_V(x)\, F(L_i T^\beta x)\, d\lambda(x) = \langle F \circ L_i \circ T^\beta, 1_V\rangle_\lambda. \tag{5}$$

In $L^2(\nu_i)$, write $F = F_c + F_w$, where $F_c = \mathbb{E}_{\nu_i}(F \mid \iota_i) = G \circ \iota_i$ with $G\colon K_i \to [0,1]$ Borel, and $F_w \perp H_c(\nu_i)$. Let $r_i$ be the constant of Lemma 2.18(a) for $L = L_i$ and $U = U_i$, so that $\iota_i \circ U_i = \iota_i \circ L_i + r_i$ $\lambda$-a.e.

*Positivity.* Let $\Delta = \{F_c > 0\}$, a $\sigma(\iota_i)$-measurable set. Then $\int_{\Delta^c} F\, d\nu_i = \int_{\Delta^c} F_c\, d\nu_i = 0$, so $\nu_i(U_i(V)\setminus\Delta) = 0$ [24, Remark 2.2]. Since $(U_i)_*\lambda = \nu_i$ and $V \subset U_i^{-1}(U_i(V))$, we get $F_c(U_i x) > 0$ for $\lambda$-a.e. $x \in V$. Hence

$$\eta_V := \int_V F_c(U_i x)\, d\lambda(x) > 0.$$

*The Kronecker part.* For every $\beta$ and $\lambda$-a.e. $x$ we have $F_c(T^\beta L_i x) = G(\iota_i(L_i x) + \alpha_i(\beta))$ and $F_c(U_i x) = G(\iota_i(L_i x) + r_i)$. Here we use that $L_i$ and $U_i$ push $\lambda$ to $\nu_i$, that $\iota_i$ is a.e. equivariant, and the choice of $r_i$. Since $(\iota_i \circ L_i)_*\lambda = m_{K_i}$,

$$\big|\langle F_c \circ L_i \circ T^\beta, 1_V\rangle_\lambda - \eta_V\big| \le \int_{K_i} \big|G(z+\alpha_i(\beta)) - G(z+r_i)\big|\, dm_{K_i}(z) = \big\|G(\cdot+\alpha_i(\beta)-r_i) - G\big\|_{L^1(m_{K_i})}.$$

By continuity of translation in $L^1(m_{K_i})$, there is an open $O \ni 0$ in $K_i$ such that the right-hand side is $< \eta_V/4$ whenever $\alpha_i(\beta) \in r_i + O$.

*The weak mixing part.* Let $h = F_w \circ L_i \in L^2(\lambda)$, so that $F_w \circ L_i \circ T^\beta = \mathcal{U}_\beta h$. Since $L_i$ is a factor map, $\langle \mathcal{U}_\beta h, h\rangle_\lambda = \langle \mathcal{U}_\beta F_w, F_w\rangle_{\nu_i}$, so $\sigma_h = \sigma_{F_w}$, which has no atoms by Lemma 2.8(c). By Lemma 2.6(d), for every Følner sequence $\Theta$,

$$\mathop{\mathbb{E}}_{\beta\in\Theta_M} |\langle \mathcal{U}_\beta h, 1_V\rangle_\lambda| \to 0.$$

*Assembly.* Let $\mathcal{N}_\delta$ be the open $\delta$-ball around $U_i(w)$, and let

$$R_\delta = \{\beta :\ T^\beta L_i(w) \in \mathcal{N}_\delta,\ \alpha_i(\beta) \in r_i + O\}.$$

By (D4) and Lemma 2.18(b), applied with $L = L_i$, $U = U_i$ and $\nu = \nu_i$, there are a Følner sequence $\Omega$ and $c_0 > 0$ with $|R_\delta \cap \Omega_M| \ge c_0|\Omega_M|$ for all large $M$. Let $R'_\delta = \{\beta \in R_\delta : |\langle \mathcal{U}_\beta h, 1_V\rangle| \ge \eta_V/4\}$. By the weak mixing part, $|R'_\delta \cap \Omega_M| \le (4/\eta_V)\sum_{\beta\in\Omega_M} |\langle \mathcal{U}_\beta h, 1_V\rangle| = o(|\Omega_M|)$. Hence $|(R_\delta \setminus R'_\delta) \cap \Omega_M| \ge (c_0/2)|\Omega_M|$ for all large $M$, and $R_\delta \setminus R'_\delta$ is infinite by Lemma 2.5. For $\beta \in R_\delta \setminus R'_\delta$, by (5),

$$\lambda(W_i(V,\beta)) = \langle F_c \circ L_i \circ T^\beta, 1_V\rangle + \langle \mathcal{U}_\beta h, 1_V\rangle \ge \eta_V - \tfrac{\eta_V}{4} - \tfrac{\eta_V}{4} > 0.$$

This proves Claim B.

*Nested boxes.* Every open subset of $X^{[m]}$ is a countable union of open boxes whose closures lie in it and whose coordinates have arbitrarily small diameter. So every open set of positive $\lambda$-measure contains such a box of positive measure. Let $V_0 = \{x \in X^{[m]} : x_{\vec{1}} \in E\}$. This is an open box, and $\lambda(V_0) = \mu(E) > 0$ by (D2). Let $i_n \in \{1,\dots,m+1\}$ with $i_n \equiv n \pmod{m+1}$.

Suppose an open box $V_n$ with $\lambda(V_n) > 0$ has been chosen. Choose $\beta_n \notin \{\beta_0, \dots, \beta_{n-1}\}$ as follows.

- If $i_n = m+1$, let $\beta_n$ satisfy $T^{\beta_n} w \in V_n$ (Claim A), and put $W^* = V_n$.
- If $i_n = i \le m$, let $\beta_n$ satisfy $d(T^{\beta_n} L_i(w), U_i(w)) < 1/(n+1)$ and $\lambda(W_i(V_n, \beta_n)) > 0$ (Claim B), and put $W^* = W_i(V_n, \beta_n)$.

Then choose an open box $V_{n+1}$ with $\overline{V_{n+1}} \subset W^*$, $\lambda(V_{n+1}) > 0$, and all coordinate diameters $< 1/(n+1)$.

The compact sets $\overline{V_n}$ are nested and their diameters tend to 0. So $\bigcap_n \overline{V_n} = \{x\}$, and $x \in V_n$ for all $n$; in particular $x_{\vec{1}} \in E$. Let $y = (w, x)$.

- If $i_n = m+1$, then $T^{\beta_n} w$ and $x$ both lie in $V_n$, so $d(T^{\beta_n} L_{m+1}(y), U_{m+1}(y)) = d(T^{\beta_n} w, x) \le \operatorname{diam} V_n$.
- If $i_n = i \le m$, then $x \in W_i(V_n, \beta_n)$. So for every $\eta \in [m-1]$, both $T^{\beta_n} x_{\eta^{(i,0)}}$ and $x_{\eta^{(i,1)}}$ lie in $V_{n,\eta^{(i,1)}}$, and
$$d\big(T^{\beta_n} L_i(x), U_i(x)\big) \le \operatorname{diam} V_n.$$
Together with $d(T^{\beta_n} L_i(w), U_i(w)) < 1/(n+1)$, and $L_i(y) = (L_i w, L_i x)$, $U_i(y) = (U_i w, U_i x)$, this gives $T^{\beta_n} L_i(y) \to U_i(y)$ along these $n$.

The $\beta_n$ are pairwise distinct, so $y$ is an Erdős cube. Moreover $y_{\vec{0}} = w_{\vec{0}} = a$ and $y_{\vec{1}} = x_{\vec{1}} \in E$. □

## 2.8. **Building the cube level by level.**

**Proposition 2.23** (Induction step)**.** *If an $m$-dimensional cube datum exists, then an $(m+1)$-dimensional cube datum exists.*

*Proof.* Let $(w, \lambda, \Psi)$ be an $m$-dimensional cube datum, with face measures $\nu_i$ and Kronecker maps $\iota_i \colon X^{[m-1]} \to K_i$ (rotations $\alpha_i$), $1 \le i \le m$. Let $\kappa \colon X^{[m]} \to K$ be a Kronecker map for $\lambda$, with rotation $\alpha$. For $i \le m$, apply Lemma 2.14 to $P = U_i$ and to $P = L_i$. Both push $\lambda$ to $\nu_i$, so they give the same homomorphism $\bar{\chi}_i \colon K \to K_i$, the unique one with $\bar{\chi}_i \circ \alpha = \alpha_i$, together with constants $c_i^U, c_i^L \in K_i$ such that

$$\iota_i \circ U_i = \bar{\chi}_i \circ \kappa + c_i^U, \qquad \iota_i \circ L_i = \bar{\chi}_i \circ \kappa + c_i^L \qquad \lambda\text{-a.e.} \tag{6}$$

Moreover, for every $s \in K$,

$$(U_i \times U_i)_* J_s^\lambda = (L_i \times L_i)_* J_s^\lambda = J^{\nu_i}_{\bar{\chi}_i(s)}. \tag{7}$$

*Step 1: re-basing.* This is the graph construction of [20, Proposition 3.20] and [8, Proposition 3.7], applied to the current measure $\lambda$. Let $W$ be the orbit closure of $w$ in $X^{[m]}$, so that $\lambda(W) = 1$ by (G2). By Lemma 2.16 there are $d \in K$ and a Følner sequence $\Psi'$ with $(w, d) \in \operatorname{gen}(\hat{\lambda}, \Psi')$, where $\hat{\lambda} = (\mathrm{id}, \kappa)_* \lambda$. Let $\widehat{W} \subset W \times K$ be the orbit closure of $(w, d)$; then $\hat{\lambda}(\widehat{W}) = 1$. The projection $\mathrm{pr} \colon \widehat{W} \to X^{[m]}$ is a measure-theoretic isomorphism from $\hat{\lambda}$ to $\lambda$, with inverse $x \mapsto (x, \kappa(x))$. So $\pi(x, t) = t$ is a *continuous* Kronecker map for $\hat{\lambda}$, it is equivariant everywhere, and $\kappa \circ \mathrm{pr} = \pi$ $\hat{\lambda}$-a.e. By Lemma 2.14(c) (with $\bar{\chi} = \mathrm{id}$ and $c = 0$),

$$(\mathrm{pr} \times \mathrm{pr})_* J_s^{\hat{\lambda}} = J_s^\lambda \qquad \text{for every } s \in K. \tag{8}$$

*Step 2: the new top face.* Apply Lemma 2.15 to $(\widehat{W}, \hat{\lambda}, \pi)$ and $y_0 = (w, d)$. This gives a subsequence $\Psi''$ of $\Psi'$ and a $\hat{\lambda}$-conull set $\widehat{W}_0$ such that, for $\hat{x} \in \widehat{W}_0$,

$$((w, d), \hat{x}) \in \operatorname{gen}\big(J^{\hat{\lambda}}_{\pi(\hat{x}) - d}, \Psi''\big).$$

For $x \in X^{[m]}$ put $\hat{x} = (x, \kappa(x))$ and $s(x) = \kappa(x) - d$. Since $\hat{\lambda} = (\mathrm{id}, \kappa)_*\lambda$, we have $\hat{x} \in \widehat{W}_0$ for $\lambda$-a.e. $x$. For such $x$, projecting with (G3) and (8) gives

$$w' := (w, x) \in \mathrm{gen}\big(\lambda' := J^{\lambda}_{s(x)},\ \Psi''\big) \qquad \text{in } X^{[m]} \times X^{[m]} = X^{[m+1]}. \tag{9}$$

Note that $s(x)$ is $m_K$-distributed when $x \sim \lambda$. By Lemma 2.13(c), $\lambda' = J^{\lambda}_{s(x)}$ is ergodic for $\lambda$-a.e. $x$.

*Step 3: (D1)–(D3) for $(w', \lambda', \Psi'')$.*

- (D1): $\lambda'$ is invariant and (for a.e. $x$) ergodic, $w'$ is generic by (9), and $w'_{\vec{0}} = w_{\vec{0}} = a$.
- (D2): by Lemma 2.13(b) both marginals of $\lambda'$ on $X^{[m]}$ equal $\lambda$. So the vertex marginals of $\lambda'$ are those of $\lambda$, all equal to $\mu$.
- (D3), direction $m+1$: $(L_{m+1})_*\lambda'$ and $(U_{m+1})_*\lambda'$ are the two marginals of $\lambda'$, both equal to $\lambda$.
- (D3), direction $i \le m$: $L_i(y', y'') = (L_i y', L_i y'')$, and similarly for $U_i$. By (7), $(L_i)_*\lambda' = (U_i)_*\lambda' = J^{\nu_i}_{t_i}$ with $t_i := \bar{\chi}_i(s(x))$.

*Step 4: (D4) for $(w', \lambda', \Psi'')$.*

*Direction $m+1$.* We need $U_{m+1}(w') = x \in \mathrm{supp}\,\lambda$, which holds for $\lambda$-a.e. $x$.

*Direction $i \le m$.* Let $p = U_i(w)$ and $\theta = \bar{\chi}_i(d) + c_i^U$. The map $(z, u) \mapsto (U_i z, \bar{\chi}_i(u) + c_i^U)$ from $X^{[m]} \times K$ to $X^{[m-1]} \times K_i$ is continuous. It is equivariant because $\bar{\chi}_i \circ \alpha = \alpha_i$. By (6) it pushes $\hat{\lambda}$ to $(U_i, \iota_i \circ U_i)_*\lambda = (\mathrm{id}, \iota_i)_*\nu_i = \rho_{\nu_i}$. By (G3), $(p, \theta)$ is generic for $\rho_{\nu_i}$ along $\Psi'$. Since $p \in \mathrm{supp}\,\nu_i$ by (D4), Lemma 2.17 gives $(p, \theta) \in \mathrm{supp}\,\rho_{\nu_i}$. Now let

$$(q, \varphi) := \big(U_i x,\ \iota_i(U_i x)\big).$$

When $x \sim \lambda$, the pair $(q, \varphi)$ is distributed according to $\rho_{\nu_i}$. By (6), for $\lambda$-a.e. $x$,

$$\varphi - \theta = \bar{\chi}_i(\kappa(x)) - \bar{\chi}_i(d) = \bar{\chi}_i(s(x)) = t_i.$$

By Lemma 2.19, applied to $(p, \theta) \in \mathrm{supp}\,\rho_{\nu_i}$, for $\lambda$-a.e. $x$ we have

$$U_i(w') = (U_i w, U_i x) = (p, q) \in \mathrm{supp}\, J^{\nu_i}_{\varphi - \theta} = \mathrm{supp}\, J^{\nu_i}_{t_i} = \mathrm{supp}(U_i)_*\lambda'.$$

*Conclusion.* The requirements on $x$ in Steps 2–4 are finitely many: $\hat{x} \in \widehat{W}_0$, ergodicity of $J^{\lambda}_{s(x)}$, $x \in \mathrm{supp}\,\lambda$, and the $m$ conditions from Lemma 2.19. Each holds for $\lambda$-almost every $x$. The data they depend on, namely $d$, $\Psi'$, $\Psi''$, $\widehat{W}_0$ and the points $(p, \theta)$, were all fixed before $x$ was chosen. So some $x$ satisfies all of them, and $(w', \lambda', \Psi'')$ is an $(m+1)$-dimensional cube datum. □

2.9. **Proof of Theorem 2.1.** Let $A \subset \Gamma$ with $\mathtt{d}^*(A) > 0$, and let $k \ge 1$. Choose a Følner sequence along which $|A \cap \Phi_N|/|\Phi_N|$ has a positive limsup, and pass to a subsequence along which the limit $\mathtt{d}_\Phi(A) > 0$ exists. Theorem 2.9 gives an ergodic $(X, \mu, T)$, a clopen $E$ with $\mu(E) > 0$, a Følner sequence $\Psi$, and $a \in \mathrm{gen}(\mu, \Psi)$ with $A = \{\gamma : T^\gamma a \in E\}$.

(1) $(a, \mu, \Psi)$ is a 0-dimensional cube datum.
(2) By Proposition 2.23, applied $k-1$ times, there is a $(k-1)$-dimensional cube datum.
(3) By Proposition 2.22, there is a $k$-dimensional Erdős cube $y$ with $y_{\vec{0}} = a$ and $y_{\vec{1}} \in E$.
(4) By Lemma 2.11, there are infinite $B_1, \dots, B_k$ with $T^{b_1 + \cdots + b_k} a \in E$, that is, $b_1 + \cdots + b_k \in A$, for all $b_i \in B_i$.

□

*Formalization.* Theorem 2.1 is formalized in the fourteen files `Kmrr25/ThmK*.lean`; the main statement is `thmK` in `Kmrr25/ThmK.lean`. All lemmas and propositions of this section are proved there, with four classical results stated as explicit hypotheses (the structure `ClassicalInputs` in

`Kmrr25/ThmKDefs.lean`): the correspondence principle (Theorem 2.9); the existence of generic points (G4); the discrete spectrum theorem (Lemma 2.8(a)); and, for vectors orthogonal to all eigenfunctions, the decay of averaged correlations given by the spectral theorem and Wiener's lemma (Lemma 2.6(c),(d)). Disintegrations are taken from Mathlib. Some arguments are organised differently in the formal proof: for instance, the Kronecker group is realised as a closed subgroup of $(\mathbb{R}/\mathbb{Z})^{\mathbb{N}}$, which avoids Pontryagin duality, and Lemma 2.15 splits $g$ into its Kronecker and weak mixing parts instead of using the spectral theorem for $g$.

### 2.10. **Comments on the proof.**

*Structure of the induction.* The only information passed from one level to the next is the cube datum (D1)–(D4). The decoration $d$ and the Kronecker maps are used within a single step and are rebuilt for the new measure. The face measures $J_{t_i}^{\nu_i}$ of the new measure may have much larger Kronecker factors than the $\nu_i$, but nothing about them is needed: at the next level they enter only through (D3) and (D4). The measures $\lambda_m$ need not be invariant under face transformations or permutations of digits, since the cube criterion treats each direction separately. Each direction produces its own sequence of return times, so the method gives $k$ unrelated sets $B_i$. It says nothing about sumsets $B \oplus B + t$ built from a single set, which can fail when $\{g + g : g \in G\}$ has infinite index (see the remark after Conjecture 5.15 in [22]).

*Ingredients.* Besides elementary facts about Følner sequences, the proof uses:

- Furstenberg's correspondence principle;
- the existence of generic points for ergodic measures (via Lindenstrauss' pointwise theorem);
- the mean ergodic theorem, Wiener's lemma and the van der Corput inequality;
- the discrete spectrum theorem for the Kronecker factor and Pontryagin duality;
- disintegration of measures;
- re-basing for rotations.

It uses no characteristic factors beyond the Kronecker factor, no Host–Kra or Conze–Lesigne structure theory, no nilsystems, and no equidistribution theorem. Nor does it use finite generation, or any assumption on the subgroups $n\Gamma$. Commutativity of $\Gamma$ is used throughout: in the spectral theory on $\widehat{\Gamma}$, in describing the Kronecker factor as a rotation of a compact abelian group, and in the fact that translates $\Phi_N + h_N$ of a Følner sequence are again Følner. It enters the combinatorics in Lemma 2.11 and in the diagonal action on faces. Infiniteness of $\Gamma$ is used to obtain infinitely many distinct return times (Lemma 2.5).

*Relation to earlier work.* The relative products, their ergodicity, the generic-pairs lemma and the graph construction used for re-basing are known in this context [20, 8]; see the references given with Definition 2.12, Lemmas 2.15 and 2.16 and Step 1 of Proposition 2.23. For $k = 2$ our argument differs from that of [24] in three respects: re-basing replaces topological eigenfunctions, phase returns (Lemmas 2.17 and 2.18) replace the equicontinuity estimate (4.2) of [24], and an ergodic relative product $J_s^{\mu}$ replaces an unspecified joining. The new ingredients are the treatment of the Kronecker part of the face measures through phase returns (Proposition 2.22), the support lemmas (Lemmas 2.17 to 2.19), and the level-by-level induction (Proposition 2.23).

## 3. Admissible translates for $B \oplus B + t$ in abelian groups

In this section we answer Question 5.16 of [22] and the related Question §5.3(a). For a countably infinite abelian group $G$ in which $2G = \{g + g : g \in G\}$ has finite index (by work of Charamaras–Mountakis and Ackelsberg these are exactly the groups in which Conjecture 5.15

of [22] holds), we show that a set $T \subset G$ of translates suffices if and only if $T$ meets every coset of $2G$. In particular, the transversals of $2G$ are exactly the minimal sets of translates that suffice. We also determine exactly which sets arise as the set of admissible translates of some $A$ of positive upper Banach density, and we give two-sided bounds for the set of admissible translates of each individual $A$. The new input is small: the existence of some translate is a theorem of Charamaras–Mountakis [8], and the two elementary observations we use have already appeared in the literature in closely related settings; see Remark 3.9.

3.1. **Setting and the survey's questions.** Let $(G,+)$ be a countable abelian group. A *Følner sequence* in $G$ is a sequence $\Phi = (\Phi_N)_{N\in\mathbb{N}}$ of finite nonempty subsets of $G$ such that $|\Phi_N \cap (g+\Phi_N)|/|\Phi_N| \to 1$ for every $g \in G$. (Since $G$ is abelian, left, right and two-sided Følner sequences coincide.) Every countable abelian group is amenable, so Følner sequences exist. For $A \subset G$ we write $\overline{\mathsf{d}}_\Phi(A) = \limsup_{N\to\infty} |A\cap\Phi_N|/|\Phi_N|$, and we define the *upper Banach density* as

$$\mathsf{d}^*(A) = \sup_\Phi \overline{\mathsf{d}}_\Phi(A),$$

the supremum being taken over all Følner sequences in $G$. Thus $\mathsf{d}^*(A) > 0$ if and only if $\overline{\mathsf{d}}_\Phi(A) > 0$ for some Følner sequence $\Phi$; this is the notion of positive upper Banach density used in [8]. For $B \subset G$ let $B \oplus B = \{b + b' : b, b' \in B,\ b \neq b'\}$.

Conjecture 5.15 of [22] asserts that in every countable abelian group $G$, every set $A \subset G$ with $\mathsf{d}^*(A) > 0$ contains $B \oplus B + t$ for some infinite $B \subset G$ and some $t \in G$. The survey then asks:

**Question 5.16** ([22])**.** Assuming that Conjecture 5.15 holds, what values of the shift $t$ suffice?

**Question §5.3(a)** ([22])**.** The authors write: "We suspect that there is a direct relation between coset representatives of the subgroup $\{g + g : g \in G\}$ and the restrictions that can be placed on the shift $t$." Make this relation precise.

The sentence "We would like to know if similar restrictions can be placed in other groups", which precedes Question 5.16 in [22], is the motivation for Question 5.16 itself. By Question §5.3(a) we mean the suspicion quoted above, which the survey states right after Question 5.16.

As recorded in [22], Conjecture 5.15 is known to hold precisely when $[G : 2G] < \infty$. We recall the two results behind this statement.

**Theorem 3.1** (Charamaras–Mountakis [8, Corollary 1.13])**.** *Let $(G,+)$ be a countable abelian group such that $2G$ is a subgroup of finite index, and let $A \subset G$ have positive upper Banach density. Then there exist an infinite set $B \subset A$ and some $t \in G$ such that $B \oplus B \subset A - t$.*

(In [8] the group is countable throughout, and, as the conclusion shows, infinite. The result is cited as Corollary 1.12 in [22] and in [2, Theorem 1.1], following an earlier arXiv version of [8].)

**Theorem 3.2** (Ackelsberg [2, Theorem 1.2])**.** *Suppose $\Gamma$ is a countable abelian group and $2\Gamma$ is a subgroup of infinite index. Then for any $\varepsilon > 0$ there exists $A \subset \Gamma$ such that $\mathsf{d}^*(A) > 1 - \varepsilon$ and such that if $B \subset \Gamma$ and $t \in \Gamma$ satisfy $B \oplus B + t \subset A$, then $B$ is finite.*

Ackelsberg also records [2, footnote 2] that T. Tao observed independently that the condition $[\Gamma : 2\Gamma] < \infty$ in Theorem 3.1 is necessary.

Conjecture 5.15 fails trivially for finite groups. Hence "assuming that Conjecture 5.15 holds" in Question 5.16 means that $G$ is a countably infinite abelian group with $[G : 2G] < \infty$. We

fix such a group for the rest of this section. For $A \subset G$ we let

$$T(A) = \{t \in G : \ B \oplus B + t \subset A \text{ for some infinite } B \subset G\}$$

be the set of *admissible translates* for $A$. We say that a set $T \subset G$ *suffices* if $T(A) \cap T \neq \emptyset$ for every $A \subset G$ with $\mathsf{d}^*(A) > 0$.

3.2. **Three elementary lemmas.**

**Lemma 3.3.** *Let $G$ be a countably infinite abelian group, and let $H \leq G$ be a subgroup of finite index $\ell$.*

(i) *For every Følner sequence $\Phi$ and every $r \in G$ we have $|(r+H) \cap \Phi_N|/|\Phi_N| \to 1/\ell$. In particular $\mathsf{d}^*(r+H) = 1/\ell > 0$.*
(ii) *For all $X, Y \subset G$ we have $\mathsf{d}^*(X \cup Y) \leq \mathsf{d}^*(X) + \mathsf{d}^*(Y)$.*

*Proof.* (i) Let $C \subset G$ and $x \in G$. Since $|(x+C) \cap \Phi_N| = |C \cap (\Phi_N - x)|$, we get

$$\big||(x+C) \cap \Phi_N| - |C \cap \Phi_N|\big| \leq |(\Phi_N - x) \triangle \Phi_N| = 2\big(|\Phi_N| - |\Phi_N \cap (\Phi_N - x)|\big) = o(|\Phi_N|).$$

Let $r_1 + H, \dots, r_\ell + H$ be the cosets of $H$. Applying this with $C = r_1 + H$ and $x = r_i - r_1$ shows that the $\ell$ quantities $|(r_i + H) \cap \Phi_N|$ differ pairwise by $o(|\Phi_N|)$. They add up to $|\Phi_N|$, so each of them equals $|\Phi_N|/\ell + o(|\Phi_N|)$.

(ii) For each Følner sequence $\Phi$ we have $\overline{\mathsf{d}}_\Phi(X \cup Y) \leq \overline{\mathsf{d}}_\Phi(X) + \overline{\mathsf{d}}_\Phi(Y) \leq \mathsf{d}^*(X) + \mathsf{d}^*(Y)$. Now take the supremum over $\Phi$. □

**Lemma 3.4** (Pigeonhole)**.** *If $[G : 2G] < \infty$ and $B \subset G$ is infinite, then $(B \oplus B) \cap 2G \neq \emptyset$. Consequently, if $B \oplus B + t \subset A$, then $t \in A + 2G$. Moreover, if $B \oplus B + t$ is contained in a single coset $r + 2G$, then $t \in r + 2G$.*

*Proof.* Since $B$ is infinite and $2G$ has finitely many cosets, there are distinct $b, b' \in B$ in the same coset $x + 2G$, say $b = x + 2u$ and $b' = x + 2v$. Then $b + b' = 2(x + u + v) \in 2G$ and $b + b' \in B \oplus B$. If $B \oplus B + t \subset A$, then $t + b + b' \in A$, so $t \in A - (b + b') \subset A + 2G$. If $B \oplus B + t \subset r + 2G$, then $t + b + b' \in r + 2G$, so $t \in r + 2G$. □

**Lemma 3.5** (Moving the translate)**.** *For $B \subset G$ and $s \in G$ we have $(B - s) \oplus (B - s) = B \oplus B - 2s$. Consequently $T(A) + 2G = T(A)$ for every $A \subset G$, i.e. $T(A)$ is a union of cosets of $2G$.*

*Proof.* The map $b \mapsto b - s$ is a bijection $B \to B - s$, and $b - s \neq b' - s$ if and only if $b \neq b'$; this gives the identity. If $B \oplus B + t \subset A$ with $B$ infinite, then $B - s$ is infinite and $(B - s) \oplus (B - s) + (t + 2s) = B \oplus B + t \subset A$. Thus $t + 2s \in T(A)$ for every $s \in G$. □

3.3. **The admissible translates.**

**Theorem 3.6.** *Let $G$ be a countably infinite abelian group with $[G : 2G] < \infty$. For $A \subset G$ put $R(A) = \{r \in G : \ \mathsf{d}^*(A \cap (r + 2G)) > 0\}$.*

(a) *$T(A)$ is a union of cosets of $2G$.*
(b) *$R(A) \subset T(A) \subset A + 2G$. If $\mathsf{d}^*(A) > 0$, then $R(A) \neq \emptyset$, and hence $T(A)$ contains a full coset of $2G$.*
(c) *For every nonempty union $U$ of cosets of $2G$ we have $\mathsf{d}^*(U) > 0$ and $T(U) = U$. Consequently, the family $\{T(A) : A \subset G, \ \mathsf{d}^*(A) > 0\}$ is exactly the family of nonempty unions of cosets of $2G$.*
(d) *A set $T \subset G$ suffices if and only if $T$ meets every coset of $2G$. The minimal sets that suffice are exactly the transversals of $2G$ in $G$; they all have exactly $[G : 2G]$ elements.*

*Proof.* (a) This is Lemma 3.5.

(b) The inclusion $T(A) \subset A + 2G$ is Lemma 3.4. Let $r \in R(A)$ and put $A' = A \cap (r + 2G)$. By Theorem 3.1 applied to $A'$, there are an infinite $B \subset G$ and $t \in G$ with $B \oplus B + t \subset A' \subset A$. Thus $t \in T(A)$. Since $B \oplus B + t \subset r + 2G$, Lemma 3.4 gives $t \in r + 2G$. By part (a), $r \in t + 2G \subset T(A)$. Hence $R(A) \subset T(A)$. If $\mathsf{d}^*(A) > 0$, then since $A$ is the union of the finitely many sets $A \cap (r + 2G)$, Lemma 3.3(ii) shows that $\mathsf{d}^*(A \cap (r + 2G)) > 0$ for some $r$, i.e. $R(A) \neq \emptyset$. Since $R(A)$ is a union of cosets of $2G$, the last assertion follows.

(c) By Lemma 3.3(i), $\mathsf{d}^*(U) \geq 1/[G : 2G] > 0$. For every $r \in U$ we have $U \cap (r+2G) = r+2G$, which has positive upper Banach density; hence $U \subset R(U)$. By part (b), $U \subset R(U) \subset T(U) \subset U + 2G = U$. The final assertion follows from this and parts (a) and (b).

(d) Suppose $T$ meets every coset of $2G$, and let $\mathsf{d}^*(A) > 0$. By part (b), $T(A)$ contains a coset of $2G$, and this coset meets $T$. Conversely, suppose $T \cap (r + 2G) = \emptyset$ for some $r$. Let $A = r + 2G$. Then $\mathsf{d}^*(A) > 0$, and by part (c), $T(A) = r + 2G$, which is disjoint from $T$. So $T$ does not suffice. A transversal meets every coset, so it suffices, and removing any of its points leaves a coset unmet; hence transversals are minimal. If a sufficient set $T$ is not a transversal, then it contains two points of one coset, and removing one of them leaves a sufficient set. Hence $T$ is not minimal. □

Theorem 3.6(d) answers Question 5.16, and it confirms the suspicion in Question §5.3(a) in the strongest possible form: the translates can be restricted to any prescribed set of coset representatives of $2G$, and no smaller set of translates works. Parts (a)–(c) concern an individual set $A$. The admissible translates form a union of cosets of $2G$. This union contains every coset in which $A$ has positive upper Banach density, and it is contained in the union of the cosets that $A$ meets. Every nonempty union of cosets arises in this way. In general neither bound is attained, as the next proposition shows. The only exact statement for an individual $A$ is the tautological reduction to a single coset: $r \in T(A)$ if and only if $A \cap (r + 2G) \supset B \oplus B + t$ for some infinite $B$ and some $t$. (If $B \oplus B + r \subset A$, pass to an infinite $B' \subset B$ inside one coset of $2G$. Then $B' \oplus B' \subset 2G$, so $B' \oplus B' + r \subset A \cap (r + 2G)$. The converse follows from Lemmas 3.4 and 3.5.)

**Proposition 3.7.** *Let $G = \mathbb{Z}$.*

(i) *If $A = 2\mathbb{Z} \cup \{3^n : n \in \mathbb{N}\}$, then $T(A) = 2\mathbb{Z}$, while $A + 2\mathbb{Z} = \mathbb{Z}$.*

(ii) *If $A = 2\mathbb{Z} \cup X$ with $X = \{2^i + 2^j + 1 : \ 1 \leq i < j\}$, then $R(A) = 2\mathbb{Z}$, while $T(A) = \mathbb{Z}$.*

*So both inclusions in Theorem 3.6(b) can be strict.*

*Proof.* (i) By Theorem 3.6(b), $2\mathbb{Z} \subset R(A) \subset T(A)$. Since $3 \in A$, we have $A + 2\mathbb{Z} = \mathbb{Z}$. Suppose that $t$ is odd and $B \oplus B + t \subset A$ with $B$ infinite. Choose an infinite $B' \subset B$ whose elements all have the same parity, and fix $b_1 < b_2$ in $B'$. For every $b \in B' \setminus \{b_1, b_2\}$ the numbers $b_1 + b + t$ and $b_2 + b + t$ are odd elements of $A$, hence powers of 3, and they differ by $b_2 - b_1$. The equation $3^m - 3^n = b_2 - b_1$ with $m > n \geq 1$ has only finitely many solutions, because $3^n \leq b_2 - b_1$ and $n$ determines $m$. But $b_1 + b + t$ takes infinitely many values as $b$ ranges over $B'$. This is a contradiction, so $T(A) = 2\mathbb{Z}$.

(ii) For $B = \{2^i : i \in \mathbb{N}\}$ we have $B \oplus B + 1 = X \subset A$, so $1 \in T(A)$, and $T(A) = \mathbb{Z}$ by Lemma 3.5. Clearly $2\mathbb{Z} \subset R(A)$. It remains to show that $\mathsf{d}^*(X) = 0$. We claim that if every interval of $L$ consecutive integers contains at most $c$ elements of a set $Y \subset \mathbb{Z}$, then $\mathsf{d}^*(Y) \leq c/L$. Indeed, for a Følner sequence $\Phi$ we have

$$\sum_{x=0}^{L-1} |Y \cap (\Phi_N + x)| = \sum_{y \in \Phi_N} |Y \cap \{y, \ldots, y + L - 1\}| \leq c|\Phi_N|,$$

and $|Y \cap (\Phi_N + x)| \geq |Y \cap \Phi_N| - |\Phi_N \setminus (\Phi_N + x)| = |Y \cap \Phi_N| - o(|\Phi_N|)$. Hence $L\,\overline{\mathsf{d}}_\Phi(Y) \leq c$.

Now let $I$ be an interval of $L \geq 2$ consecutive integers. An element $2^i + 2^j + 1$ of $X$ with $i < j$ lies in $(2^j, 2^{j+1})$. The ranges $[2^j, 2^{j+1})$ with $2^j \geq L$ have length at least $L$, so at most two of them meet $I$. There are at most $\log_2 L + 1$ ranges with $2^j < L$. For fixed $j$, the elements of $X \cap I$ correspond to distinct powers $2^i$ in the interval $I - 2^j - 1$ of length $L$. Such an interval contains at most one power of 2 that is $\geq L$, and at most $\log_2 L + 1$ powers of 2 that are $< L$. Hence $|X \cap I| \leq (\log_2 L + 3)^2$, and so $\mathsf{d}^*(X) \leq (\log_2 L + 3)^2/L$ for every $L \geq 2$. Thus $\mathsf{d}^*(X) = 0$ and $R(A) = 2\mathbb{Z}$. □

*Formalization.* All four parts of Theorem 3.6 are formalized in the file `Kmrr25/Q516.lean` as `thm_translates`, with Theorem 3.1 stated as an explicit hypothesis. The statements that do not use it, and Proposition 3.7 (`prop_strict_i`, `prop_strict_ii`), are proved unconditionally.

*Remark* 3.8. We compare Theorem 3.6 with the examples in [22, §5.3].

(1) For $G = \mathbb{Z}$ we have $2G = 2\mathbb{Z}$, and the transversal $\{0, 1\}$ suffices. This matches the case $k = 2$ of [22, Proposition 2.6], which is stated in $\mathbb{N}$. Taking $r = 0$ in part (b), we get that $t = 0$ is admissible whenever $\mathsf{d}^*(A \cap 2G) > 0$. This is the group analogue of item 3 of [22, Proposition 2.6] for $k = 2$. That proposition is an equivalence between statements for all $k$, but its proof works for each $k$ separately. So for $k = 2$ its items hold unconditionally, by [22, Theorem 2.2].
(2) If $G = \bigoplus_{i \in \mathbb{N}} \mathbb{Z}/3\mathbb{Z}$, then $2G = G$, so $T = \{0\}$ suffices, as observed in [22].
(3) If $G = \bigoplus_{i \in \mathbb{N}} \mathbb{Z}/2\mathbb{Z}$, then $2G = \{0\}$ has infinite index. This group lies outside the scope of Question 5.16, because by Theorem 3.2 Conjecture 5.15 fails there. In groups with $1 < [G : 2G] < \infty$, the phenomenon pointed out in [22] for this group (one cannot always take $t = 0$) is explained by cosets: for $A = r + 2G$ with $r \notin 2G$, every admissible translate lies in $r + 2G$.

*Remark* 3.9 (Credit). The only deep ingredient is Theorem 3.1. The two elementary devices in Lemmas 3.4 and 3.5 are not new. For unrestricted sumsets $t + B + B$, Charamaras, Kousek, Mountakis and Radić [7, Lemmas 2.1 and 2.2] use pigeonholing into cosets of $2G$ and write $t = 2s + g_i$ with $g_i$ a coset representative, replacing $B$ by $B + s$. Ackelsberg and Jamneshan [3, discussion after Theorem 1.9, and Corollary 1.10] use the same two arguments for $B \oplus B \oplus B$ and the cosets of $3\Gamma$, under the assumption $[\Gamma : 6\Gamma] < \infty$. For $G = \mathbb{Z}$ the reduction appears in [22, Proposition 2.6]. Our contribution consists of the exact characterisation of the sets of translates that suffice, the minimality of transversals, the description of the possible sets $T(A)$, and the two-sided bounds for $T(A)$ for $B \oplus B$.

## 4. Cartesian products in $\mathbb{N}^2$: a negative answer to Question 5.10

In this section we answer Question 5.10 of [22] negatively. We construct sets $A \subset \mathbb{N}^2$ all of whose fibres have positive density along a fixed Følner sequence (in one version, every fibre has natural density exactly $1/4$), such that no translate of $A$ contains the Ramsey configuration $\{(b_1, b_2) : b_1, b_2 \in B,\ b_1 < b_2\}$ for an infinite $B \subset \mathbb{N}$.

4.1. **The question.** Throughout, $\mathbb{N} = \{1, 2, \dots\}$. Following [22], a *Følner sequence in* $\mathbb{N}$ is a sequence $\Phi = (\Phi_N)$ of finite subsets of $\mathbb{N}$ with $|(\Phi_N + t) \cap \Phi_N|/|\Phi_N| \to 1$ for every $t \in \mathbb{N}$. The upper and lower densities along $\Phi$ are $\overline{\mathsf{d}}_\Phi$ and $\underline{\mathsf{d}}_\Phi$. When they are equal, their common value is denoted $\mathsf{d}_\Phi$. For $A \subset \mathbb{N}^2$ and $n \in \mathbb{N}$ the *fibre* of $A$ over $n$ is $A_n = \{m \in \mathbb{N} : (n, m) \in A\}$. The configuration in question is

$$\{(b_1, b_2) :\ b_1, b_2 \in B,\ b_1 < b_2\} \subset A - t, \tag{10}$$

which is equation (21) of [22]. It says that $(b_1 + t_1, b_2 + t_2) \in A$ for all $b_1 < b_2$ in $B$, where $t = (t_1, t_2)$.

**Question 5.10** ([22])**.** Suppose $A \subset \mathbb{N}^2$ satisfies

$$\liminf_{n\in\mathbb{N}} \mathsf{d}_\Phi(A_n) > 0 \tag{11}$$

for some Følner sequence $\Phi$ on $\mathbb{N}$. Does there exist an infinite set $B \subset \mathbb{N}$ and an element $t \in \mathbb{N}^2$ satisfying (10)?

The survey points out that (10) can only hold if infinitely many fibres $A_n$ have the property that every finite subfamily has infinite intersection. Example 5.11 of [22], due to F. Hernández Castro, takes $A = X \times Y$ with $X = \bigcup_n [4^n, c4^n)$ and $Y = \bigcup_n [c'4^n, c''4^n)$ (intersected with $\mathbb{N}$), where $1 < c < c' < c'' < 2$. Then $X \cap (Y - t)$ is finite for every $t \geq 0$, and hence no translate of $A$ contains a configuration (10). In that example the fibre $A_n$ is $Y$ for $n \in X$ and empty for $n \notin X$. So it shows that the lim inf in (11) cannot be replaced by a lim sup. It does not satisfy (11). We modify the example so that *every* fibre has positive density. Two products of this type are glued along a partition of the rows, and the second coordinate of each product is offset from its own row set.

### 4.2. **A criterion.**

**Lemma 4.1.** *Let $\mathbb{N} = P_0 \sqcup P_1$ be a partition, and let $S_0, S_1 \subset \mathbb{N}$ be sets such that*

$$S_c \cap (P_c + s) \text{ is finite for every } s \in \mathbb{Z} \text{ and } c \in \{0, 1\}. \tag{12}$$

*Let $A = (P_0 \times S_0) \cup (P_1 \times S_1)$. Then $A_n = S_c$ for every $n \in P_c$. Moreover, there are no infinite set $B \subset \mathbb{N}$ and $t = (t_1, t_2) \in \mathbb{Z}^2$ such that $(b_1 + t_1, b_2 + t_2) \in A$ for all $b_1 < b_2$ in $B$. In particular, $A$ satisfies (10) for no infinite $B \subset \mathbb{N}$ and no $t \in \mathbb{N}^2$.*

*Proof.* The identity $A_n = S_c$ for $n \in P_c$ holds because $P_0 \cap P_1 = \emptyset$. Suppose that $B$ and $t$ exist. Every $b \in B$ is the smaller element of some pair in $B$, so $b + t_1 \in \mathbb{N}$. By the pigeonhole principle there are $c \in \{0, 1\}$ and an infinite $B' \subset B$ with $b + t_1 \in P_c$ for all $b \in B'$. Let $b_0 = \min B'$. For every $b \in B' \setminus \{b_0\}$ we have $(b_0 + t_1, b + t_2) \in A$ and $b_0 + t_1 \in P_c$, so $b + t_2 \in A_{b_0+t_1} = S_c$. Also $b + t_1 \in P_c$, so $b + t_2 \in P_c + (t_2 - t_1)$. Hence the infinitely many distinct numbers $b + t_2$, for $b \in B' \setminus \{b_0\}$, lie in $S_c \cap (P_c + (t_2 - t_1))$. This contradicts (12). □

### 4.3. **Two examples.**

**Example 4.2** (Scale $4^k$)**.** Let

$$P_0 = \bigcup_{k\geq 0} [4^k, 2 \cdot 4^k), \qquad P_1 = \bigcup_{k\geq 0} [2 \cdot 4^k, 4^{k+1}),$$

$$S_0 = \bigcup_{k\geq 1} [9 \cdot 4^{k-1}, 15 \cdot 4^{k-1}), \qquad S_1 = \bigcup_{k\geq 1} [5 \cdot 4^{k-1}, 7 \cdot 4^{k-1}),$$

all intervals being intervals of integers. So $\mathbb{N} = P_0 \sqcup P_1$. Each block of $S_0$ lies in the middle of the block $[2 \cdot 4^k, 4^{k+1})$ of $P_1$, and each block of $S_1$ lies in the middle of the block $[4^k, 2 \cdot 4^k)$ of $P_0$. Put $A = (P_0 \times S_0) \cup (P_1 \times S_1)$.

**Example 4.3** (Square blocks)**.** For $k \in \mathbb{N}$ let $J_k = [k^2, (k+1)^2) \cap \mathbb{N}$, so that $\mathbb{N} = \bigsqcup_{k\geq 1} J_k$ and $|J_k| = 2k + 1$. Let $P'_c = \bigcup_{k\equiv c \pmod 2} J_k$ for $c \in \{0, 1\}$, and let $S'_c$ consist of the *middle parts* of the blocks of the opposite parity:

$$S'_c = \{k^2 + r : \ k \in \mathbb{N},\ k \equiv 1 - c \pmod 2,\ r \in \mathbb{Z},\ k \leq 2r < 3k\}.$$

Since $k \le 2r < 3k$ forces $0 \le r \le 2k$, we have $S'_c \cap J_k = \{k^2 + r : k \le 2r < 3k\}$ when $k \equiv 1 - c$, and this set has exactly $k$ elements. Put $A' = (P'_0 \times S'_0) \cup (P'_1 \times S'_1)$.

**Proposition 4.4.** (i) *The sets of Example 4.2 satisfy (12). More precisely, every element of $S_c \cap (P_c + s)$ is smaller than $15|s|$. Along the Følner sequence $\Phi_N = [1, 4^N]$ we have $\mathsf{d}_\Phi(S_0) = 1/2$ and $\mathsf{d}_\Phi(S_1) = 1/6$. Hence every fibre of $A$ has density $1/2$ or $1/6$ along $\Phi$, and $A$ satisfies (11).*

(ii) *The sets of Example 4.3 satisfy (12). More precisely, every element of $S'_c \cap (P'_c + s)$ is smaller than $(2|s|+1)^2$. Both $S'_0$ and $S'_1$ have natural density $1/4$. Hence every fibre of $A'$ has natural density $1/4$, and $A'$ satisfies (11) along $\Phi_N = [1, N]$.*

*Proof.* (i) Let $m \in S_0$, say $9 \cdot 4^{k-1} \le m < 15 \cdot 4^{k-1}$, and suppose $|s| < 4^{k-1}$. Then $2 \cdot 4^k = 8 \cdot 4^{k-1} < m - s < 16 \cdot 4^{k-1} = 4^{k+1}$, so $m - s \in P_1$ and $m - s \notin P_0$. Similarly, if $m \in S_1$, say $5 \cdot 4^{k-1} \le m < 7 \cdot 4^{k-1}$, and $|s| < 4^{k-1}$, then $4^k < m - s < 2 \cdot 4^k$, so $m - s \in P_0$ and $m - s \notin P_1$. Hence every $m \in S_c \cap (P_c + s)$ lies in a block with $4^{k-1} \le |s|$, and so $m < 15 \cdot 4^{k-1} \le 15|s|$. (For $s = 0$ the intersection is empty.)

The sets $[1, 4^N]$ form a Følner sequence in $\mathbb{N}$, since $|([1, M]+t) \cap [1, M]| = M - t$ for $0 \le t \le M$. For $N \ge 2$ the blocks of $S_0$ and $S_1$ with $k \le N - 1$ lie in $[1, 4^N]$, since $15 \cdot 4^{N-2} < 4^N$. The blocks with $k \ge N$ start at $9 \cdot 4^{N-1}$ and $5 \cdot 4^{N-1}$, both larger than $4^N$, so they are disjoint from $[1, 4^N]$. Therefore

$$|S_0 \cap [1, 4^N]| = \sum_{k=1}^{N-1} 6 \cdot 4^{k-1} = 2(4^{N-1} - 1), \qquad |S_1 \cap [1, 4^N]| = \sum_{k=1}^{N-1} 2 \cdot 4^{k-1} = \tfrac{2}{3}(4^{N-1} - 1),$$

and dividing by $4^N$ gives the limits $1/2$ and $1/6$. Every fibre $A_n$ equals $S_0$ or $S_1$ by Lemma 4.1, so $\inf_n \mathsf{d}_\Phi(A_n) = 1/6 > 0$.

(ii) Let $m = k^2 + r \in S'_c$ with $k \le 2r < 3k$ and $k \equiv 1 - c \pmod 2$. Suppose $|s| < k/2$. Then $r - s \ge r - |s| > r - k/2 \ge 0$ and $r - s \le r + |s| < 3k/2 + k/2 = 2k$. Hence $m - s = k^2 + (r - s) \in J_k \subset P'_{1-c}$, so $m - s \notin P'_c$. Consequently every element of $S'_c \cap (P'_c + s)$ lies in a block $J_k$ with $k \le 2|s|$, and hence is smaller than $(2|s| + 1)^2$.

For the density, let $N \ge 1$ and let $K \ge 1$ be such that $K^2 \le N < (K + 1)^2$. The blocks $J_k$ with $k < K$ lie in $[1, N]$, and $J_K$ is the only block that meets $[1, N]$ without lying inside it. Hence

$$|S'_c \cap [1, N]| = \sum_{\substack{1 \le k < K \\ k \equiv 1 - c \ (2)}} k + O(K) = \frac{K^2}{4} + O(K) = \frac{N}{4} + O(\sqrt{N}).$$

So $\mathsf{d}(S'_c) = 1/4$. Every fibre of $A'$ is $S'_0$ or $S'_1$, and $[1, N]$ is a Følner sequence. □

**Theorem 4.5.** *The sets $A$ of Example 4.2 and $A'$ of Example 4.3 satisfy (11). In fact every fibre has a density along the indicated Følner sequence: $1/2$ or $1/6$ for $A$, and natural density exactly $1/4$ for $A'$. Nevertheless, there is no infinite $B \subset \mathbb{N}$ and no $t \in \mathbb{Z}^2$ such that $(b_1 + t_1, b_2 + t_2)$ lies in $A$ (respectively $A'$) for all $b_1 < b_2$ in $B$. In particular, the answer to Question 5.10 is negative.*

*Proof.* Combine Proposition 4.4 with Lemma 4.1. □

*Formalization.* Lemma 4.1, Proposition 4.4 and Theorem 4.5 are formalized in `Kmrr25/Q510.lean` (`criterion`, `q510_example_4k`, `q510_example_blocks`, `q510_negative`).

*Remark* 4.6. (1) The set $A'$ shows that the answer is negative already for the standard Følner sequence $[1, N]$, that is, with natural densities: every fibre has natural density

1/4. It does not satisfy (11) along every Følner sequence. Each $S'_c$ misses the blocks $J_k$ with $k \equiv c$, and these have length $2k+1 \to \infty$. So along $\Phi_N = J_{2N}$, for instance, the fibres $S'_0$ have density 0. The hypothesis that (11) holds along every Følner sequence, i.e. that the fibres have uniformly positive lower Banach density, is not addressed here; compare (4). The set $A$ of Example 4.2 has fibres without natural density. Its fibres have lower natural densities 2/9 and 2/15, which are approached at the integers just below the left endpoints of the blocks of $S_0$ and $S_1$, respectively.

(2) The survey explains that a positive answer to Question 5.10 would imply Theorem 2.2 and Conjecture 3.10 of [22]. A negative answer has no bearing on either statement.

(3) The survey's Theorem 5.12 shows that a weaker hypothesis gives infinite $B, C$ with $\{(b,c) : b \in B,\ c \in C,\ b < c\} \subset A$. This is consistent with Theorem 4.5: our sets contain even the full products $P_0 \times S_0$ and $P'_0 \times S'_0$. What fails is the diagonal coupling in (10): every element of $B$, except the smallest, must serve both as a row index and as a column index.

(4) The construction in Lemma 4.1 cannot work if every fibre is required to be syndetic with a gap bound that does not depend on the fibre. Indeed, suppose that for each $c$ every $n \in \mathbb{N}$ satisfies $n + s \in S_c$ for some $0 \le s \le L$. Then $P_c \subset \bigcup_{0 \le s \le L}(S_c - s)$, and (12) fails for the class $c$ with $P_c$ infinite. We do not know the answer to Question 5.10 under this stronger hypothesis.

## 5. Corners in $G^2$ and $G^d$: Questions 5.23 and §5.4(b)

In this section we answer Question 5.23 of [22] and its $G^d$ version, Question §5.4(b), negatively. The negative answer to the question as stated is elementary. The hypothesis controls only the fibres in the last coordinate, while the conclusion requires infinitely many points of $A$ on a single line in the first coordinate direction. So this part of the answer depends on the formulation. We therefore also treat the natural repaired version in which these points are dropped. Using Section 4, we show that it fails as well for $G = \mathbb{Z}$ (and for $G = \mathbb{N}$). The repair that adds a positive-density hypothesis on the columns remains open.

5.1. **The question.** Let $(G, \cdot)$ be a group. Following [22, §5.4], a *left Følner sequence* is a sequence $(\Phi_N)$ of finite nonempty subsets of $G$ with $|\Phi_N \cap g\Phi_N|/|\Phi_N| \to 1$ for every $g \in G$. Austin's theorem on corners in amenable groups [4], quoted in [22, Theorem 5.22], motivates the following question.

**Question 5.23** ([22])**.** Let $(G, \cdot)$ be an amenable group and let $N \mapsto \Phi_N$ be a left Følner sequence in $G$. If $A \subset G^2$ satisfies

$$\liminf_{g \to \infty} \liminf_{M \to \infty} \frac{1}{|\Phi_M|} \sum_{h \in \Phi_M} \mathbf{1}_A(g,h) > 0, \tag{13}$$

where $g \to \infty$ means leaving all finite sets, do there exist $(t,s) \in G^2$ and an injective sequence $b\colon \mathbb{N} \to G$ such that

$$\{(t,s),\ (b(i)t,s),\ (b(i)t, b(j)s) :\ i < j\} \subset A\,? \tag{14}$$

The survey adds that if $G = \mathbb{N}$, an affirmative answer to Question 5.23 implies an affirmative answer to Question 5.10. Two remarks on the setting are in order. First, a group with a left Følner sequence is countable. Indeed, let $H$ be the subgroup generated by $\bigcup_N \Phi_N$, which is countable. If $g \notin H$, then $g\Phi_N \subset gH$ is disjoint from $\Phi_N \subset H$, so the Følner condition fails for $g$. Second, if $G$ is finite, there is no injective $b\colon \mathbb{N} \to G$. So we assume, as the survey intends, that $G$ is countably infinite.

For $A \subset G^2$ and $g \in G$ write $A_g = \{h : (g,h) \in A\}$ for the *row fibre* and $A^h = \{g : (g,h) \in A\}$ for the *column*. Condition (13) concerns only the row fibres. The points $(b(i)t, s)$, $i \in \mathbb{N}$, in (14) are pairwise distinct (since $b$ is injective) and lie in the single column $A^s$.

**Lemma 5.1.** *If $G$ is an infinite group and $(\Phi_M)$ is a left Følner sequence, then $|\Phi_M| \to \infty$.*

*Proof.* Suppose that $|\Phi_M| \le K$ for infinitely many $M$. Choose a finite $S \subset G$ with $|S| > K$. For each $x \in S$ the number $|\Phi_M \setminus x\Phi_M|$ is a nonnegative integer, and $|\Phi_M \setminus x\Phi_M|/|\Phi_M| \to 0$. Along the $M$ with $|\Phi_M| \le K$ we therefore have $|\Phi_M \setminus x\Phi_M| = 0$ for all $x \in S$ once $M$ is large. Then $\Phi_M \subset x\Phi_M$, and since $|x\Phi_M| = |\Phi_M|$, we get $x\Phi_M = \Phi_M$. Pick such an $M$ and some $y \in \Phi_M$. Then $Sy \subset \Phi_M$, so $|\Phi_M| \ge |S| > K$, a contradiction. □

5.2. **The question as stated, in $G^2$ and $G^d$.** The survey states that Question 5.23 "can be generalized to $G^d$ for $d > 2$ in a straightforward manner", but does not write out the generalisation. Modelled on Austin's $d$-dimensional corners [22, Theorem 5.22], the natural conclusion asks for $t \in G^d$ and an injective $b\colon \mathbb{N} \to G$ such that, for all $i_1 < \cdots < i_d$, the $d+1$ points

$$(t_1, \dots, t_d),\ (b(i_1)t_1, t_2, \dots, t_d),\ (b(i_1)t_1, b(i_2)t_2, t_3, \dots, t_d),\ \dots,\ (b(i_1)t_1, \dots, b(i_d)t_d) \tag{15}$$

lie in $A$. The natural analogues of (13) constrain the fibres in the last coordinate,

$$\liminf_{(x_1,\dots,x_{d-1})\to\infty} \liminf_{M\to\infty} \frac{|\{h \in \Phi_M : (x_1, \dots, x_{d-1}, h) \in A\}|}{|\Phi_M|} > 0, \tag{16}$$

or the row fibres over the first coordinate,

$$\liminf_{x_1\to\infty} \liminf_{M\to\infty} \frac{|\{y \in \Phi_M^{d-1} : (x_1, y) \in A\}|}{|\Phi_M|^{d-1}} > 0. \tag{17}$$

**Question §5.4(b)** ([22])**.** Let $d \ge 3$, let $G$ be a countably infinite amenable group with a left Følner sequence $\Phi$, and suppose $A \subset G^d$ satisfies (16) or (17). Do there exist $t \in G^d$ and an injective $b\colon \mathbb{N} \to G$ such that the points (15) lie in $A$ for all $i_1 < \cdots < i_d$?

For $d = 2$ both (16) and (17) reduce to (13), and Question §5.4(b) reduces to Question 5.23.

**Theorem 5.2.** *Let $G$ be a countably infinite group, enumerated as $G = \{g_1, g_2, \dots\}$ without repetitions, and let $d \ge 2$. Put*

$$E_d = \{x \in G^d :\ x_1 = g_n,\ x_d = g_m \text{ for some } n < m\}.$$

*Let $\Phi$ be any left Følner sequence in $G$. Then:*

- (i) *for every $(x_1, \dots, x_{d-1}) \in G^{d-1}$, the last-coordinate fibre $\{h : (x_1, \dots, x_{d-1}, h) \in E_d\}$ is cofinite, and its density along $\Phi$ equals 1;*
- (ii) *for every $x_1 \in G$, the set $\{y \in G^{d-1} : (x_1, y) \in E_d\}$ has density 1 along $\Phi^{d-1} = (\Phi_M^{d-1})_M$;*
- (iii) *$|E_d \cap \Phi_M^d|/|\Phi_M|^d \to 1/2$;*
- (iv) *for every $t \in G^d$, the set $\{g \in G : (gt_1, t_2, \dots, t_d) \in E_d\}$ is finite.*

*In particular $E_d$ satisfies (16) and (17), and (13) when $d = 2$, with all limits equal to 1. But there is no $t \in G^d$ and no injective $b$ such that $(b(i)t_1, t_2, \dots, t_d) \in E_d$ for infinitely many $i$. Hence the answers to Question 5.23 and Question §5.4(b) are negative, for every countably infinite amenable group $G$ and every left Følner sequence.*

*Proof.* If $x_1 = g_n$, then the last-coordinate fibre is $G \setminus \{g_1, \dots, g_n\}$, whatever $x_2, \dots, x_{d-1}$ are. Its density along $\Phi$ is at least $1 - n/|\Phi_M|$, which tends to 1 by Lemma 5.1. This proves (i). Part (ii) follows similarly: the set in question is $G^{d-2} \times (G \setminus \{g_1, \dots, g_n\})$, and its relative density in $\Phi_M^{d-1}$ is at least $1 - n/|\Phi_M|$. For (iii), note that among the $|\Phi_M|^2 - |\Phi_M|$ pairs $(x, y) \in \Phi_M^2$ with $x \neq y$, exactly one of $(x, y)$ and $(y, x)$ has the first entry earlier in the enumeration. Hence $|E_d \cap \Phi_M^d| = \frac{1}{2}(|\Phi_M|^2 - |\Phi_M|)\,|\Phi_M|^{d-2}$, and (iii) follows from Lemma 5.1. For (iv), let $t_d = g_m$. If $(gt_1, t_2, \dots, t_d) \in E_d$, then $gt_1 \in \{g_1, \dots, g_{m-1}\}$, so $g$ lies in the finite set $\{g_1, \dots, g_{m-1}\}t_1^{-1}$. Since $b$ is injective, $(b(i)t_1, t_2, \dots, t_d) \in E_d$ can hold for at most $m - 1$ values of $i$. □

*Formalization.* Theorem 5.2 and the resulting negative answers to Questions 5.23 and §5.4(b) are formalized in `Kmrr25/Q523.lean` (`q523literal`, `q523_negative`). Theorem 5.5 below is not formalized.

For $G = \mathbb{Z}$ with $\Phi_M = [1, M]$, the simpler set $\{(g, h) \in \mathbb{Z}^2 : h > |g|\}$ also works. For $G = \mathbb{N}$ one can take $\{(n, m) \in \mathbb{N}^2 : n < m\}$. In the semigroup $\mathbb{N}$, where the survey's implication to Question 5.10 is stated, this means that the answer to the literal question is negative for a trivial reason.

*Remark* 5.3 (The answer depends on the formulation). Theorem 5.2 has nothing to do with sumsets: the conclusion (14) contains infinitely many distinct points of a single column, and (13) does not even force columns to be infinite. The survey's comment that, for $G = \mathbb{N}$, Question 5.23 implies Question 5.10 shows that the authors are mainly interested in the points $(b(i)t, b(j)s)$. The points $(t, s)$ and $(b(i)t, s)$ come from the shape of Austin's corners. There are two natural repairs:

(a) drop the points $(t, s)$ and $(b(i)t, s)$, and ask only for $\{(b(i)t, b(j)s) : i < j\} \subset A$ (and, in $G^d$, only for the top points $(b(i_1)t_1, \dots, b(i_d)t_d)$ with $i_1 < \cdots < i_d$);
(b) keep (14) but add to (13) the analogous condition on the columns, that is,

$$\liminf_{h\to\infty} \liminf_{M\to\infty} \frac{|A^h \cap \Phi_M|}{|\Phi_M|} > 0.$$

In Theorem 5.5 below we show, using Section 4, that repair (a) still has a negative answer for $G = \mathbb{Z}$. Repair (b), with positive-density hypotheses on both the row fibres and the columns, is to our knowledge open. Neither the examples of Theorem 5.2 nor those of Theorem 5.5 satisfy it: for instance, the columns of $\tilde{A}$ below are empty for $|h| \notin S_0' \cup S_1'$.

5.3. **The repaired question (a).** For $G = \mathbb{N}$, repair (a) of Question 5.23 is answered by Theorem 4.5 directly. Suppose that $t, s \in \mathbb{N}$ and an injective $b\colon \mathbb{N} \to \mathbb{N}$ satisfy $(b(i)+t, b(j)+s) \in A'$ for all $i < j$. Since $b$ is injective, it is unbounded, so it has a strictly increasing subsequence. Its range $B$ then satisfies (10) with the translate $(t, s)$. This is impossible. Also, $A'$ satisfies (13) with $\Phi_M = [1, M]$, because every fibre of $A'$ has natural density $1/4$. To treat the group $\mathbb{Z}$ and the $d$-dimensional version, we transfer $A'$ to $\mathbb{Z}^2$. For $x \in \mathbb{Z}$ let $|x|_+ = \max\{|x|, 1\}$, and let

$$\tilde{A} = \{(g, h) \in \mathbb{Z}^2 :\ h \neq 0,\ (|g|_+, |h|) \in A'\},$$

where $A' = (P_0' \times S_0') \cup (P_1' \times S_1')$ is the set of Example 4.3.

**Lemma 5.4.** *For every $g \in \mathbb{Z}$, the row fibre $\{h : (g, h) \in \tilde{A}\}$ meets $[1, M]$ in $|S_c' \cap [1, M]|$ points, where $c$ is the class with $|g|_+ \in P_c'$. Hence it has density $1/4$ along $\Phi_M = [1, M]$. There are no $(t, s) \in \mathbb{Z}^2$ and injective $b\colon \mathbb{N} \to \mathbb{Z}$ with $(b(i) + t, b(j) + s) \in \tilde{A}$ for all $i < j$.*

*Proof.* The fibre over $g$ is $\{h \neq 0 : |h| \in S_c'\}$, and its intersection with $[1, M]$ is $S_c' \cap [1, M]$. The density statement now follows from Proposition 4.4(ii).

Suppose that $t, s, b$ exist. Since $b$ is injective, for every $L$ only finitely many $i$ satisfy $|b(i)| \le L$. Choose $\epsilon \in \{1, -1\}$ such that $\epsilon b(i) > 0$ for infinitely many $i$. Then choose $i_1 < i_2 < \cdots$ inductively so that the numbers $\beta_k = \epsilon b(i_k)$ are positive and strictly increasing, with $\beta_1 > |t| + |s| + 1$. For $k < l$ we have $(b(i_k) + t, b(i_l) + s) \in \tilde{A}$. Now $|b(i_k) + t|_+ = |\epsilon\beta_k + t| = \beta_k + \epsilon t$, because $\beta_k > |t| + 1$, and similarly $|b(i_l) + s| = \beta_l + \epsilon s$. So $(\beta_k + \epsilon t, \beta_l + \epsilon s) \in A'$ for all $k < l$. Then the infinite set $B = \{\beta_k\} \subset \mathbb{N}$ and the translate $(\epsilon t, \epsilon s) \in \mathbb{Z}^2$ contradict Theorem 4.5. □

**Theorem 5.5.** *Let $d \ge 2$ and $\tilde{A}^{(d)} = \{x \in \mathbb{Z}^d : (x_1, x_d) \in \tilde{A}\}$, and let $\Phi_M = [1, M]$.*

(i) *For every $(x_1, \dots, x_{d-1}) \in \mathbb{Z}^{d-1}$, the last-coordinate fibre of $\tilde{A}^{(d)}$ has density $1/4$ along $\Phi$. For every $x_1 \in \mathbb{Z}$, the fibre $\{y \in \mathbb{Z}^{d-1} : (x_1, y) \in \tilde{A}^{(d)}\}$ has density $1/4$ along $\Phi^{d-1}$. In particular $\tilde{A}^{(d)}$ satisfies (16) and (17), and $\tilde{A}^{(2)} = \tilde{A}$ satisfies (13).*

(ii) *There are no $t \in \mathbb{Z}^d$ and injective $b\colon \mathbb{N} \to \mathbb{Z}$ such that $(b(i_1) + t_1, \dots, b(i_d) + t_d) \in \tilde{A}^{(d)}$ for all $i_1 < \cdots < i_d$.*

*Consequently repair (a) of Question 5.23 has a negative answer for $G = \mathbb{Z}$, and so does its analogue for Question §5.4(b), in which only the top points of (15) are required.*

*Proof.* (i) The last-coordinate fibre over $(x_1, \dots, x_{d-1})$ is the row fibre of $\tilde{A}$ over $x_1$. The fibre over $x_1$ is $\mathbb{Z}^{d-2} \times \{h : (x_1, h) \in \tilde{A}\}$, and its relative density in $[1, M]^{d-1}$ equals that of the row fibre in $[1, M]$. Now apply Lemma 5.4.

(ii) Suppose that $t$ and $b$ exist. If $i < j$ and $j - i \ge d - 1$, there are indices $i < i_2 < \cdots < i_{d-1} < j$, and so $(b(i) + t_1, b(j) + t_d) \in \tilde{A}$. Let $b'(k) = b((d-1)k)$. Then $b'$ is injective, and for $k < l$ we have $(d-1)l - (d-1)k \ge d - 1$. Hence $(b'(k) + t_1, b'(l) + t_d) \in \tilde{A}$ for all $k < l$, which contradicts Lemma 5.4. □

To summarise: the configurations (14) and (15) cannot be forced by the fibre-density hypotheses (13), (16) or (17). For $G = \mathbb{Z}$ (and for $G = \mathbb{N}$) this remains true after the lower corner points are dropped. Whether a uniform hypothesis on the row fibres, for example syndeticity with a bound on the gaps that does not depend on the fibre, suffices is open even for $G = \mathbb{N}$; see Remark 4.6(4). So is repair (b) of Remark 5.3, which also requires the columns to have positive density.

## Part 2. Sumsets in the integers

### 6. Unrestricted sums $B + B$ at the lower density threshold

In this section we answer Question §3.2(a) of [22], negatively, under each of its two possible readings. Throughout this part of the paper $B+B = \{b+b' : b, b' \in B\}$ denotes the *unrestricted* sumset, which contains the diagonal elements $2b$, and $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$.

Kousek and Radić [19, Theorem 1.2] proved that a set $A \subset \mathbb{N}$ contains $B + B$ for some infinite $B \subset \mathbb{N}$ as soon as $\overline{\mathrm{d}}(A) > 5/6$ or $\underline{\mathrm{d}}(A) > 3/4$, and showed that both thresholds are sharp [19, Propositions 4.1 and 4.2]. Right after reporting this, the survey asks the following.

**Question §3.2(a)** ([22])**.** What occurs if we only assume, for example, that $|A \cap [N]| > \frac{3}{4}N + \log N$ for infinitely many $N$? That is: must such a set $A \subset \mathbb{N}$ contain $B + B$ for some infinite set $B \subset \mathbb{N}$?

Read literally ("for infinitely many $N$"), the hypothesis is implied by $\overline{\mathrm{d}}(A) > 3/4$, so it lies far below the upper density threshold $5/6$. The question is presumably meant as a boundary case of the lower density threshold, that is, with "for all sufficiently large $N$". We answer both readings.

**Theorem 6.1.** (a) *There is a set $A \subset \mathbb{N}$ such that for every $C > 0$ we have $|A \cap [N]| > \frac{3}{4}N + C\log N$ for infinitely many $N$, and such that $B+B \not\subset A$ for every infinite $B \subset \mathbb{N}$.*

(b) *There are a set $A \subset \mathbb{N}$ and a non-decreasing unbounded function $\psi\colon \mathbb{N} \to [0,\infty)$ such that*

$$|A \cap [N]| \;\geq\; \tfrac{3}{4}N + \psi(N)\log N \qquad \textit{for all sufficiently large } N,$$

*and $B + B \not\subset A$ for every infinite $B \subset \mathbb{N}$.*

Since $\psi \to \infty$, part (b) gives $|A \cap [N]| > \frac{3}{4}N + C\log N$ for all large $N$, for every constant $C$ and every base of the logarithm.

*Remark* 6.2. (i) Part (a) is essentially contained in [19]: the set $A'$ of [19, Proposition 4.1] has upper density $5/6 > 3/4$ and contains no $B+B$, so it satisfies the literal hypothesis of Question §3.2(a). We give an explicit variant with an exact count, to make the size of the excess visible. The literal reading is therefore answered by the known sharpness examples, and the natural question is the one in part (b).

(ii) The set in part (b) has lower density exactly $3/4$: at least $3/4$ by construction, and at most $3/4$ by [19, Theorem 1.2], since it contains no $B+B$. The proof uses the diagonal elements $2b \in B+B$ in an essential way (see Step (iv) in the proof of Proposition 6.5).

(iii) We do not know which functions $g(N) = o(N)$ have the property that $|A \cap [N]| \geq \frac{3}{4}N + g(N)$ for all large $N$ forces $B + B \subset A$ for some infinite $B$; part (b) shows that $g(N) = \psi(N)\log N$ does not, for a suitable $\psi \to \infty$.

## 6.1. **The literal reading.**

*Proof of Theorem 6.1(a).* Let $S = \bigcup_{n\geq 1} \left[4^n,\, 2\cdot 4^n - 2n\right) \cap \mathbb{Z}$ and $A = (2\mathbb{N}-1) \cup (S \cap 2\mathbb{N})$.

*$S$ contains no $D+D$ with $D$ infinite.* Suppose $D + D \subset S$, let $d' = \min D$ and let $d \in D$ be large. Since $2d \in S$, there is $m$ with $4^m \leq 2d < 2\cdot 4^m - 2m$, and $m \to \infty$ as $d \to \infty$; take $d$ so large that $m \geq d'$. Then $2\cdot 4^{m-1} \leq d \leq d + d' < 4^m - m + d' \leq 4^m$. But $S \cap [2\cdot 4^{m-1}, 4^m) = \varnothing$, because the block of index $m-1$ ends below $2\cdot 4^{m-1}$ and the block of index $m$ starts at $4^m$. This contradicts $d + d' \in S$.

*$A$ contains no $B+B$ with $B$ infinite.* If $B+B \subset A$, choose an infinite $B' \subset B$ whose elements all have the same parity. Then $B' + B'$ consists of even numbers, so $B' + B' \subset A \cap 2\mathbb{N} \subset S$, which is impossible.

*Counting.* Let $N_n = 2\cdot 4^n - 2n - 1$. The odd numbers in $[N_n]$ number $(N_n+1)/2 = 4^n - n$. The block $[4^m, 2\cdot 4^m - 2m)$ has even length and starts at an even number, so it contains $4^m/2 - m$ even numbers; the blocks with $m \leq n$ lie in $[N_n]$ and the block with $m = n+1$ does not meet it. Hence

$$|A \cap [N_n]| = 4^n - n + \sum_{m=1}^{n} \left(\frac{4^m}{2} - m\right) = 4^n - n + \frac{4^{n+1} - 4}{6} - \frac{n(n+1)}{2},$$

and subtracting $\frac{3}{4}N_n = \frac{3}{2}4^n - \frac{3}{2}n - \frac{3}{4}$ gives

$$|A \cap [N_n]| - \tfrac{3}{4}N_n = \frac{4^n}{6} - \frac{n^2}{2} + \frac{1}{12}.$$

This exceeds $C\log N_n = O(n)$ for every fixed $C$ once $n$ is large (for $C = 1$ and the natural logarithm already for $n \geq 3$). □

6.2. **Rotation blocks.** For the second reading we first reduce to a statement about a set $E$ of density $1/2$. For $E \subset \mathbb{N}$ write $2E = \{2e : e \in E\}$.

**Lemma 6.3.** *Let $E \subset \mathbb{N}$ and $A = (2\mathbb{N}-1) \cup 2E$. If $B + B \subset A$ for some infinite $B \subset \mathbb{N}$, then there are an infinite $D \subset \mathbb{N}_0$ and $t \in \{0,1\}$ with $D + D + t \subset E$. Moreover $|A \cap [N]| = \lceil N/2 \rceil + |E \cap [1, \lfloor N/2 \rfloor]|$ for every $N \in \mathbb{N}$.*

*Proof.* Choose an infinite $B' \subset B$ of constant parity; then $B' + B' \subset A \cap 2\mathbb{N} = 2E$. If $B' \subset 2\mathbb{N}$, put $D = \{b/2 : b \in B'\}$, so that $2(D + D) = B' + B'$ and $D + D \subset E$. If $B'$ consists of odd numbers, put $D = \{(b-1)/2 : b \in B'\} \subset \mathbb{N}_0$, so that $B' + B' = 2(D + D + 1)$ and $D + D + 1 \subset E$. The counting formula is clear. □

Let $\rho(x) = \mathbf{1}_{[0,1/2)}(\{x\}) - \frac{1}{2}$ for $x \in \mathbb{R}$, so that $\rho$ is 1-periodic, right-continuous, has jumps exactly at $\frac{1}{2}\mathbb{Z}$, and satisfies $\rho(x + \frac{1}{2}) = -\rho(x)$. For $n \in \mathbb{N}$ let $\lambda(n) = \lfloor \log_2 n \rfloor$ and $\varepsilon(n) = (-1)^{\lambda(n)}$. Note that

$$\lambda(2n + t) = \lambda(n) + 1 \qquad (n \in \mathbb{N},\ t \in \{0,1\}). \tag{18}$$

**Definition 6.4.** Let $\alpha, \beta \in \mathbb{R}$ and let $(w_i)_{i \geq 1}$ be positive integers. Let $J_0 = \{x \in \mathbb{T} : \{x - \beta\} \in [0, \frac{1}{2})\}$ and $J_1 = \mathbb{T} \setminus J_0$, and put

$$R = \{n \in \mathbb{N} :\ n\alpha \bmod 1 \in J_{\lambda(n) \bmod 2}\}, \qquad G = \bigcup_{i \geq 1} [2^i - w_i,\, 2^i + w_i), \qquad E = R \setminus G.$$

So $R$ is a Bohr set on each dyadic block, using the half-circle $J_0$ on blocks $[2^j, 2^{j+1})$ with $j$ even and the complementary half-circle on blocks with $j$ odd; $E$ is obtained by removing slowly growing neighbourhoods of the powers of 2. For $\alpha \in \{0, \frac{1}{2}\}$ these are block sets of the kind used in [19, Section 4].

**Proposition 6.5.** *In Definition 6.4, assume $w_i \to \infty$. Then there are no infinite $D \subset \mathbb{N}_0$ and $t \in \{0,1\}$ with $D + D + t \subset E$.*

*Proof.* Suppose $D$ is infinite, $t \in \{0,1\}$ and $D + D + t \subset E$; removing 0 from $D$ we may assume $D \subset \mathbb{N}$.

(i) For $d \in D$ put $z_d = d\alpha + t\alpha/2 \bmod 1$. Then $(d + d' + t)\alpha \equiv z_d + z_{d'}$ and $(2d + t)\alpha \equiv 2z_d \pmod 1$.

(ii) For $k = 0, 1, 2, 3$ let $Q_k = \{z \in \mathbb{T} : \{z - \beta/2\} \in [\frac{k}{4}, \frac{k+1}{4})\}$. If $z, z' \in Q_k$, write $z \equiv \beta/2 + x$, $z' \equiv \beta/2 + x'$ with $x, x' \in [\frac{k}{4}, \frac{k+1}{4})$; then $z + z' \equiv \beta + (x + x')$ with $x + x' \in [\frac{k}{2}, \frac{k+1}{2})$, so $z + z' \in J_{k \bmod 2}$. In particular $2z \in J_{k \bmod 2}$ as well.

(iii) By the pigeonhole principle there are $\sigma \in \{0,1\}$, $k \in \{0,1,2,3\}$ and an infinite $D' \subset D$ with $\lambda(d) \equiv \sigma \pmod 2$ and $z_d \in Q_k$ for all $d \in D'$.

(iv) Let $d \in D'$. Since $2d + t \in E \subset R$ and $\lambda(2d + t) = \lambda(d) + 1$ by (18), we have $(2d + t)\alpha \in J_{1-\sigma}$. By (i) and (ii), $(2d + t)\alpha \equiv 2z_d \in J_{k \bmod 2}$. As $J_0 \cap J_1 = \varnothing$, this forces $k \equiv 1 - \sigma \pmod 2$. (This is the only place where a diagonal element is used.)

(v) Let $d_0 = \min D'$ and let $d' \in D'$, $d' > d_0$. Then $n := d_0 + d' + t \in E \subset R$ and, by (i) and (ii), $n\alpha \equiv z_{d_0} + z_{d'} \in J_{k \bmod 2} = J_{1-\sigma}$. Since $n \in R$, this means $\lambda(n) \equiv 1 - \sigma \not\equiv \lambda(d') \pmod 2$. As $n > d'$, some power $2^i$ lies in $(d', n]$. Since $n \notin G$ and $n \geq 2^i$, we get $n \geq 2^i + w_i$, so $w_i \leq n - 2^i < n - d' = d_0 + t$.

(vi) Letting $d' \to \infty$ through $D'$, the index $i$ in (v) tends to infinity (because $2^i > d'$), while $w_i < d_0 + t$ stays bounded. This contradicts $w_i \to \infty$. □

**Lemma 6.6.** *In Definition 6.4, for every $M \in \mathbb{N}$,*

$$|E \cap [1, M)| \;\geq\; \frac{M-1}{2} + S_{\alpha,\beta}(M) - \sum_{i \geq 1:\ 2^i - w_i < M} 2w_i,$$

*where $S_{\alpha,\beta}(M) = \sum_{1 \leq n < M} \varepsilon(n)\, \rho(n\alpha - \beta)$.*

*Proof.* If $\lambda(n)$ is even, then $\mathbf{1}_R(n) = \mathbf{1}_{[0,1/2)}(\{n\alpha - \beta\})$, so $\mathbf{1}_R(n) - \frac{1}{2} = \rho(n\alpha - \beta)$; if $\lambda(n)$ is odd, then $\mathbf{1}_R(n) - \frac{1}{2} = -\rho(n\alpha - \beta)$. Thus $\mathbf{1}_R(n) - \frac{1}{2} = \varepsilon(n)\rho(n\alpha - \beta)$ and $|R \cap [1, M)| = \frac{M-1}{2} + S_{\alpha,\beta}(M)$. Finally $|G \cap [1, M)|$ is at most the sum of the lengths $2w_i$ of the intervals $[2^i - w_i, 2^i + w_i)$ that meet $[1, M)$. □

Thus everything hinges on making the twisted discrepancy $S_{\alpha,\beta}(M)$ grow faster than $\log M$ for *all* large $M$: the gaps cost $\sum 2w_i$, which grows faster than $\log M$ because $w_i \to \infty$.

**Proposition 6.7.** *There are an irrational $\alpha$ and integers $1 \leq N_1 \leq N_2 \leq \cdots$ such that*

$$S_{\alpha,1/6}(M) \;\geq\; i \log M \qquad \text{for all } i \geq 1 \text{ and all } M \geq N_i.$$

We prove Proposition 6.7 in Section 6.3, and first deduce part (b) of the theorem.

*Proof of Theorem 6.1(b).* Let $\alpha$ and $(N_i)$ be as in Proposition 6.7; we may assume $N_1 \geq 2$. Since $|S_\alpha(M)| \leq (M-1)/2$, for each $M \geq 2$ the inequality $S_\alpha(M) \geq i \log M$ fails for large $i$, so $N_i \to \infty$. Let $\psi_0(M) = \max\{i : N_i \leq M\}$ for $M \geq N_1$ and $\psi_0(M) = 0$ for $M < N_1$. Then $\psi_0$ is non-decreasing and unbounded, and $S_{\alpha,1/6}(M) \geq \psi_0(M) \log M$ for $M \geq N_1$. Take $\beta = \frac{1}{6}$ and

$$w_i = \max\Big(1,\ \min\big(\lfloor 2^i/4 \rfloor,\ \lfloor \psi_0(2^{i-1})/16 \rfloor\big)\Big) \qquad (i \geq 1),$$

so that $w_i \to \infty$, and let $E$ be as in Definition 6.4.

We bound the gap term of Lemma 6.6. We have $w_1 = 1$. For $i \geq 2$ we have $w_i \leq 2^i/4$, so $2^i - w_i < M$ implies $\frac{3}{4}2^i < M$, hence $2^{i-1} < M$, hence $\psi_0(2^{i-1}) \leq \psi_0(M)$ and $w_i \leq \max(1, \psi_0(M)/16)$. The indices $i \geq 2$ with $\frac{3}{4}2^i < M$ satisfy $i < \log_2 M + 1$, so there are at most $\log_2 M + 2$ indices in the sum. Hence, once $\psi_0(M) \geq 16$,

$$\sum_{2^i - w_i < M} 2w_i \leq \frac{\psi_0(M)}{8}\,(\log_2 M + 2) = \psi_0(M)\Big(\frac{\log M}{8 \log 2} + \frac{1}{4}\Big) \leq \psi_0(M)\big(0.19 \log M + 0.25\big).$$

By Lemma 6.6, for all large $M$,

$$|E \cap [1, M)| - \frac{M-1}{2} \;\geq\; \psi_0(M)\big(0.81 \log M - 0.25\big) \;\geq\; \tfrac{1}{2}\psi_0(M) \log M + 1.$$

Let $A = (2\mathbb{N} - 1) \cup 2E$. By Lemma 6.3, with $M = \lfloor N/2 \rfloor + 1$,

$$|A \cap [N]| = \lceil N/2 \rceil + |E \cap [1, M)| \;\geq\; \frac{N}{2} + \frac{N-1}{4} + \tfrac{1}{2}\psi_0(M) \log M + 1 \;\geq\; \frac{3}{4}N + \tfrac{1}{2}\psi_0(M) \log(N/2)$$

for all large $N$. With $\psi(N) = \frac{1}{4}\psi_0(\lfloor N/2 \rfloor + 1)$, which is non-decreasing and unbounded, we get $|A \cap [N]| \geq \frac{3}{4}N + \psi(N) \log N$ for all large $N$. Finally, if $B + B \subset A$ with $B$ infinite, Lemma 6.3 produces an infinite $D \subset \mathbb{N}_0$ and $t \in \{0, 1\}$ with $D + D + t \subset E$, contradicting Proposition 6.5. □

*Formalization.* Both parts of Theorem 6.1, together with Propositions 6.5 and 6.7 and the lemmas of Section 6.3, are formalized in `Kmrr25/S32a.lean` (`thm_u32a_a`, `thm_u32a_b`).

6.3. **A twisted discrepancy estimate.** For $\theta \in \mathbb{R}$ and $n \in \mathbb{Z}$ let $r_\theta(n) = \rho(n\theta - \frac{1}{6})$ and $S_\theta(M) = \sum_{1 \le n < M} \varepsilon(n) r_\theta(n)$, so that $S_\theta = S_{\theta,1/6}$. Let $\mathcal{Q}$ be the set of rationals $\theta = a/q$ with

$$q = 3 \cdot 2^p \ (p \ge 1), \qquad 0 < a < q, \qquad \gcd(a,q) = 1.$$

For $\theta = a/q \in \mathcal{Q}$ (always written in lowest terms) put

$$\Delta(\theta) = \sum_{2^p \le n < 2^{p+1}} r_\theta(n), \qquad G(\theta) = (-1)^p \Delta(\theta), \qquad \mathcal{D}(\theta) = \sup_{u \le v} \Big| \sum_{u \le n < v} r_\theta(n) \Big|.$$

Note that $q$ is even and $a$ is odd, and that $2^p = q/3$.

**Lemma 6.8.** *Let $\theta = a/q \in \mathcal{Q}$.*

(i) $\sum_{n=u}^{u+q-1} r_\theta(n) = 0$ *for every $u \in \mathbb{Z}$; in particular $\mathcal{D}(\theta) \le q/2$.*
(ii) *For every $j \ge p$, $\sum_{2^j \le n < 2^{j+1}} \varepsilon(n) r_\theta(n) = G(\theta)$.*
(iii) $S_\theta(2^j) = S_\theta(2^p) + (j-p)G(\theta)$ *for $j \ge p$, and $|S_\theta(M) - S_\theta(2^j)| \le \mathcal{D}(\theta)$ whenever $2^j \le M \le 2^{j+1}$.*

*Proof.* (i) We have $n\theta - \frac{1}{6} = (na - 2^{p-1})/q$ with $2^{p-1} \in \mathbb{Z}$. As $n$ runs through $q$ consecutive integers, $na - 2^{p-1}$ runs through all residues modulo $q$, and $\rho(k/q) = \frac{1}{2}$ for exactly $q/2$ residues $k$ and $-\frac{1}{2}$ for the others. Hence the sum vanishes, the function $D(N) = \sum_{0 \le n < N} r_\theta(n)$ is $q$-periodic, every interval sum equals a sum over fewer than $q$ consecutive integers, and $\mathcal{D}(\theta) \le q/2$.

(ii) For $j \ge p$ let $u_j \in \{1,2\}$ with $u_j \equiv 2^{j-p} \pmod 3$; thus $u_j = 1$ if $j-p$ is even and $u_j = 2$ otherwise. Since $q = 3 \cdot 2^p$ divides $2^p(2^{j-p} - u_j)$, periodicity gives $D(2^j) = D(2^p u_j)$. Therefore $\sum_{2^j \le n < 2^{j+1}} r_\theta(n) = D(2^p u_{j+1}) - D(2^p u_j) = (-1)^{j-p}\big(D(2^{p+1}) - D(2^p)\big) = (-1)^{j-p}\Delta(\theta)$. On $[2^j, 2^{j+1})$ we have $\varepsilon = (-1)^j$, which gives (ii).

(iii) The first claim follows from (ii). If $2^j \le M \le 2^{j+1}$, then

$$S_\theta(M) - S_\theta(2^j) = (-1)^j \sum_{2^j \le n < M} r_\theta(n).$$

□

**Lemma 6.9.** *Let $\theta = a/q \in \mathcal{Q}$, let $d \ge 1$, $L = 2^d$, let $b$ be an odd integer with $0 < b < L$, and suppose that $\theta' = \theta + \frac{b}{qL} = \frac{aL+b}{qL}$ belongs to $\mathcal{Q}$ (with denominator $qL$). Then for every set $I$ of at most $qL$ consecutive integers,*

$$\Big| \sum_{n \in I} r_{\theta'}(n) \Big| \le \frac{|I|\, b}{L} + q, \qquad \text{and consequently} \qquad \mathcal{D}(\theta') \le q(b+1).$$

*Proof.* Let $n_0 = \min I$ and $K = \lfloor |I|/q \rfloor$. Split $I$ into the $K$ blocks $\{n_0 + kq + s : 0 \le s < q\}$, $0 \le k < K$, and a remainder of fewer than $q$ integers, which contributes at most $q/2$ in absolute value. Put $c = b/(2L) \in (0, \frac{1}{2})$. Since $a$ is odd, $\frac{q}{2}\theta' = \frac{a}{2} + c \equiv \frac{1}{2} + c \pmod 1$, so for every $n$,

$$r_{\theta'}(n) + r_{\theta'}(n + \tfrac{q}{2}) = \rho(x) + \rho(x + \tfrac{1}{2} + c) = \rho(x) - \rho(x + c), \qquad x = n\theta' - \tfrac{1}{6}.$$

Within each block pair $s$ with $s + \frac{q}{2}$, $0 \le s < \frac{q}{2}$. Fix $s$, and let $x_k = (n_0 + kq + s)\theta' - \frac{1}{6}$. Since $q\theta' = a + 2c$ with $a \in \mathbb{Z}$, we have $\rho(x_k) - \rho(x_k + c) = \rho(y_k) - \rho(y_k + c)$ with $y_k = x_0 + 2ck$. This difference is 0 unless the interval $(y_k, y_k + c]$ contains a point of $\frac{1}{2}\mathbb{Z}$, in which case it is $\pm 1$. The intervals $(y_k, y_k + c]$, $0 \le k < K$, are pairwise disjoint and contained in an interval of length $2cK$, which contains at most $4cK + 1$ points of $\frac{1}{2}\mathbb{Z}$. Summing over the $q/2$ values of $s$, the blocks contribute at most $\frac{q}{2}(4cK + 1) = \frac{qKb}{L} + \frac{q}{2} \le \frac{|I|b}{L} + \frac{q}{2}$. This proves the first

bound. By Lemma 6.8(i) applied to $\theta'$, every interval sum of $r_{\theta'}$ equals a sum over fewer than $qL$ consecutive integers, so $\mathcal{D}(\theta') \le qL \cdot b/L + q$. □

**Lemma 6.10.** *Let $\theta = a/q \in \mathcal{Q}$, $\theta' \ge \theta$ and $n \ge 0$ with $n(\theta' - \theta) < 1/q$. Then $r_{\theta'}(n) = r_\theta(n)$.*

*Proof.* $n\theta - \frac{1}{6}$ lies in $\frac{1}{q}\mathbb{Z}$, and $\rho$ is constant on each interval $[\frac{k}{q}, \frac{k+1}{q})$ because its jumps lie in $\frac{1}{2}\mathbb{Z} \subset \frac{1}{q}\mathbb{Z}$. The point $n\theta' - \frac{1}{6}$ lies in the same interval as $n\theta - \frac{1}{6}$. □

The next lemma produces, from $\theta \in \mathcal{Q}$, a nearby $\theta' \in \mathcal{Q}$ with a much larger gain $G(\theta')$ per dyadic block.

**Lemma 6.11.** *Let $\theta = a/q \in \mathcal{Q}$ with $q = 3 \cdot 2^p$, and put $h = q - a \ge 1$. Let $m \ge p$ with $m \equiv p \pmod 2$, let $b = h2^m + 1$ and let $R \ge 1$ be the order of 2 modulo $3b$. For $t \in \mathbb{N}$ put $d = m + 2Rt$, $L = 2^d$ and $\theta' = \frac{aL+b}{qL}$. Then $0 < b < L$, $\theta' \in \mathcal{Q}$ with denominator $qL = 3 \cdot 2^{p+d}$, $p + d$ is even, and*

$$G(\theta') = \Delta(\theta') \ \ge \ \frac{q2^m}{12} - h - q - 2.$$

*Proof.* Since $2^R \equiv 1 \pmod{3b}$ we have $2^R > 3b$, hence $L \ge 2^{2R} > b$. Moreover $2^d = 2^m(2^R)^{2t} \equiv 2^m \pmod{3b}$.

(1) $hL + 1 \equiv h2^m + 1 = b \equiv 0 \pmod b$, so $U = (hL + 1)/b$ is an integer.

(2) $U - 1 = h(2^d - 2^m)/b$. Here $(2^d - 2^m)/b$ is an integer divisible by 3 (as $3b \mid 2^d - 2^m$) and by $2^m$ (as $b$ is odd). Hence $q = 3 \cdot 2^p$ divides $U - 1$; in particular $U \equiv 1 \pmod 6$, say $U = 6v + 1$.

(3) Put $a' = aL + b$ and $q' = qL$. Since $aU + h \equiv a + h = q \equiv 0 \pmod q$, the number $\tau = (aU + h)/q$ is an integer, and

$$a'U = aLU + bU = aLU + hL + 1 = L(aU + h) + 1 = \tau q' + 1.$$

Hence $\gcd(a', q') = 1$ and $\gcd(\tau, U) = 1$. Also $0 < a' < aL + hL = q'$, so $\theta' = a'/q' \in \mathcal{Q}$; the exponent of 2 in $q'$ is $p + d = p + m + 2Rt$, which is even. Moreover $\theta' = \frac{\tau}{U} + \frac{1}{Uq'}$.

(4) Let $N = q'/3 = 2^{p+d}$, so that $\Delta(\theta') = \sum_{N \le n < 2N} r_{\theta'}(n)$. For $n$ in this range put $z = n/q' \in [\frac{1}{3}, \frac{2}{3})$ and let $m' \in \{0, \dots, U-1\}$ be the residue of $n\tau$ modulo $U$. Then $n\theta' - \frac{1}{6} \equiv \frac{m'+z}{U} - \frac{1}{6} \pmod 1$ with $\frac{m'+z}{U} \in [0, 1)$, so $r_{\theta'}(n) = \frac{1}{2}$ if and only if $\frac{m'+z}{U} \in [\frac{1}{6}, \frac{2}{3})$, that is, $m' + z \in [v + \frac{1}{6}, 4v + \frac{2}{3})$. Since $z \in [\frac{1}{3}, \frac{2}{3})$, this holds if and only if $v \le m' \le 4v$. Hence on any $U$ consecutive integers in $[N, 2N)$, on which $m'$ takes every value once (as $\gcd(\tau, U) = 1$), there are $3v + 1$ values $+\frac{1}{2}$ and $3v$ values $-\frac{1}{2}$, and the sum is $\frac{1}{2}$.

(5) Split $[N, 2N)$ into $\lfloor N/U \rfloor$ blocks of $U$ consecutive integers and a remainder of fewer than $U \le hL + 1 \le q'$ integers. By Lemma 6.9, the remainder contributes at most $Ub/L + q = h + \frac{1}{L} + q$ in absolute value. Hence

$$\Delta(\theta') \ge \frac{1}{2}\Big(\frac{N}{U} - 1\Big) - h - 1 - q.$$

Finally $\frac{N}{2U} = \frac{qLb}{6(hL+1)} \ge \frac{qb}{6(h+1)} \ge \frac{q2^m}{12}$, using $b = h2^m + 1 \ge (h + 1)2^{m-1}$. As $p + d$ is even, $G(\theta') = \Delta(\theta')$. □

**Lemma 6.12.** *Let $\theta = a/q \in \mathcal{Q}$ with $q = 3 \cdot 2^p$, let $c \ge 0$, and let $N_0, X_0 \in \mathbb{N}$. Suppose that $G(\theta) > c \log 2$ and $S_\theta(M) \ge c \log M$ for all $M \ge N_0$. Let $m, b, R$ be as in Lemma 6.11, chosen so that $\frac{q2^m}{12} - h - q - 2 \ge c \log 2$, and let $\theta'_t$ denote the number $\theta'$ of Lemma 6.11 for the parameter $t$. Then for all sufficiently large $t$:*

(i) *$r_{\theta'_t}(n) = r_\theta(n)$ for $0 \le n < X_0$;*

(ii) $S_{\theta'_t}(M) \geq c \log M$ *for all* $M \geq N_0$.

*Proof.* Write $\theta' = \theta'_t$, $d = m + 2Rt$, $P' = p + d$, and let $\ell = \lceil \log_2 b \rceil + 1$ and $J = d - \ell$. For $0 \leq n < 2^J$ we have $n(\theta' - \theta) = \frac{nb}{q2^d} < \frac{b2^{-\ell}}{q} \leq \frac{1}{2q}$, so $r_{\theta'}(n) = r_\theta(n)$ by Lemma 6.10. In particular (i) holds once $2^J \geq X_0$, and $S_{\theta'}(M) = S_\theta(M)$ for $M \leq 2^J$.

Let $B_0 = q(b+1)$; by Lemma 6.9 (with $\theta' \in \mathcal{Q}$ by Lemma 6.11), $\mathcal{D}(\theta') \leq B_0$, and $B_0$ does not depend on $t$. By Lemma 6.8(iii) for $\theta$,

$$\Phi(t) := S_\theta(2^p) + (J-p)G(\theta) - (p+\ell)B_0 - c(P'+1)\log 2 - B_0 = d\left(G(\theta) - c\log 2\right) + C_0,$$

where $C_0$ does not depend on $t$. Since $G(\theta) > c \log 2$ and $d \to \infty$ as $t \to \infty$, we may take $t$ so large that $\Phi(t) \geq 0$, $J \geq p$ and $2^J \geq \max(X_0, N_0)$. We check (ii) for this $t$.

If $N_0 \leq M \leq 2^J$, then $S_{\theta'}(M) = S_\theta(M) \geq c \log M$.

If $2^J \leq M \leq 2^{P'}$, the interval $[2^J, M)$ meets at most $P' - J = p + \ell$ dyadic blocks, on each of which $\varepsilon$ is constant; hence $S_{\theta'}(M) \geq S_{\theta'}(2^J) - (p+\ell)B_0 = S_\theta(2^J) - (p+\ell)B_0 \geq c(P'+1)\log 2 + B_0 \geq c \log M$, using $S_\theta(2^J) = S_\theta(2^p) + (J-p)G(\theta)$ and $\Phi(t) \geq 0$. In particular $S_{\theta'}(2^{P'}) \geq c(P'+1)\log 2 + B_0$.

If $M > 2^{P'}$, let $2^j \leq M < 2^{j+1}$ with $j \geq P'$. By Lemma 6.8(iii) for $\theta'$ (whose denominator is $3 \cdot 2^{P'}$) and $G(\theta') \geq c \log 2$,

$$S_{\theta'}(M) \geq S_{\theta'}(2^{P'}) + (j-P')G(\theta') - B_0 \geq c(P'+1)\log 2 + (j-P')c\log 2 = c(j+1)\log 2 \geq c \log M.$$

□

*Proof of Proposition 6.7.* Let $\theta_1 = \frac{1}{12}$, so $q = 12$, $p = 2$. For $4 \leq n < 8$ we have $\frac{n}{12} - \frac{1}{6} = \frac{n-2}{12} \in [0, \frac{1}{2})$, so $\Delta(\theta_1) = 2$ and $G(\theta_1) = 2 > 2\log 2$. By Lemma 6.8(iii), $S_{\theta_1}(M) \geq S_{\theta_1}(4) + 2(j-2) - 6$ for $2^j \leq M < 2^{j+1}$, $j \geq 2$, so there is $N_1$ with $S_{\theta_1}(M) \geq \log M$ for $M \geq N_1$.

Suppose that for some $k \geq 1$ we have constructed $\theta_1, \ldots, \theta_k$ with $\theta_i = a_i/q_i \in \mathcal{Q}$, and $N_1 \leq \cdots \leq N_k$, such that

(H1) $G(\theta_k) > (k+1)\log 2$, and
(H2) $S_{\theta_k}(M) \geq i \log M$ for all $1 \leq i \leq k$ and all $M \geq N_i$.

(For $k = 1$ this was just checked.) Choose $m \geq p_k$ with $m \equiv p_k \pmod 2$ and $\frac{q_k 2^m}{12} - h_k - q_k - 2 > (k+2)\log 2$, where $q_k = 3 \cdot 2^{p_k}$, $h_k = q_k - a_k$. Apply Lemma 6.12 with $\theta = \theta_k$, $c = k$ and $N_0 = X_0 = N_k$; its hypotheses hold by (H1) and (H2). Taking $t$ large enough we obtain $\theta_{k+1} = \theta'_t \in \mathcal{Q}$ such that $r_{\theta_{k+1}} = r_{\theta_k}$ on $[0, N_k)$, $S_{\theta_{k+1}}(M) \geq k \log M$ for $M \geq N_k$, and in addition

$$\eta_k := \theta_{k+1} - \theta_k < \min\left(\frac{1}{8q_k^2}, \frac{\eta_{k-1}}{4}\right) \qquad (\eta_0 := 1), \tag{19}$$

which is possible because $\theta'_t - \theta_k = b/(q_k 2^{m+2Rt}) \to 0$. By Lemma 6.11, $G(\theta_{k+1}) > (k+2)\log 2$. Property (H2) for $\theta_{k+1}$ and $i \leq k$ holds: for $N_i \leq M \leq N_k$ we have $S_{\theta_{k+1}}(M) = S_{\theta_k}(M) \geq i \log M$, and for $M \geq N_k$ we have $S_{\theta_{k+1}}(M) \geq k \log M \geq i \log M$. For $i = k+1$, Lemma 6.8(iii) and $G(\theta_{k+1}) > (k+2)\log 2$ give some $N_{k+1} \geq N_k$ with $S_{\theta_{k+1}}(M) \geq (k+1)\log M$ for all $M \geq N_{k+1}$.

The sequence $(\theta_k)$ is increasing, and by (19) its limit $\alpha$ satisfies $0 < \alpha - \theta_k < \frac{4}{3}\eta_k < \frac{1}{6q_k^2}$. Since $q_{k+1} = q_k 2^d > q_k$, we have $q_k \to \infty$. If $\alpha = c'/s$ were rational, then $\frac{1}{sq_k} \leq |\alpha - \frac{a_k}{q_k}| < \frac{1}{6q_k^2}$ for all $k$, which is impossible for large $k$; so $\alpha$ is irrational. For each $n \geq 1$ the point $n\alpha - \frac{1}{6}$ is irrational, hence not in $\frac{1}{2}\mathbb{Z}$, so $\rho$ is continuous there and $r_{\theta_k}(n) \to r_\alpha(n)$. Consequently $S_{\theta_k}(M) \to S_\alpha(M)$ for each $M$. Given $i$ and $M \geq N_i$, we have $S_{\theta_k}(M) \geq i \log M$ for all $k \geq i$ by (H2), and letting $k \to \infty$ gives $S_\alpha(M) \geq i \log M$. □

## 7. Ordered polynomial sumsets

In this section we answer two unnumbered questions from [22, §3.3]. In Section 7.1 we determine exactly which ordered polynomial sums $\{f(b_1)+g(b_2) : b_1 < b_2\}$ appear, up to a shift, in every set of positive density: those with $\deg f \le 1$ and $\deg g \le 1$. In Section 7.2 we observe that the natural generalisations of Question 3.18 of [22] have negative answers. Both answers rely on work of Ackelsberg [2] and Kousek [17]; our contribution is small and we say precisely what it is.

### 7.1. Which ordered polynomial sums are unavoidable.

For functions $f, g$ and $B \subset \mathbb{N}$ write

$$f(B) \oplus_< g(B) = \{f(b_1)+g(b_2) : \ b_1, b_2 \in B, \ b_1 < b_2\}.$$

The survey considers these "ordered sums" for polynomials $f, g$ "over $\mathbb{N}$" [22, (5)], notes that they are partition regular by Ramsey's theorem, and asks:

**Question §3.3(a)** ([22])**.** Let $f$ and $g$ be polynomials over $\mathbb{N}$. Can instances of $f(B) \oplus_< g(B)$ be found, up to a shift, in every set with positive density? That is, does every $A \subset \mathbb{N}$ of positive density contain $f(B) \oplus_< g(B) + t$ for some infinite $B \subset \mathbb{N}$ and some $t \ge 0$?

The survey does not specify the density; its default is upper Banach density. Special cases are Theorem 2.2 of [22] ($f = g = x$, positive, in the restricted form), Conjecture 3.10 ($f, g$ linear), Question 3.13 ($f = x^2$, $g = x$) and Example 3.14 ($g = x^2$). We interpret "polynomials over $\mathbb{N}$" in the widest sensible way and let

$$\mathcal{P} = \{f \in \mathbb{Q}[x] : \ f(\mathbb{N}) \subset \mathbb{N}_0\};$$

this contains all polynomials with coefficients in $\mathbb{N}_0$.

**Theorem 7.1.** *Let $f, g \in \mathcal{P}$.*

(a) *If $\deg f \le 1$ and $\deg g \le 1$, then for every $A \subset \mathbb{N}$ with $\mathsf{d}^*(A) > 0$ there are an infinite $B \subset \mathbb{N}$ and $t \ge 0$ with $f(B) \oplus_< g(B) + t \subset A$.*

(b) *If $\max(\deg f, \deg g) \ge 2$, then for every $\varepsilon > 0$ there is a set $A \subset \mathbb{N}$ with $\underline{\mathsf{d}}(A) > 1 - \varepsilon$ such that $f(B) \oplus_< g(B) + t \not\subset A$ for every infinite $B \subset \mathbb{N}$ and every $t \in \mathbb{Z}$. Moreover, one set $A$ works simultaneously for all such pairs $(f, g)$.*

Since the hypothesis in (a) is the weakest density condition and the conclusion in (b) concerns the strongest one, the classification does not depend on which notion of density is used in Question §3.3(a).

*Remark* 7.2 (Attribution). Part (a) is a direct consequence of Kousek's theorem [17, Theorem 1.2] (which proved Conjecture 3.10 of [22]); we only add the bookkeeping for constant terms and zero coefficients. Part (b) was known for $\deg g \ge 2$ in the reversed or sparse forms of [22, Example 3.14], and for $g(x) = x$ by Ackelsberg's theorem [2, Theorem 1.5]. Our proof of (b) is Ackelsberg's argument with $P(x) + x$ replaced by $f(x) + g(x)$; the only cases not covered explicitly in the literature that we know of are those with $\deg f \ge 2$ and $g(x) = mx + c$, $m \ge 2$.

*Remark* 7.3 (Scope).

(i) The argument for (b) does not use the order $b_1 < b_2$: the same set $A$ also avoids $\{f(b_1)+g(b_2) : b_1 > b_2\} + t$ and $\{f(b_1)+g(b_2) : b_1 \ne b_2\} + t$.

(ii) Integer polynomials that are not $\mathbb{N}_0$-valued on $\mathbb{N}$ are outside Question §3.3(a), and the classification does not extend to them. For instance, for $f(x) = -x$ and $g(x) = x$ the configuration is the infinite set $\{b_2 - b_1 : b_1 < b_2\}$ and $f + g = 0$; since $|R_k(0)| = 1$, the proof of (b) below shows that it can be avoided by sets of lower density $> 1 - \varepsilon$, although both polynomials are linear.

(iii) The survey remarks that it is immaterial whether one uses $f(B) \oplus_< g(B)$ or $\{r+s : r \in f(B), s \in g(B), r < s\}$. This is not accurate in general. If $g$ is non-constant and $f(b) \geq g(b)$ for all large $b$, the two formulations are equivalent: refining $B$ we may assume $f(b_1) < g(b_2)$ for all $b_1 < b_2$ in $B$, and also that $f$ is non-decreasing and $f \geq g$ on $B$, so that $f(b_1) < g(b_2)$ forces $b_1 < b_2$. If instead $f(b) < g(b)$ for all large $b$, the second set contains the diagonal elements $f(b)+g(b)$, and the answer can change even for linear $f, g$: for $f(x) = x$, $g(x) = 2x$ the second set contains $\{b_1 + 2b_2 : b_1 \leq b_2\}$, and Kousek [17, Proposition 5.1] (with $m = 1$, $\ell = 2$) gives a set of positive upper density containing no shift $\{b_1 + 2b_2 : b_1 \leq b_2\} + t$, $t \in \mathbb{N}$, with $B$ infinite (the case $t = 0$ reduces to $t = 3$ by replacing $B$ with $B - 1$). If $g \equiv c$ and $\deg f \geq 2$, the second set is finite, while $f(B) \oplus_< g(B) = f(B \setminus \{\max B\}) + c$ is infinite. Theorem 7.1 uses the formulation [22, (5)].

(iv) In $\mathbb{Q}$ the situation is different: Ackelsberg [1, Theorem 1.4] proved that if $P \in \mathbb{Q}[x]$ and $P(x)+x$ are non-constant, then every $A \subset \mathbb{Q}$ of positive upper Banach density contains $\{b_i,\ P(b_i) + b_j : i < j\} + t$ for some infinite $\{b_n\} \subset \mathbb{Q}$ and some $t \in \mathbb{Q}$.

We collect elementary facts about $\mathcal{P}$.

**Lemma 7.4.** *Let $f, g \in \mathcal{P}$.*

(i) *$f(\mathbb{Z}) \subset \mathbb{Z}$, and there is $D \in \mathbb{N}$ with $Df \in \mathbb{Z}[x]$. For such $D$ and every $k \in \mathbb{N}$, $n \equiv n' \pmod{Dk}$ implies $f(n) \equiv f(n') \pmod{k}$.*

(ii) *If $f$ is non-constant, its leading coefficient is positive. Consequently $\deg(f+g) = \max(\deg f, \deg g)$.*

(iii) *If $\deg f \leq 1$, then $f(x) = mx + c$ with $m \in \mathbb{N}_0$ and $c \in \mathbb{Z}$.*

*Proof.* (i) Write $f(x) = \sum_i \Delta^i f(1) \binom{x-1}{i}$ (Newton's forward difference formula); the numbers $\Delta^i f(1)$ are integers because $f(\mathbb{N}) \subset \mathbb{Z}$, and $\binom{x-1}{i} \in \mathbb{Z}$ for $x \in \mathbb{Z}$. A common denominator $D$ of the coefficients exists since $f \in \mathbb{Q}[x]$. If $n - n' = Dks$, then $D(f(n) - f(n')) = (n-n')Q$ for some $Q \in \mathbb{Z}$ (as $Df \in \mathbb{Z}[x]$), so $f(n) - f(n') = ksQ$. (ii) A non-constant polynomial with negative leading coefficient tends to $-\infty$. If $\deg f = \deg g$, the leading coefficients are both positive and do not cancel. (iii) $m = f(2) - f(1) \in \mathbb{Z}$, $c = f(1) - m \in \mathbb{Z}$, and $m \geq 0$ by (ii). □

For an integer-valued $h \in \mathbb{Q}[x]$ and $k \in \mathbb{N}$ let $R_k(h) = \{h(n) \bmod k : n \in \mathbb{Z}\} \subset \mathbb{Z}/k\mathbb{Z}$.

**Lemma 7.5** (Ackelsberg)**.** *Let $h$ be a polynomial in $\mathbb{Q}[x]$ of degree at least 2 such that $h(\mathbb{Z}) \subset \mathbb{Z}$. Then $\inf_{k \in \mathbb{N}} |R_k(h)|/k = 0$.*

*Proof.* Let $\mathbb{P}_h$ be the set of primes $p$ such that $h(\mathbb{Z})$ meets every residue class modulo $p$. By Ackelsberg's Lemma 4.3 in [2], $\sum_{p \in \mathbb{P} \setminus \mathbb{P}_h} 1/p = \infty$, so $\prod_{p \in \mathbb{P} \setminus \mathbb{P}_h} (1 - 1/p) = 0$. If $p_1, \dots, p_s \in \mathbb{P} \setminus \mathbb{P}_h$ are distinct and $k = p_1 \cdots p_s$, then by the Chinese remainder theorem every element of $R_k(h)$ reduces modulo $p_i$ to an element of $R_{p_i}(h)$, so $|R_k(h)| \leq \prod_i |R_{p_i}(h)| \leq \prod_i (p_i - 1)$ and $|R_k(h)|/k \leq \prod_i (1 - 1/p_i)$. This is the argument of the proof of [2, Theorem 1.5], applied to $h$ instead of $P(x) + x$. □

**Lemma 7.6.** *Let $E_1, E_2, \dots \subset \mathbb{N}$ have natural densities $\delta_1, \delta_2, \dots$. There is $S \subset \mathbb{N}$ such that $E_j \setminus S$ is finite for every $j$ and $\overline{\mathrm{d}}(S) \leq \sum_j \delta_j$.*

*Proof.* This is the natural-density version of [2, Lemma 3.1]. Choose $M_1 < M_2 < \cdots$ such that $|E_i \cap [N]| \leq (\delta_i + 2^{-J})N$ for all $J$, all $i \leq J$ and all $N \geq M_J$, and let $S = \bigcup_j \big(E_j \cap (M_j, \infty)\big)$. Then $E_j \setminus S \subset [M_j]$. If $M_J \leq N < M_{J+1}$, then $S \cap [N] \subset \bigcup_{j \leq J} (E_j \cap [N])$, so $|S \cap [N]| \leq (\sum_j \delta_j + J2^{-J})N$. □

*Proof of Theorem 7.1(b).* The set of pairs $(f,g)\in\mathcal{P}^2$ with $\max(\deg f,\deg g)\ge 2$ is countable; enumerate the triples $(f,g,t)$ with such $(f,g)$ and $t\in\mathbb{Z}$ as $(f_j,g_j,t_j)_{j\ge1}$. Put $h_j=f_j+g_j$. By Lemma 7.4, $h_j$ is integer-valued on $\mathbb{Z}$ and $\deg h_j\ge 2$, so by Lemma 7.5 there is $k_j$ with $|R_{k_j}(h_j)|/k_j<\varepsilon 2^{-j}$. Let

$$E_j=\{x\in\mathbb{N}:\ (x-t_j)\bmod k_j\in R_{k_j}(h_j)\},$$

a union of residue classes of natural density $\delta_j=|R_{k_j}(h_j)|/k_j$. Let $S$ be given by Lemma 7.6 and $A=\mathbb{N}\setminus S$. Then $\underline{\mathtt{d}}(A)=1-\overline{\mathtt{d}}(S)\ge 1-\sum_j\delta_j>1-\varepsilon$.

Suppose that $f_j(B)\oplus_< g_j(B)+t_j\subset A$ for some $j$ and some infinite $B$. Let $k=k_j$ and let $D$ be such that $Df_j,Dg_j\in\mathbb{Z}[x]$. By the pigeonhole principle there is $n$ such that $B'=B\cap(n+Dk\mathbb{Z})$ is infinite. By Lemma 7.4(i), $f_j(b_1)+g_j(b_2)\equiv f_j(n)+g_j(n)=h_j(n)\pmod k$ for all $b_1,b_2\in B'$, so $f_j(B')\oplus_< g_j(B')+t_j\subset A\cap E_j\subset E_j\setminus S$, which is finite. But $f_j(B')\oplus_< g_j(B')$ is infinite: if $g_j$ is non-constant, fix $b_1=\min B'$ and let $b_2\to\infty$; if $g_j$ is constant, then $\deg f_j\ge 2$ and $f_j(b_1)\to\infty$ as $b_1\to\infty$ in $B'$. This contradiction proves (b) for every $(f,g)$ and every $t\in\mathbb{Z}$ simultaneously. □

*Proof of Theorem 7.1(a).* By Lemma 7.4(iii), $f(x)=mx+c$ and $g(x)=\ell x+c'$ with $m,\ell\in\mathbb{N}_0$ and $c,c'\in\mathbb{Z}$. Let $\mathtt{d}^*(A)>0$; in particular $A$ is infinite.

If $m,\ell\ge1$, Kousek's theorem [17, Theorem 1.2] gives an infinite $B$ and $t\in\mathbb{N}$ with $\{mb_1+\ell b_2:\ b_1,b_2\in B,\ b_1<b_2\}+t\subset A$. For $j\in\mathbb{N}$ let $B_j=\{b-j:b\in B,\ b>j\}$. Then

$$f(B_j)\oplus_< g(B_j)+t'_j=\{mb_1+\ell b_2:\ b_1,b_2\in B,\ j<b_1<b_2\}+t\subset A,\qquad t'_j=t-c-c'+(m+\ell)j,$$

and $t'_j\ge0$ for $j$ large.

If $m=0<\ell$, then $f(B)\oplus_< g(B)=\{\ell b+c+c':b\in B\setminus\{\min B\}\}$. Choose $r$ such that $A\cap(r+\ell\mathbb{Z})$ is infinite, then $r'\equiv r\pmod\ell$ with $r'\ge c+c'$, and put $t=r'-c-c'$ and $B=\{(x-r')/\ell:\ x\in A,\ x\equiv r\pmod\ell,\ x>r'\}$. Then $\ell b+c+c'+t\in A$ for all $b\in B$. The case $\ell=0<m$ is the same, since then $f(B)\oplus_< g(B)=\{mb+c+c':b\in B\}$ (every element of an infinite $B$ has a larger one). If $m=\ell=0$, the configuration is the single point $c+c'$; take $t=x-c-c'$ for some $x\in A$ with $x\ge c+c'$. □

*Formalization.* Theorem 7.1 is formalized in `Kmrr25/S33a.lean`. Part (b) (`thm_u33a_neg`) is unconditional; there Ackelsberg's Lemma 4.3, used in the proof of Lemma 7.5, is proved by a different argument (Hermite's criterion and Dirichlet's theorem). Part (a) (`thm_u33a_pos`) takes Kousek's theorem [17, Theorem 1.2] as an explicit hypothesis.

7.2. **Generalisations of Question 3.18.** The survey asks whether shifts of $\{b_i^2+b_j:i<j\}$ appear in every set of positive upper Banach density [22, Question 3.13], and generalises this by adding the base point $b_j$ [22, Question 3.15] and by replacing $x^2$ with arbitrary polynomials $p_1,\dots,p_k\in\mathbb{Z}[x]$ with $p_\ell(0)=0$ [22, Question 3.16]. Its Remark 3.17 records that the answers to Questions 3.15 and 3.16 are negative by Ackelsberg's work; indeed, both configurations contain that of Question 3.13, which Ackelsberg showed can be avoided [2, Theorem 1.5]. Then it asks:

**Question 3.18** ([22])**.** If $A\subset\mathbb{N}$ has positive upper Banach density, must there exist an infinite set $B=\{b_1<b_2<\cdots\}$ and $t\ge0$ such that $\{b_ib_j+b_k:\ i<j<k\}$ is contained in $A-t$?

**Question §3.3(e)** ([22])**.** Similar to the way Question 3.13 is generalised in Questions 3.15 and 3.16, one can inquire about analogous generalisations of Question 3.18.

The faithful analogues are:

(F1) (base point, as in Question 3.15) must every $A$ with $\mathsf{d}^*(A) > 0$ contain $\{b_k,\ b_ib_j + b_k : i < j < k\} + t$ for some infinite $B$ and $t \geq 0$?

(F2) (polynomial family, as in Question 3.16) given $p_1, \dots, p_r \in \mathbb{Z}[x, y]$ with $p_\ell(0, 0) = 0$, must every $A$ with $\mathsf{d}^*(A) > 0$ contain $\{p_\ell(b_i, b_j) + b_k :\ 1 \leq \ell \leq r,\ i < j < k\} + t$ for some infinite $B$ and $t \geq 0$?

As in Remark 3.17 of [22], (F2) is a question about all such families.

**Proposition 7.7.** *Both (F1) and (F2) have negative answers. More precisely, for every $\varepsilon > 0$ there is $A \subset \mathbb{N}$ with $\mathsf{d}^*(A) > 1-\varepsilon$ such that for no $t \in \mathbb{Z}$ and no infinite $B = \{b_1 < b_2 < \cdots\} \subset \mathbb{N}$ does $A - t$ contain the configuration in (F1), or the configuration in (F2) for the one-element family $p_1(x, y) = xy$.*

*Proof.* Ackelsberg [2, Theorem 1.7] proved that for every $\varepsilon > 0$ there is $A \subset \mathbb{N}$ with $\mathsf{d}^*(A) > 1-\varepsilon$ such that whenever $t \in \mathbb{Z}$ and $\{b_1b_2 + b_3 + t :\ b_1, b_2, b_3 \in B,\ b_1 < b_2 < b_3\} \subset A$, the set $B$ is finite; this answers Question 3.18 negatively. The configuration in (F1), and the configuration in (F2) for $p_1 = xy$, both contain $\{b_ib_j + b_k : i < j < k\}$, so this set $A$ has the required property. □

*Formalization.* Proposition 7.7 and the remark following it are formalized in `Kmrr25/S33e.lean` (`prop_u33e`, `F2_sq`), including proofs of the two quoted theorems of Ackelsberg in the form used here.

For (F2) one does not even need Question 3.18: for the one-element family $p_1(x, y) = x^2$, if $\{b_i^2 + b_k : i < j < k\} + t \subset A$, then $C = \{b_1, b_3, b_5, \dots\}$ satisfies $\{c^2 + c' : c, c' \in C,\ c < c'\} + t \subset A$ (between $b_{2i-1}$ and $b_{2i'-1}$, $i < i'$, lies $b_{2i}$), which is the configuration of Question 3.13 of [22]; by Ackelsberg's theorem [2, Theorem 1.5], recorded in [22], there are sets of upper Banach density arbitrarily close to 1 in which no translate contains it.

*Remark* 7.8. This answer is easy: the generalisations contain the configuration of Question 3.18 itself, which was already answered negatively by Ackelsberg [2, Theorem 1.7] (the second arXiv version of [22] does not record this), in exactly the way that Questions 3.15 and 3.16 are answered by Remark 3.17 of [22]. For (F2) the survey itself already suffices, through the family $\{x^2\}$. We include it for completeness. A more interesting question is which single polynomials $p \in \mathbb{Z}[x, y]$ have the property that every set of positive upper Banach density contains a translate of $\{p(b_i, b_j) + b_k : i < j < k\}$ for some infinite $B$; we do not know the answer, for instance, for $p(x, y) = x(y - x)$.

## 8. Sumsets of the form $B + mB$

This section concerns Subsection 3.4 of [22]. For $B \subset \mathbb{N}$ and $m \in \mathbb{N}$ the survey writes $B + mB = \{b + mb' : b, b' \in B\}$; this unrestricted sumset contains the diagonal elements $(m + 1)b$. We write

$$B \oplus mB = \{b + mb' :\ b, b' \in B,\ b \neq b'\}$$

for the restricted version. We answer Question 3.20 negatively, for two and for three or more colours, even after excluding $m = 1$ (Section 8.1); Question 3.23 negatively as stated, the answer depending on the diagonal (Section 8.2); and Question 3.24 negatively (Section 8.3), where a counterexample is in fact already implicit in the work of Kousek and Radić [19].

8.1. **Colourings and Question 3.20.** A finite colouring of $\mathbb{N}$ is a map $\chi\colon \mathbb{N} \to \{0, 1, \dots, r-1\}$. The survey gives, for each $m$, a 3-colouring with no monochromatic $B + mB$, $B$ infinite [22, Example 3.9]. Two colours already suffice: Hindman [13, Theorem 2.11] proved that for every $m \geq 2$ the 2-colouring $n \mapsto \lfloor \log_m n \rfloor \bmod 2$ admits no infinite $B$ with $\{mx+y : x, y \in B,\ x \neq y\}$ monochromatic (see also [15, Remark 1.5]). Observing that such colourings depend heavily on $m$, the survey asks:

**Question 3.20** ([22])**.** Is it true that for any finite colouring of $\mathbb{N}$ there exists $q \in \mathbb{N}$ such that for all $m \in \mathbb{N}$ with $\gcd(m, q) = 1$ there is an infinite set $B \subset \mathbb{N}$ such that $B + mB$ is monochromatic?

The motivation given is that "the set of problematic $m$" might be "small in a multiplicative sense". Since $\gcd(1, q) = 1$ for every $q$, a positive answer for a colouring $\chi$ would require an infinite $B$ with $B+B$ monochromatic. Hindman [13, Theorem 2.4] constructed a 3-colouring of $\mathbb{N}$ with no such $B$, so Question 3.20 has a negative answer as stated, for three or more colours. This hinges on the literal formulation: for two colours, $m = 1$ is harmless by the recent solution of Owings' problem [27] by Huang, Lian, Shao, Xiao, Xu and Zhang [15, Theorem 1.1], and the heuristic behind the question concerns $m \geq 2$. We show that the answer is negative also for $m \geq 2$ and for two colours (for two colours, with the unrestricted $B + mB$ of the survey). What is new is that a *single* colouring is bad for infinitely many $m$ simultaneously, including $m$ in every reduced residue class and infinitely many primes; for each individual $m$, Hindman's theorem already provides a 2-colouring for which $m$ is strongly bad.

For a colouring $\chi$ call $m \in \mathbb{N}$ *bad* if no infinite $B \subset \mathbb{N}$ has $B + mB$ monochromatic, and *strongly bad* if no infinite $B$ has $B \oplus mB$ monochromatic. Strongly bad integers are bad.

**Theorem 8.1.** *Let $r \geq 3$ and $\chi_r(n) = \lfloor \log_2 n \rfloor \bmod r$. Every integer $m$ with $2^k < m < 2^{k+1}$ for some $k \geq 1$ with $k \not\equiv 0, -1 \pmod r$ is strongly bad for $\chi_r$. The set of these $m$ has logarithmic density $1 - 2/r$ and contains infinitely many primes.*

**Theorem 8.2.** *For $\rho > 1$ let $\chi^{(\rho)}(n) = \lfloor \log_\rho n \rfloor \bmod 2$.*

(i) *Let $\varphi = \frac{1+\sqrt{5}}{2}$ and let $L_1 = 1$, $L_2 = 3$, $L_{n+1} = L_n + L_{n-1}$ be the Lucas numbers. For every odd $n \geq 3$, $L_n$ is bad for $\chi^{(\varphi)}$. For every $q \in \mathbb{N}$ there are infinitely many odd $n \geq 3$ with $L_n \equiv 1 \pmod q$.*

(ii) *There are $\rho \in (\frac{3}{2}, 2)$ and infinitely many primes that are bad for $\chi^{(\rho)}$.*

**Corollary 8.3.** *For every $r \geq 2$ there is an $r$-colouring of $\mathbb{N}$ such that for every $q \in \mathbb{N}$ some prime $m \geq 2$ with $\gcd(m, q) = 1$ is bad. In particular Question 3.20 has a negative answer for every number $r \geq 2$ of colours, also when the condition is imposed only for $m \geq 2$ or only for prime $m$.*

*Proof.* For $r \geq 3$ use Theorem 8.1: infinitely many primes are bad, and $q$ has finitely many prime factors. For $r = 2$ use Theorem 8.2(ii) (or, for not necessarily prime $m$, Theorem 8.2(i), where $L_n \equiv 1 \pmod q$ gives $\gcd(L_n, q) = 1$). □

By Theorem 8.1, the set of bad $m$ need not be small in any multiplicative sense: for $r$ colours it can have logarithmic density $1-2/r$. Question 3.19 of [22], whether for every finite colouring some $m$ is not bad, remains open for $r \geq 3$ colours; for $r = 2$ it is answered positively with $m = 1$ by [15, Theorem 1.1].

*Remark* 8.4 (The diagonal and two colours). For $m \geq 2$, $B + mB$ is the union of the diagonal $\{(m + 1)b : b \in B\}$ and the two ordered pieces $\{mx + y : x, y \in B,\ x < y\}$ and $\{mx + y : x, y \in B,\ x > y\}$. Huang et al. [15, Theorem 1.6] proved that for every 2-colouring of $\mathbb{N}$ and all

$m, \ell \in \mathbb{N}$ there is an infinite $B$ with $(m+\ell)B \cup \{mx+\ell y : x, y \in B,\ x < y\}$ monochromatic. Applied with $(m,1)$ and with $(1,m)$, this shows that for two colours the diagonal together with either ordered piece can always be made monochromatic. So a 2-colouring for which some $m$ is bad must use both ordered pieces; the proof of Theorem 8.2 uses one element from each of the three parts. Hindman's theorem [13, Theorem 2.11] shows that each single $m \geq 2$ is strongly bad for some 2-colouring, namely $\lfloor \log_m n \rfloor \bmod 2$ (this is the colouring $\chi^{(\rho)}$ of Theorem 8.2 with $\rho = m$). Whether a single 2-colouring can make infinitely many pairwise coprime $m$ strongly bad, that is, whether Hindman's theorem has a version uniform in $m$ (equivalently, the version of Question 3.20 with $B \oplus mB$ and two colours), remains open. For related 2- and 3-colour obstructions to weighted ordered sums see also [15, Propositions 5.1 and 5.2].

*Remark* 8.5. Di Nasso, Luperi Baglini, Mamino, Mennuni and Ragosta [9, §1] answer negatively the shift-free versions of several questions of Subsections 3.3 and 3.4 of [22] (in the numbering of its published version) in a strengthened "Ramsey partition regular" form. These are different statements from Question 3.20.

For $\rho > 1$ and $n \in \mathbb{N}$ put $\lambda_\rho(n) = \lfloor \log_\rho n \rfloor$, so that $\lambda_\rho(n) = k$ if and only if $\rho^k \leq n < \rho^{k+1}$.

**Lemma 8.6.** *If $x, y \in \mathbb{R}$ and $y - x \in (k, k+1)$ for some $k \in \mathbb{Z}$, then $\lfloor y \rfloor - \lfloor x \rfloor \in \{k, k+1\}$.*

*Proof.* $\lfloor y \rfloor - \lfloor x \rfloor$ is an integer in $(y - x - 1, y - x + 1) \subset (k-1, k+2)$. □

**Lemma 8.7.** *Let $\rho > 1$, $m \geq 2$ and $b, b' \in \mathbb{N}$. Put $\delta = \log_\rho(1 + mb/b')$ and $\delta' = \log_\rho(1 + b/(mb'))$. Then $\log_\rho(b' + mb) = \log_\rho b' + \delta$, $\log_\rho(mb' + b) = \log_\rho b' + \log_\rho m + \delta'$, and $0 < \delta' < \delta$; moreover $\delta \to 0$ as $b' \to \infty$ with $b$ fixed.*

*Proof.* The identities are immediate, and $b/(mb') < mb/b'$ because $m \geq 2$. □

**Proposition 8.8.** *Let $\rho > 1$, $r \geq 3$ and $\chi(n) = \lambda_\rho(n) \bmod r$. Let $m \geq 2$ be such that $L = \log_\rho m \notin \mathbb{Z}$ and $k = \lfloor L \rfloor$ satisfies $k \not\equiv 0$ and $k + 1 \not\equiv 0 \pmod r$. Then $m$ is strongly bad for $\chi$.*

*Proof.* Let $B$ be infinite and fix $b \in B$. By Lemma 8.7 there is $b' \in B$, $b' \neq b$, with $\delta < L - k$. Put $X = b' + mb$ and $Y = b + mb'$, both in $B \oplus mB$. By Lemma 8.7, $\log_\rho Y - \log_\rho X = L + \delta' - \delta \in (L - \delta, L) \subset (k, k+1)$, so by Lemma 8.6 $\lambda_\rho(Y) - \lambda_\rho(X) \in \{k, k+1\}$. Neither number is divisible by $r$, so $\chi(X) \neq \chi(Y)$. □

*Proof of Theorem 8.1.* For $2^k < m < 2^{k+1}$ we have $\log_2 m \notin \mathbb{Z}$ and $\lfloor \log_2 m \rfloor = k$, so Proposition 8.8 with $\rho = 2$ applies. Let $S$ be the set of these $m$. For each $k$, $\sum_{2^k < m < 2^{k+1}} 1/m = \log 2 + O(2^{-k})$, and among any $r$ consecutive values of $k$ exactly $r - 2$ are admissible. Hence $\sum_{m \in S,\, m < 2^{K+1}} 1/m = \frac{r-2}{r} K \log 2 + O(1)$, and for $2^K \leq N < 2^{K+1}$ the partial sum up to $N$ differs from this by $O(1)$. So $S$ has logarithmic density $1 - 2/r$. By Bertrand's postulate each interval $(2^k, 2^{k+1})$, $k \geq 1$, contains a prime. □

**Proposition 8.9.** *Let $m \geq 2$ and $\rho > 1 + \frac{1}{m}$, and suppose there is an odd integer $j$ with $m < \rho^j < m + 1$. Then $m$ is bad for $\chi(n) = \lambda_\rho(n) \bmod 2$.*

*Proof.* Let $L = \log_\rho m$ and $L' = \log_\rho(m+1)$. Then $L < j < L'$ and $L' - L = \log_\rho(1 + \frac{1}{m}) < 1$. Let $\eta = \min(j - L, L' - j) > 0$. Let $B$ be infinite, fix $b \in B$, and by Lemma 8.7 choose $b' \in B$, $b' \neq b$, with $\delta < \eta$. Consider the elements

$$X = b' + mb, \qquad Y = b + mb', \qquad Z = b' + mb' = (m+1)b'$$

of $B+mB$, and let $u=\log_\rho b'$, $x=\log_\rho X$, $y=\log_\rho Y$, $z=\log_\rho Z$. By Lemma 8.7, $x=u+\delta$, $y=u+L+\delta'$ and $z=u+L'$ with $0<\delta'<\delta<\eta$. Hence $y<u+L+\eta\le u+j<x+j$ and $z=u+L'\ge u+j+\eta>x+j$, so

$$y<x+j<z,\qquad 0<z-y<L'-L<1. \tag{20}$$

Suppose $X,Y,Z$ have the same colour. Since $0<z-y<1$, the integer $\lambda_\rho(Z)-\lambda_\rho(Y)$ is 0 or 1, and it is even; so $\lambda_\rho(Y)=\lambda_\rho(Z)=:N$. Then $N\le y<z<N+1$, and by (20) $N<x+j<N+1$, so $N=\lfloor x\rfloor+j=\lambda_\rho(X)+j$. As $j$ is odd, $\chi(X)\ne\chi(Y)$, a contradiction. □

**Lemma 8.10.** *There are $\rho\in(\frac32,2)$, odd integers $j_1<j_2<\cdots$ and primes $p_1<p_2<\cdots$ with $p_k<\rho^{j_k}<p_k+1$ for all $k$.*

*Proof.* We construct closed intervals $[\frac32,2]=I_0\supset I_1\supset\cdots$, odd $j_k$ and primes $p_k$ with $p_k<s^{j_k}<p_k+1$ for all $s\in I_k$; any $\rho\in\bigcap_k I_k$ then works. Let $p_0=1$, $j_0=0$, and suppose $I_{k-1}=[\alpha,\beta]$ with $\frac32\le\alpha<\beta$ is constructed. Choose an odd $j>j_{k-1}$ so large that $x:=\alpha^j\ge p_{k-1}+2$ and $\beta^j\ge 4x$, put $n=\lceil x\rceil$, and let $p$ be a prime with $n<p<2n$ (Bertrand's postulate). Then $\alpha^j\le n<p$, $p>p_{k-1}$, and $p+1\le 2n\le 2x+2<4x\le\beta^j$. Let $I_k=[(p+\frac13)^{1/j},(p+\frac23)^{1/j}]\subset(\alpha,\beta)$, $j_k=j$ and $p_k=p$. □

*Proof of Theorem 8.2.* (i) Let $\psi=\frac{1-\sqrt5}{2}=-\varphi^{-1}$, so $L_n=\varphi^n+\psi^n$. For odd $n$, $\psi^n\in(-1,0)$, so $L_n<\varphi^n<L_n+1$. For $n\ge3$, $L_n\ge4$ and $\varphi>1+\frac{1}{L_n}$, so Proposition 8.9 applies with $m=L_n$, $\rho=\varphi$, $j=n$. For the second claim, $(L_n \bmod q)$ satisfies an invertible linear recurrence over $\mathbb{Z}/q\mathbb{Z}$ and is therefore purely periodic, with some period $P$; then $L_n\equiv L_1=1 \pmod q$ for all $n\equiv1 \pmod{2P}$, and these $n$ are odd.

(ii) Take $\rho$, $(j_k)$, $(p_k)$ from Lemma 8.10. Then $\rho>\frac32\ge1+\frac{1}{p_k}$ and $p_k<\rho^{j_k}<p_k+1$ with $j_k$ odd, so Proposition 8.9 applies to $m=p_k$. □

*Formalization.* Theorems 8.1 and 8.2, Corollary 8.3 and the propositions and lemmas they rely on are formalized in `Kmrr25/Q320.lean` (`thm320A`, `thm320B`, `cor320`). Hindman's 3-colouring, quoted for $m=1$, is not formalized.

8.2. **Logarithmic density and Question 3.23.** The survey shows that a density version of Question 3.19 needs care: there is a thick set of positive upper density containing no $B\oplus mB+t$ with $m\ge2$ and $B$ infinite [22, Example 3.21]. That set has zero upper logarithmic density, which motivates:

**Question 3.23** ([22])**.** Does every set $A\subset\mathbb{N}$ with $\overline{\mathrm{d}}_{\log}(A)>0$ contain a sumset $B+mB+t$ for some $m\ge2$, $t\ge0$ and some infinite set $B\subset\mathbb{N}$?

As written, $B+mB$ is the unrestricted sumset: the survey writes $B\oplus B$ or states $b_1\ne b_2$ explicitly when it means the restricted one, and calls $B+B$ the unrestricted sumset. The answer to Question 3.23 as stated is negative.

For each *fixed* pair $(\ell,m)$, unrestricted sums $mB+\ell B$ have been studied by Kousek [17]. He showed that $\underline{\mathrm{d}}(A)>1/2$ forces $\{mb_1+\ell b_2:b_1\le b_2\}+t\subset A$, and that a suitable upper density condition depending on $m/\ell$ does too [17, Theorems 1.6 and 1.7]; these are sharp, and for distinct $\ell,m$ there are sets of positive lower density containing no shift of $mB+\ell B=\{mb_1+\ell b_2:b_1,b_2\in B\}$ with $B$ infinite [17, Propositions 5.1, 6.5, 7.6 and 7.7]; see also his Questions 7.8 and 7.9. Those examples depend on $(\ell,m)$, whereas Question 3.23 allows $m$ to depend on $A$. What is new below is a single set of positive logarithmic density that avoids $\ell B+mB+t$ for all $\ell,m\ge1$ and $t\in\mathbb{Z}$ at once. As in Kousek's results (and as in Remark 8.4

for colourings), the diagonal together with one ordered piece cannot be avoided by sets of lower density $> 1/2$, and our proof uses the diagonal and both ordered pieces.

Fix $\eta \in (0, \frac{1}{6})$ and let $U_\eta = (\frac{1}{3}+\eta, \frac{2}{3}-\eta)$. Let $T_k = 2^{2^k}$, $X_k = kT_k$ and $Y_k = T_{k+1}/k = T_k^2/k$ for $k \geq 1$; note $T_k \leq X_k < Y_k \leq T_{k+1}$. Let

$$R_k = \{s \in \{0,\dots,k-1\} :\ s/k \in U_\eta\}, \qquad A_\eta = \bigcup_{k\geq 1} \{n \in \mathbb{N} :\ X_k \leq n < Y_k,\ n \bmod k \in R_k\}.$$

Thus on the block $[X_k, Y_k)$ the set $A_\eta$ is the periodic set $\{n : \{n/k\} \in U_\eta\}$, and the transition intervals $[Y_{k-1}, X_k)$ are empty.

**Theorem 8.11.** (a) *The logarithmic density of $A_\eta$ exists and equals $\frac{1}{3} - 2\eta$.*

(b) *For all integers $\ell, m \geq 1$, all $t \in \mathbb{Z}$ and every infinite $B \subset \mathbb{N}$, we have $\ell B + mB + t \not\subset A_\eta$. More precisely, for every $c \in B$ only finitely many $b \in B$ satisfy*

$$\ell b + mc + t, \quad \ell c + mb + t, \quad (\ell+m)b + t \ \in A_\eta.$$

*In particular (with $\ell = 1$) $A_\eta$ is a negative answer to Question 3.23, simultaneously for all $m$.*

*Remark* 8.12. (i) The answer hinges on the diagonal: the contradiction comes from the element $(\ell+m)b + t$. The example motivating Question 3.23, [22, Example 3.21], concerns the restricted sumset, and for it $A_\eta$ is no counterexample: for every odd $m \geq 3$ there is an infinite $B$ with $B \oplus mB \subset A_\eta$. Indeed, let $\varepsilon = \frac{1}{2}(\frac{1}{6} - \eta)$, so that $[\frac{1}{2}, \frac{1}{2}+\varepsilon) \subset U_\eta$. Choose $b_1 < b_2 < \cdots$ inductively: given $b_1, \dots, b_{j-1}$, choose an even $k$ so large that $(m+1)b_{j-1} < \varepsilon k$ and $Y_k/(m+1) - X_k > k$, and choose $b_j \in [X_k, Y_k/(m+1))$ with $b_j \equiv k/2 \pmod k$. For $i < j$ the numbers $b_j + mb_i$ and $mb_j + b_i$ lie in $[X_k, Y_k)$ and are congruent modulo $k$ to $\frac{k}{2} + mb_i$ and to $\frac{k}{2} + b_i$ respectively (as $m$ is odd, $\frac{mk}{2} \equiv \frac{k}{2}$), both of which lie in $[\frac{k}{2}, \frac{k}{2} + \varepsilon k)$. Hence both belong to $A_\eta$.

(ii) The restricted version of Question 3.23 (does every $A$ with $\overline{\mathsf{d}}_{\log}(A) > 0$ contain $B \oplus mB + t$ for some $m \geq 2$, $t \geq 0$ and infinite $B$?) remains open.

**Lemma 8.13.** *Let $k \geq 1$, $s_1, s_2, s_3 \in R_k$ and $s \in \mathbb{Z}$ with $|s| < \eta k$. Then $s_1 + s_2 - s_3 \not\equiv s \pmod k$.*

*Proof.* From $s_i/k \in U_\eta$ we get $s_1 + s_2 - s_3 - s \in \big(k(\frac{2}{3}+2\eta) - k(\frac{2}{3}-\eta) - \eta k,\ k(\frac{4}{3}-2\eta) - k(\frac{1}{3}+\eta) + \eta k\big) = (2\eta k,\ k(1-2\eta)) \subset (0, k)$, which contains no multiple of $k$. □

**Lemma 8.14.** *If $n, n' \in A_\eta$ lie in the blocks of indices $k < k'$, then $n'/n > k(k+1)$. If $n \in A_\eta$ and $n \geq T_{K+1}$, then the block of $n$ has index at least $K+1$.*

*Proof.* $n < Y_k = T_{k+1}/k$ and $n' \geq X_{k+1} = (k+1)T_{k+1}$. The block of index $k$ lies in $[T_k, T_{k+1})$. □

**Lemma 8.15.** *Let $k \in \mathbb{N}$, $R \subset \{0,\dots,k-1\}$, $\rho = |R|/k$, and $1 \leq P < Q$ real. Let $g\colon \mathbb{Z} \cap [P, Q) \to \{0,1\}$ be constant on each of $J$ consecutive subintervals partitioning $[P, Q)$. Then*

$$\Big| \sum_{\substack{P \leq n < Q \\ n \bmod k \in R}} \frac{g(n)}{n} - \rho \sum_{P \leq n < Q} \frac{g(n)}{n} \Big| \leq \frac{2Jk}{P}.$$

*Proof.* It suffices to show that the error on one subinterval $[P_1, Q_1)$, $P_1 \geq P$, on which $g \equiv 1$, is at most $2k/P$. Let $n_0 = \lceil P_1 \rceil$ and split the integers of $[P_1, Q_1)$ into complete periods $\{n_0 + ik, \dots, n_0 + (i+1)k - 1\}$, $0 \leq i < I$, and a remainder of fewer than $k$ integers. On the $i$-th period both sums lie in $[\rho k/(n_0 + (i+1)k),\ \rho k/(n_0 + ik)]$, so the errors telescope to at most $\rho k/n_0 \leq k/P$. On the remainder both sums lie in $[0, k/n_0]$. □

*Proof of Theorem 8.11(b).* Fix $\ell, m \geq 1$, $t \in \mathbb{Z}$ and $c \in B$, and put $s = (\ell+m)c + t$ and $C = \ell + m + 1$. For $b \in B$ let

$$u = \ell b + mc + t, \qquad v = \ell c + mb + t, \qquad w = (\ell+m)b + t,$$

all three elements of $\ell B + mB + t$, and note the identity $u + v - w = s$. If $b \geq 2(|t| + (\ell+m)c)$, then $u, v, w \in [b/2, Cb]$, so their pairwise ratios are at most $2C$. Choose $K$ with $K(K+1) > 2C$ and $K \geq |s|/\eta$, and suppose $b \geq \max\big(2(|t| + (\ell+m)c),\, 2T_{K+1}\big)$ and $u, v, w \in A_\eta$. By Lemma 8.14 the blocks of $u, v, w$ have indices $\geq K+1$, and they coincide, since otherwise two of the numbers would have ratio $> (K+1)(K+2) > 2C$. Let $k > K$ be the common index. The residues of $u, v, w$ modulo $k$ lie in $R_k$ and satisfy $(u \bmod k) + (v \bmod k) - (w \bmod k) \equiv s \pmod k$ with $|s| < \eta k$, contradicting Lemma 8.13. So only finitely many $b \in B$ have $u, v, w \in A_\eta$, and as $B$ is infinite, $\ell B + mB + t \not\subset A_\eta$. □

*Proof of Theorem 8.11(a).* All logarithms are natural. Let $H(P, Q) = \sum_{P \leq n < Q} 1/n$, so that $H(P, Q) = \log(Q/P) + O(1/P)$ for $1 \leq P < Q$. Let $\rho_k = |R_k|/k$ and $\mu = |U_\eta| = \frac{1}{3} - 2\eta$. The open interval $(k(\frac{1}{3} + \eta), k(\frac{2}{3} - \eta))$ has length $k\mu$, so $|\rho_k - \mu| \leq 1/k$.

Let $T_K \leq N < T_{K+1}$ and $S(N) = \sum_{n \leq N,\, n \in A_\eta} 1/n$. By Lemma 8.15 with $J = 1$,

$$S(N) = \sum_{k<K} \rho_k H(X_k, Y_k) + \rho_K H\big(X_K, \min(N+1, Y_K)\big) + O\Big(\sum_k \frac{2k}{X_k}\Big),$$

where the middle term is read as 0 if $N < X_K$, and $\sum_k 2k/X_k = \sum_k 2/T_k < \infty$. We have $H(X_k, Y_k) = 2^k \log 2 - 2\log k + O(1/T_k)$, and $\log N = 2\log 2 + \sum_{1 \leq k < K} 2^k \log 2 + \log(N/T_K)$. Moreover the middle term differs from $\mu \log(N/T_K)$ by at most $|\rho_K - \mu| 2^K \log 2 + O(\log K)$: if $N < X_K$ then $\log(N/T_K) < \log K$; if $X_K \leq N < Y_K$ then $H(X_K, N+1) = \log(N/T_K) - \log K + O(1)$; and if $N \geq Y_K$ then both $H(X_K, Y_K)$ and $\log(N/T_K)$ lie within $2\log K + O(1)$ of $2^K \log 2$. Altogether

$$\big|S(N) - \mu \log N\big| \leq \sum_{k \leq K} \frac{2^k \log 2}{k} + O(K \log K) = O(2^K/K),$$

while $\log N \geq 2^K \log 2$. Hence $S(N)/\log N \to \mu$. □

*Formalization.* Both parts of Theorem 8.11 are formalized in `Kmrr25/Q323.lean` (`thm323`); Remark 8.12 is not.

8.3. **Logarithmic equidistribution and Question 3.24.** All examples of the survey of sets of positive density avoiding some unrestricted sumset $B + B + t$ are built at a logarithmic scale; to test whether this is the only obstruction, the survey asks:

**Question 3.24** ([22])**.** If $A \subset \mathbb{N}$ has $\mathsf{d}_{\log}(A) > 0$ and for every interval $(a, b) \subset [0, 1)$ and every $\theta > 0$

$$\mathsf{d}_{\log}\big(\{n \in A :\ \{\theta \log n\} \in (a, b)\}\big) = \mathsf{d}_{\log}(A) \cdot (b - a), \tag{21}$$

does $A$ contain a sumset $B + B + t$ for some $t \geq 0$ and some infinite set $B \subset \mathbb{N}$?

The answer is negative, and a counterexample is already implicit in the work of Kousek and Radić [19]: the set from the first part of [19, Proposition 4.2], which has natural density $\frac{1}{2}$ and contains no $B + B + t$, also satisfies (21), as we check in Proposition 8.17 (this observation is not made in [19]). By [19, Theorem 1.3], every set of lower density $> \frac{1}{2}$ contains some $B + B + t$, so the value $\frac{1}{2}$ is optimal among sets having a natural density; we do not know whether $\mathsf{d}_{\log}(A) > \frac{1}{2}$ together with (21) forces $B + B + t \subset A$. That example couples a logarithmic

phase with residue classes modulo 2. In Theorem 8.18 we give an alternative example with no residue-class coupling at all, in which a single logarithmic phase drifts slowly; it shows that the test (21), which involves only linear phases $\theta \log n$, does not detect logarithmic-scale obstructions. Recall that by Kra, Moreira, Richter and Robertson [21, Theorem 1.2] every set of positive upper Banach density contains $B \oplus B + t$ for some infinite $B$ and $t \in \mathbb{N}$; so all these examples necessarily exploit the diagonal elements $2b$.

**Lemma 8.16.** *Let $F \subset [0,\infty)$ be a union of pairwise disjoint intervals with left endpoints $\alpha_i$ such that $\#\{i : \alpha_i \le L\} \le C(L+1)$ for all $L \ge 0$. Let $k \in \mathbb{N}$ and $j \in \mathbb{Z}$. Then, as $N \to \infty$,*

$$\sum_{\substack{n \le N \\ \log_2 n \in F}} \frac{1}{n} = \log 2 \cdot \big|F \cap [0, \log_2 N]\big| + O(1) \qquad \text{and} \qquad \sum_{\substack{n \le N,\ n \equiv j\ (k) \\ \log_2 n \in F}} \frac{1}{n} = \frac{1}{k} \sum_{\substack{n \le N \\ \log_2 n \in F}} \frac{1}{n} + O(1),$$

*where $|\cdot|$ is Lebesgue measure.*

*Proof.* If $1 \le x \le y$ and $I$ is an interval with endpoints $x, y$, then $|\sum_{n \in I} 1/n - \log(y/x)| \le 1/x$, by comparison with $\int dt/t$. Each component of $F \cap [0, \log_2 N]$ with left endpoint $\alpha_i$ corresponds to such an interval $I$ with $x = 2^{\alpha_i}$, and $\log(y/x) = \log 2$ times the length of the component. The total error is at most $\sum_i 2^{-\alpha_i} \le \sum_{L \ge 0} C(L+1) 2^{1-L} < \infty$. The second claim follows in the same way from Lemma 8.15 ($J = 1$), applied to each component, with total error at most $\sum_i 2k\, 2^{-\alpha_i} < \infty$. □

**Proposition 8.17** (set of Kousek–Radić)**.** *Let $A^{(1)} = \bigcup_{n \ge 1} [4^n, (2 - \frac{1}{n})4^n)$, $A^{(2)} = \bigcup_{n \ge 1} [(2 + \frac{1}{n})4^n, 4^{n+1})$ and $A_{\mathrm{KR}} = (A^{(1)} \cap 2\mathbb{N}) \cup (A^{(2)} \cap (2\mathbb{N}+1))$ (intersected with $\mathbb{N}$). Then $A_{\mathrm{KR}}$ has natural density $\frac{1}{2}$, contains no $B + B + t$ with $t \ge 0$ and $B$ infinite, and satisfies (21) with $\mathsf{d}_{\log}(A_{\mathrm{KR}}) = \frac{1}{2}$.*

*Proof.* The first two assertions are [19, Proposition 4.2], stated there for $t \in \mathbb{N}$; the case $t = 0$ follows, because if $B + B \subset A_{\mathrm{KR}}$ then $B' = \{b - 1 : b \in B,\ b \ge 2\}$ satisfies $B' + B' + 2 \subset A_{\mathrm{KR}}$.

For (21), let $V_0 = [0, \frac{1}{2})$, $V_1 = [\frac{1}{2}, 1)$ and $A^* = \{n \in 2\mathbb{N} : \{\frac{1}{2}\log_2 n\} \in V_0\} \cup \{n \in 2\mathbb{N}+1 : \{\frac{1}{2}\log_2 n\} \in V_1\}$. Since $\{\frac{1}{2}\log_2 n\} \in V_0$ if and only if $n \in [4^j, 2 \cdot 4^j)$ for some $j \ge 0$, the symmetric difference of $A_{\mathrm{KR}}$ and $A^*$ is contained in $[1,4) \cup W$, where $W = \bigcup_{n \ge 1} [(2 - \frac{1}{n})4^n, (2 + \frac{1}{n})4^n]$. For $N < 4^{K+1}$ we have $\sum_{n \le N,\, n \in W} \frac{1}{n} \le \sum_{n \le K} \big(\log \frac{2+1/n}{2-1/n} + 4^{-n}\big) = O(\log K) = o(\log N)$, so it suffices to verify (21) for $A^*$. Fix $\theta > 0$ and $(a,b) \subset [0,1)$, put $\theta' = \theta \log 2$, so that $\{\theta \log n\} = \{\theta' \log_2 n\}$, and let $F_i = \{x \ge 0 : \{x/2\} \in V_i,\ \{\theta' x\} \in (a,b)\}$ for $i \in \{0,1\}$. These sets satisfy the hypothesis of Lemma 8.16, and $F_0 \sqcup F_1 = F := \{x \ge 0 : \{\theta' x\} \in (a,b)\}$. By Lemma 8.16,

$$\sum_{\substack{n \le N,\ n \in A^* \\ \{\theta \log n\} \in (a,b)}} \frac{1}{n} = \sum_{i=0}^{1} \sum_{\substack{n \le N,\ n \equiv i\ (2) \\ \log_2 n \in F_i}} \frac{1}{n} = \frac{1}{2} \sum_{i=0}^{1} \sum_{\substack{n \le N \\ \log_2 n \in F_i}} \frac{1}{n} + O(1) = \frac{1}{2} \log 2\, |F \cap [0, \log_2 N]| + O(1),$$

and $|F \cap [0, L]| = (b-a)L + O(1)$. Dividing by $\log N = \log 2 \cdot \log_2 N$ gives the limit $\frac{1}{2}(b-a)$. The same computation without the condition on $\{\theta \log n\}$ (that is, with $F = [0,\infty)$) gives $\mathsf{d}_{\log}(A^*) = \frac{1}{2}$. □

The colour class $(A_0 \cap 3\mathbb{N}) \cup (A_1 \cap (3\mathbb{N}+1)) \cup (A_2 \cap (3\mathbb{N}+2))$, with $A_i = \{n : \{\frac{1}{2}\log_2 n\} \in [\frac{i}{3}, \frac{i+1}{3})\}$, from the remark after Proposition 5.1 of [19] is handled in the same way (with residues modulo 3) and gives a counterexample of logarithmic density $\frac{1}{3}$.

**Theorem 8.18.** *Let $0 < \eta < \frac{1}{2}$, $f(x) = \frac{x}{2} + \sqrt{x}$ for $x \geq 0$, and*

$$A = \big\{n \in \mathbb{N} : \ \{f(\log_2 n)\} \in [0, \tfrac{1}{2} - \eta)\big\}.$$

*Then $\mathsf{d}_{\log}(A) = \frac{1}{2} - \eta$, $A$ satisfies (21), and $A$ contains no $B + B + t$ with $t \in \mathbb{Z}$ and $B \subset \mathbb{N}$ infinite.*

**Lemma 8.19.** *Let $\theta' > 0$ and $(h, k) \in \mathbb{Z}^2 \setminus \{(0,0)\}$. Then $\frac{1}{L}\int_0^L e\big(hf(x) + k\theta' x\big)\,dx \to 0$ as $L \to \infty$, where $e(y) = e^{2\pi i y}$.*

*Proof.* The phase is $\psi(x) = \gamma x + h\sqrt{x}$ with $\gamma = \frac{h}{2} + k\theta'$. If $\gamma \neq 0$, let $x_0 = (h/\gamma)^2$. On $[x_0, L]$ the derivative $\psi'(x) = \gamma + \frac{h}{2\sqrt{x}}$ is monotone and $|\psi'| \geq |\gamma|/2$. Integrating by parts,

$$\int_{x_0}^{L} e(\psi)\,dx = \Big[\frac{e(\psi)}{2\pi i \psi'}\Big]_{x_0}^{L} - \int_{x_0}^{L} e(\psi)\,d\Big(\frac{1}{2\pi i \psi'}\Big),$$

and since $1/\psi'$ is monotone with $|1/\psi'| \leq 2/|\gamma|$, each of the two terms is at most $\frac{1}{2\pi} \cdot \frac{4}{|\gamma|}$. The integral over $[0, x_0]$ is at most $x_0$. If $\gamma = 0$, then $h \neq 0$ (otherwise $k\theta' = 0$, so $k = 0$), and $\int_0^L e(h\sqrt{x})\,dx = \int_0^{\sqrt{L}} 2y\, e(hy)\,dy = O(\sqrt{L}/|h|)$ by one integration by parts. In both cases the integral is $o(L)$. □

*Proof of Theorem 8.18.* Let $c = \frac{1}{2} - \eta$, fix $\theta > 0$, put $\theta' = \theta \log 2$, and let $\Psi(x) = (f(x) \bmod 1,\ \theta' x \bmod 1) \in \mathbb{T}^2$. By Lemma 8.19 and linearity, $\frac{1}{L}\int_0^L P(\Psi(x))\,dx \to \int_{\mathbb{T}^2} P$ for every trigonometric polynomial $P$, hence, by uniform density of trigonometric polynomials in $C(\mathbb{T}^2)$, for every continuous $P$. If $\Omega \subset \mathbb{T}^2$ is a box, then for each $\delta > 0$ there are continuous $P_- \leq \mathbf{1}_\Omega \leq P_+$ with $\int (P_+ - P_-) < \delta$, and therefore $\frac{1}{L}|\{x \in [0, L] : \Psi(x) \in \Omega\}| \to |\Omega|$. We apply this to $\Omega = [0, c) \times (a, b)$ and to $\Omega = [0, c) \times \mathbb{T}$.

Since $f$ is continuous and strictly increasing with $f(0) = 0$, the sets $F = \{x \geq 0 : \{f(x)\} \in [0, c)\} = \bigcup_{j \geq 0} [f^{-1}(j), f^{-1}(j + c))$ and $F' = F \cap \{x : \{\theta' x\} \in (a, b)\}$ are unions of disjoint intervals; the endpoints of their components in $[0, L]$ are among the at most $2(f(L) + 1) + 2(\theta' L + 1)$ points where $f(x)$ or $\theta' x$ hits $\mathbb{Z} \cup (\mathbb{Z} + c)$ or $(\mathbb{Z} + a) \cup (\mathbb{Z} + b)$, so Lemma 8.16 applies. Since $A = \{n : \log_2 n \in F\}$ and $\{n \in A : \{\theta \log n\} \in (a, b)\} = \{n : \log_2 n \in F'\}$, we obtain $\mathsf{d}_{\log}(A) = \lim |F \cap [0, L]|/L = c$ and the logarithmic density of $\{n \in A : \{\theta \log n\} \in (a, b)\}$ equals $\lim |F' \cap [0, L]|/L = c(b - a)$. This proves (21).

Now suppose $B$ is infinite, $t \in \mathbb{Z}$ and $B + B + t \subset A$. Let $b' = \min B$ and, for $b \in B$ large, put $n_1 = b + b' + t$, $n_2 = 2b + t$ (both in $B + B + t$) and $x_i = \log_2 n_i$. As $b \to \infty$, $x_2 - x_1 = \log_2 \frac{2b+t}{b+b'+t} \to 1$ and $|\sqrt{x_2} - \sqrt{x_1}| \leq \frac{|x_2 - x_1|}{2\sqrt{\min(x_1, x_2)}} \to 0$, so $\Delta_b := f(x_2) - f(x_1) - \frac{1}{2} \to 0$. Take $b$ with $|\Delta_b| < \eta/2$. Since $n_1 \in A$, $\{f(x_1)\} \in [0, c)$, so $f(x_2) \equiv \{f(x_1)\} + \frac{1}{2} + \Delta_b$ with $\{f(x_1)\} + \frac{1}{2} + \Delta_b \in [\frac{1}{2} - \frac{\eta}{2}, 1 - \frac{\eta}{2})$. Hence $\{f(x_2)\} \notin [0, c)$, that is, $n_2 \notin A$, a contradiction. □

*Formalization.* Proposition 8.17, including the two assertions quoted from [19], and Theorem 8.18 are formalized in `Kmrr25/Q324.lean` (`prop_KR`, `thm324`).

*Remark* 8.20. (i) The obstruction in Theorem 8.18 is still of logarithmic scale: the phase $f(\log_2 n)$ increases by $\frac{1}{2} + o(1)$ when $n$ is doubled. Equidistribution against all linear phases $\theta \log n$ does not rule out such phases, because the term $\sqrt{\log_2 n}$ makes $f$ asymptotically independent of every linear phase, including the resonant frequency $\theta = 1/(2 \log 2)$.

(ii) The set $A_\eta$ of Section 8.2 is a further counterexample to Question 3.24: it contains no $B+B+t$ by Theorem 8.11(b) with $\ell = m = 1$, and it satisfies (21). To see the latter, let $g(n) = \mathbf{1}[\{\theta \log n\} \in (a,b)]$. On the block $[X_k, Y_k)$, $g$ is constant on $J_k \le 2\theta \log(Y_k/X_k)+3 = O(2^k)$ subintervals, so by Lemma 8.15 the sum of $g(n)/n$ over $A_\eta \cap [X_k, Y_k)$ equals $\rho_k$ times the sum over $[X_k, Y_k)$ up to $O(2^k/T_k)$, which is summable. The rest of the proof of Theorem 8.11(a) applies verbatim with weights $g(n)$ (the transition intervals contribute $O(K \log K)$ and the errors $|\rho_k - \mu|$ contribute $O(2^K/K)$), giving $\sum_{n \le N,\, n \in A_\eta} g(n)/n = \mu \sum_{n \le N} g(n)/n + o(\log N)$; and $\sum_{n \le N} g(n)/n = (b-a)\log N + O(1)$ by Lemma 8.16. Here the obstruction comes from scale-separated blocks with periodic structure inside each block, rather than from a phase of $\log n$; the empty transition intervals $[Y_{k-1}, X_k)$ have logarithmic density zero and are invisible to (21).

## 9. Sums of blocks from a density Ramsey family

In this section we answer Question 3.31 of [22] and its $k$-fold analogue, Question §3.6(a), positively, in the strongest form one can hope for. The proof is a short modification of the proof of the Density Finite Sums Theorem of Kra, Moreira, Richter and Robertson [23]: what is needed is the quantitative (positive lower average) form of the recurrence property that is actually established in [23], combined with a version of Bergelson's intersectivity lemma along Følner sequences.

Throughout, $\mathbb{N} = \{1, 2, \dots\}$, and for $A \subset \mathbb{N}$ and $t \in \mathbb{Z}$ we write $A - t = \{n \in \mathbb{N} : n + t \in A\}$. The upper Banach density is

$$\mathsf{d}^*(A) = \limsup_{N \to \infty} \sup_{M \ge 0} \frac{|A \cap \{M+1, \dots, M+N\}|}{N}.$$

**Definition 9.1** ([22, Definition 3.29])**.** A collection $\mathcal{D} \subset \{F \subset \mathbb{N} : 0 < |F| < \infty\}$ has the *density Ramsey property* if every $A \subset \mathbb{N}$ with $\mathsf{d}^*(A) > 0$ contains some $X \in \mathcal{D}$.

By Szemerédi's theorem, the family of $\ell$-term arithmetic progressions has the density Ramsey property for every $\ell$; trivially so does the family of singletons. Theorem 3.30 of [22] states that for such a family $\mathcal{D}$ and $\mathsf{d}^*(A) > 0$ there are pairwise disjoint $B_1, B_2, \dots \in \mathcal{D}$ and pairwise disjoint $C_1, C_2, \dots \in \mathcal{D}$ with $B_i + C_j \subset A$ for all $i, j$. The survey observes that this does not contain the $B + B + t$ theorem ([21, Theorem 1.2], Theorem 2.2 of [22]) and asks:

**Question 3.31** ([22])**.** If $\mathcal{D} \subset \{F \subset \mathbb{N} : 0 < |F| < \infty\}$ has the density Ramsey property, is it true that for any $A \subset \mathbb{N}$ with $\mathsf{d}^*(A) > 0$ there exist $t \in \mathbb{N}$ and disjoint $B_1, B_2, \dots \in \mathcal{D}$ such that $B_i + B_j \subset A - t$ for all $i \ne j \in \mathbb{N}$?

Immediately afterwards the survey adds: "Similarly, one can formulate the analog for the $k$-fold sum of the sets." We read this as follows.

**Question §3.6(a)** ([22])**.** Let $\mathcal{D}$ have the density Ramsey property, $k \in \mathbb{N}$ and $\mathsf{d}^*(A) > 0$. Do there exist $t \in \mathbb{N}$ and disjoint $B_1, B_2, \dots \in \mathcal{D}$ such that $B_{i_1} + \cdots + B_{i_k} \subset A - t$ for all pairwise distinct $i_1, \dots, i_k \in \mathbb{N}$?

The indices must be distinct: with $\mathcal{D}$ the singletons and repeated indices allowed, already $k = 2$ would require $B + B + t \subset A$, which fails in general by Example 2.3 of [22].

**Theorem 9.2.** *Let $\mathcal{D} \subset \{F \subset \mathbb{N} : 0 < |F| < \infty\}$ have the density Ramsey property, let $k \in \mathbb{N}$ and let $A \subset \mathbb{N}$ with $\mathsf{d}^*(A) > 0$. Then there exist $t \in \mathbb{N}$ and $B_1, B_2, \dots \in \mathcal{D}$ with $\max B_i < \min B_{i+1}$ for all $i$, such that*

$$B_{i_1} + B_{i_2} + \cdots + B_{i_m} \subset A - t \qquad \text{for all } 1 \le m \le k \text{ and all } i_1 < i_2 < \cdots < i_m.$$

**Corollary 9.3.** *Question 3.31 and Question §3.6(a) have positive answers. In Question 3.31 one may moreover require $B_i \subset A - t$ for all $i$ and $\max B_i < \min B_{i+1}$.*

*Proof.* Take $k = 2$, respectively the given $k$, in Theorem 9.2; sums over distinct indices may be written with increasing indices. □

*Remark* 9.4. Taking $\mathcal{D}$ to be the singletons, Theorem 9.2 is the Density Finite Sums Theorem [23, Theorem 1.1] (with the shift $t \geq 1$); for $k = 2$ it contains the $B + B + t$ theorem. Taking $\mathcal{D}$ to be the $\ell$-term progressions gives a common extension of these results and Szemerédi's theorem, which is the unification sought in [22, §3.6]. Concerning novelty: the proof below combines the construction of nested boxes from support points in the proof of [23, Proposition 3.2] with the chain argument of [23, Lemma 2.2], with single return times replaced by blocks from $\mathcal{D}$; all dynamical input is taken from [23]. Our contribution is the observation that the quantitative recurrence established there, together with Lemma 9.6, suffices.

9.1. **Two lemmas on Følner sequences.** A *Følner sequence* in $\mathbb{N}$ is a sequence $\Phi = (\Phi_N)$ of finite nonempty subsets of $\mathbb{N}$ with $|(\Phi_N + 1) \cap \Phi_N|/|\Phi_N| \to 1$. For $K \subset \mathbb{N}$ put $\overline{\mathsf{d}}_\Phi(K) = \limsup_N |K \cap \Phi_N|/|\Phi_N|$.

**Lemma 9.5.** *Let $\Phi$ be a Følner sequence in $\mathbb{N}$. Then $|\Phi_N| \to \infty$, and $\mathsf{d}^*(K) \geq \overline{\mathsf{d}}_\Phi(K)$ for every $K \subset \mathbb{N}$.*

*Proof.* For a finite nonempty $F \subset \mathbb{N}$ we have $\max F + 1 \notin F$, so $|(F+1) \cap F| \leq |F| - 1$; hence the Følner property forces $|\Phi_N| \to \infty$. Decompose $\Phi_N$ into its $r_N$ maximal blocks of consecutive integers. If $x, x+1 \in \Phi_N$ then $x$ and $x+1$ lie in the same block, so $|(\Phi_N + 1) \cap \Phi_N| = |\Phi_N| - r_N$ and therefore $r_N = o(|\Phi_N|)$. Fix $L \in \mathbb{N}$ and let $\delta_L = \sup\{|K \cap I|/|I|\}$, the supremum over all intervals $I \subset \mathbb{N}$ of length at least $L$. Blocks of length $< L$ contain at most $L r_N$ points in total, and a block $I$ of length $\geq L$ contains at most $\delta_L |I|$ points of $K$. Hence $|K \cap \Phi_N| \leq L r_N + \delta_L |\Phi_N|$ and $\overline{\mathsf{d}}_\Phi(K) \leq \delta_L$. Finally $\delta_L = \sup_{L' \geq L} a_{L'}$ where $a_{L'} = \sup_M |K \cap \{M+1, \dots, M+L'\}|/L'$, so $\delta_L \to \limsup_{L'} a_{L'} = \mathsf{d}^*(K)$. □

The next lemma is a version of Bergelson's intersectivity lemma [5] in which density is measured along a Følner sequence; working directly along $\Phi$ avoids the passage through an enumeration of the index set that appears in the proof of Theorem 3.30 of [22].

**Lemma 9.6.** *Let $(\Omega, \nu)$ be a probability space, $\Phi$ a Følner sequence in $\mathbb{N}$, and $R \subset \mathbb{N}$ with $\overline{\mathsf{d}}_\Phi(R) \geq \delta > 0$. For $n \in R$ let $O_n \subset \Omega$ be measurable with $\nu(O_n) \geq \kappa > 0$. Then there is $K \subset R$ with $\mathsf{d}^*(K) \geq \delta\kappa$ such that $\nu\big(\bigcap_{n \in F} O_n\big) > 0$ for every finite nonempty $F \subset K$.*

*Proof.* Let $\mathcal{F}_0$ be the (countable) set of finite nonempty $F \subset R$ with $\nu(\bigcap_{n \in F} O_n) = 0$, and let $Z = \bigcup_{F \in \mathcal{F}_0} \bigcap_{n \in F} O_n$; then $\nu(Z) = 0$. Put $f_N = |\Phi_N|^{-1} \sum_{n \in R \cap \Phi_N} \mathbf{1}_{O_n \setminus Z}$. Then $0 \leq f_N \leq 1$ and $\int f_N \, d\nu \geq \kappa \, |R \cap \Phi_N|/|\Phi_N|$. Choose $N_1 < N_2 < \cdots$ with $|R \cap \Phi_{N_i}|/|\Phi_{N_i}| \to \overline{\mathsf{d}}_\Phi(R)$. By the reverse Fatou lemma, $\int \limsup_i f_{N_i} \, d\nu \geq \limsup_i \int f_{N_i} \, d\nu \geq \delta\kappa$, so there is $\omega \in \Omega$ with $\limsup_i f_{N_i}(\omega) \geq \delta\kappa$. Let $K = \{n \in R : \omega \in O_n \setminus Z\}$. Then $f_N(\omega) = |K \cap \Phi_N|/|\Phi_N|$, so $\overline{\mathsf{d}}_\Phi(K) \geq \delta\kappa$ and $\mathsf{d}^*(K) \geq \delta\kappa$ by Lemma 9.5. If $F \subset K$ is finite and nonempty, then $\omega \in \bigcap_{n \in F} O_n$ and $\omega \notin Z$, so $\nu(\bigcap_{n \in F} O_n) > 0$. □

9.2. **The dynamical input.** For a topological system $(Y, T)$ (a compact metric space with a homeomorphism) we write $T_\Delta = T \times \cdots \times T$ ($k+1$ factors) on $Y^{k+1}$ and $M(Y^{k+1})$ for the Borel probability measures on $Y^{k+1}$.

**Proposition 9.7** (Kra–Moreira–Richter–Robertson)**.** *Let $A \subset \mathbb{N}$ with $\mathsf{d}^*(A) > 0$ and $k \in \mathbb{N}$. There exist a topological system $(Y, T)$, a point $a \in Y$, an open set $\tilde{E} \subset Y$, a Følner sequence*

*$\Phi$ in $\mathbb{N}$ such that $a$ is generic along $\Phi$ for an ergodic $T$-invariant Borel probability measure $\tilde{\mu}$ on $Y$, a measure $\sigma \in M(Y^{k+1})$ and an integer $t \geq 0$ such that*

(i) $A = \{n \in \mathbb{N} : T^n a \in \tilde{E}\}$;
(ii) $\sigma(\{a\} \times Y^k) = 1$;
(iii) $\sigma\big(Y \times (T^{-t}\tilde{E})^k\big) > 0$;
(iv) *for all open $U_1, \dots, U_k \subset Y$ such that $V = U_1 \times \cdots \times U_k$ satisfies $\sigma(Y \times V) > 0$,*

$$\liminf_{N\to\infty} \frac{1}{|\Phi_N|} \sum_{n\in\Phi_N} \sigma\big((Y \times V) \cap T_\Delta^{-n}(V \times Y)\big) > 0.$$

All of this is contained in [23]; we indicate where. Property (iv) is stronger than the statement that $\sigma$ is "progressive" ([23, Definition 3.1], which only asks for infinitely many such $n$), but it is exactly what is proved in [23], in the proof that Theorems 6.1 and 6.2 imply Theorem 4.9; we repeat that argument.

*Proof.* By the correspondence principle [23, Theorem 2.3] there are a measure preserving system $(X, \mu, S)$, a point $x_0 \in X$ whose forward orbit closure has full measure, and an open $E \subset X$ with $\mu(E) > 0$ and $A = \{n : S^n x_0 \in E\}$. As in the proof that Theorem 2.4 implies Theorem 1.2 in [23], ergodic decomposition yields an ergodic $S$-invariant $\mu'$ with $\mu'(E) \geq \mu(E)$ and a Følner sequence along which $x_0$ is generic for $\mu'$. As in the proof that Theorems 4.8 and 4.9 imply Theorem 3.3 in [23], which relies on [20, Proposition 5.7 and Lemma 5.8], there are an ergodic system $(Y, \tilde{\mu}, T)$ with topological pronilfactors, a continuous factor map $\pi\colon Y \to X$, a Følner sequence $\Phi$ in $\mathbb{N}$ and a point $a \in Y$ generic for $\tilde{\mu}$ along $\Phi$ with $\pi(a) = x_0$. (Lemma 5.8 of [20] is stated for a transitive point; as in the proof that Theorem 5.9 implies Theorem 4.4 there, one first replaces $X$ by the orbit closure of $x_0$, which has full $\mu'$-measure, and $E$ by its intersection with it.) A continuous factor map satisfies $\pi \circ T = S \circ \pi$ *everywhere* [20, Definition 2.2]; in [20, Lemma 5.8] it is the projection to the first coordinate of an orbit closure of a product map. Hence with $\tilde{E} = \pi^{-1}(E)$, which is open, $T^n a \in \tilde{E}$ if and only if $S^n x_0 \in E$, which is (i); moreover $\tilde{\mu}(\tilde{E}) = \mu'(E) > 0$.

Let $\sigma$ be the measure of [23, Definition 4.5] for $(Y, \tilde{\mu}, T)$, $k$, $\Phi$ and $a$. Formula (4.3) there shows $\sigma = \delta_a \otimes \sigma'$ for some $\sigma'$, which gives (ii), and [23, Theorem 4.8] gives $t \geq 0$ with (iii). For (iv), let $V$ be as there. Since $V$ is open and $\sigma(Y \times V) > 0$, regularity and Urysohn's lemma give a continuous $G\colon Y^k \to [0,1]$ with $G \leq \mathbf{1}_V$ and $\int 1 \otimes G \, d\sigma > 0$. Pointwise, $\mathbf{1}_{(Y\times V)\cap T_\Delta^{-n}(V\times Y)} \geq (1 \otimes G) \cdot \big((G \otimes 1) \circ T_\Delta^n\big)$. By [23, Theorem 6.2], the averages over $n \in \Phi_N$ of $(G \otimes 1) \circ T_\Delta^n - (1 \otimes G) \circ T_\Delta^n$ tend to 0 in $L^2(\sigma)$; as $|1 \otimes G| \leq 1$, the Cauchy–Schwarz inequality gives

$$\liminf_N \frac{1}{|\Phi_N|} \sum_{n\in\Phi_N} \sigma\big((Y\times V)\cap T_\Delta^{-n}(V\times Y)\big) \;\geq\; \liminf_N \frac{1}{|\Phi_N|} \sum_{n\in\Phi_N} \int_{Y^k} G \cdot \big(G \circ (T \times \cdots \times T)^n\big)\, d\nu,$$

where $\nu$ is the projection of $\sigma$ to the last $k$ coordinates. By [23, Lemma 4.7], $\sigma$ is invariant under $\mathrm{Id} \times T \times T^2 \times \cdots \times T^k$, so $\nu$ is invariant under $T \times T^2 \times \cdots \times T^k$; since $\int G \, d\nu > 0$, [23, Theorem 6.1] shows that the right-hand side is positive. □

9.3. **Proof of Theorem 9.2.** We first prove the statement with $t \geq 0$ instead of $t \in \mathbb{N}$. Let $(Y, T)$, $a$, $\tilde{E}$, $\Phi$, $\sigma$, $t$ be as in Proposition 9.7 and put $U = T^{-t}\tilde{E}$, an open set. By (i), for $n \in \mathbb{N}$,

$$T^n a \in U \iff n \in A - t. \tag{22}$$

Since $\{a\} \times Y^k$ is a closed set of full measure, $\operatorname{supp}\sigma \subset \{a\} \times Y^k$.

By a *box* we mean a set $V = V^{(1)} \times \cdots \times V^{(k)} \subset Y^k$ with all $V^{(j)}$ open and nonempty. We construct boxes $V_0, V_1, \dots$ and sets $B_1, B_2, \dots \in \mathcal{D}$ such that for all $i \geq 1$:

(I1) $\sigma(Y \times V_i) > 0$ and $V_i^{(j)} \subset V_{i-1}^{(j)}$ for all $j$;
(I2) for every $n \in B_i$ and every $v = (v_1, \dots, v_k) \in V_i$ we have $T^n a \in V_{i-1}^{(1)}$ and $T^n v_j \in V_{i-1}^{(j+1)}$ for $1 \le j \le k-1$;
(I3) $\min B_i > \max B_{i-1}$ (with $\max B_0 := 0$).

Let $V_0 = U^k$; by (iii), $\sigma(Y \times V_0) > 0$ (in particular $U \neq \emptyset$).

Suppose $V_i$ has been constructed with $\sigma(Y \times V_i) > 0$. For $n \in \mathbb{N}$ let $O_n = (Y \times V_i) \cap T_\Delta^{-n}(V_i \times Y)$, an open subset of $Y^{k+1}$. By (iv), $c := \liminf_N |\Phi_N|^{-1} \sum_{n \in \Phi_N} \sigma(O_n) > 0$. Let $R = \{n > \max B_i : \sigma(O_n) \ge c/2\}$. As $\sigma(O_n) \le 1$,

$$\frac{1}{|\Phi_N|} \sum_{n \in \Phi_N} \sigma(O_n) \le \frac{c}{2} + \frac{|R \cap \Phi_N|}{|\Phi_N|} + \frac{\max B_i}{|\Phi_N|},$$

and the last term tends to 0 by Lemma 9.5; hence $\overline{\mathtt{d}}_\Phi(R) \ge c/2$. Lemma 9.6 (with $\nu = \sigma$, $\delta = \kappa = c/2$) gives $K \subset R$ with $\mathtt{d}^*(K) > 0$ such that $\sigma(\bigcap_{n \in F} O_n) > 0$ for every finite nonempty $F \subset K$. By the density Ramsey property there is $B_{i+1} \in \mathcal{D}$ with $B_{i+1} \subset K$; then (I3) holds for $i+1$.

The set $O = \bigcap_{n \in B_{i+1}} O_n$ is open with $\sigma(O) > 0$, so it meets $\operatorname{supp} \sigma$, say in $(a, w)$. By definition of the product topology there is a box $W \ni w$ with $\{a\} \times W \subset O$. Put $V_{i+1} = W$. Since $Y \times W$ is an open neighbourhood of a point of $\operatorname{supp} \sigma$, $\sigma(Y \times W) > 0$. Since $\{a\} \times W \subset O_n \subset Y \times V_i$ and both $W$ and $V_i$ are boxes with nonempty factors, $W^{(j)} \subset V_i^{(j)}$ for all $j$; this is (I1). If $n \in B_{i+1}$ and $v \in W$, then $(a, v) \in O_n \subset T_\Delta^{-n}(V_i \times Y)$, that is, $(T^n a, T^n v_1, \dots, T^n v_{k-1}) \in V_i$; this is (I2).

Now fix $1 \le m \le k$, indices $i_1 < \cdots < i_m$ and $b_l \in B_{i_l}$. We claim that for $h = 0, 1, \dots, m-1$,

$$y_h := T^{b_{m-h} + \cdots + b_m} a \in V_{i_{m-h}-1}^{(h+1)}. \tag{23}$$

For $h = 0$ this is (I2) for the block $B_{i_m}$. Suppose (23) holds for some $h \le m-2$. Since $i_{m-h} - 1 \ge i_{m-h-1}$, (I1) gives $y_h \in V_{i_{m-h-1}}^{(h+1)}$. As $V_{i_{m-h-1}}$ is a box with nonempty factors, there is $v \in V_{i_{m-h-1}}$ with $v_{h+1} = y_h$. Applying (I2) to the block $B_{i_{m-h-1}}$, the element $n = b_{m-h-1}$ and $j = h+1 \le m-1 \le k-1$, we get $y_{h+1} = T^{b_{m-h-1}} y_h \in V_{i_{m-h-1}-1}^{(h+2)}$, which is (23) for $h+1$. For $h = m-1$ we obtain $T^{b_1 + \cdots + b_m} a \in V_{i_1 - 1}^{(m)} \subset V_0^{(m)} = U$, so $b_1 + \cdots + b_m \in A - t$ by (22). Together with (I3) this proves the theorem with $t \ge 0$.

Finally, to get $t \in \mathbb{N}$ apply the above to $A' = A - 1 = \{n \in \mathbb{N} : n + 1 \in A\}$, which has $\mathtt{d}^*(A') = \mathtt{d}^*(A) > 0$. If all the required sums lie in $A' - t'$ with $t' \ge 0$, then they lie in $A - (t' + 1)$. □

*Formalization.* Theorem 9.2 and Corollary 9.3, Lemmas 9.5 and 9.6 and the construction above are formalized in `Kmrr25/Q331.lean` (`thm_dR`, `cor331`), with the conclusion of Proposition 9.7 stated as an explicit hypothesis; the dynamical input from [23] is not formalized.

## 10. Sumsets in the sets $A_x = \{n : \{n^\gamma x\} \le 1/\log n\}$

In this section we answer Question 4.13 of [22] completely: the answer is "exactly the non-integers $\gamma$". For $t \in \mathbb{R}$ let $\{t\}$ be its fractional part and $\|t\|$ its distance to $\mathbb{Z}$.

**Question 4.13** ([22])**.** Fix $\gamma > 0$ and, for each $x \in \mathbb{R}$, let $A_x := \{n \in \mathbb{N} : \{n^\gamma x\} \le 1/\log n\}$. For which values of $\gamma$ is it true that for almost every $x \in \mathbb{R}$, the set $A_x$ contains a sumset $B + C$ for infinite sets $B, C \subset \mathbb{N}$?

(The condition is void for $n = 1$; since every element of $B + C$ is at least 2, whether $1 \in A_x$ is irrelevant.) The survey notes that $A_x$ is thick for almost every $x$ when $\gamma < 1$, and that for integer $\gamma$ and every irrational $x$ the set $A_x$ contains no infinite sumset.

**Theorem 10.1.** *Let $\gamma > 0$.*

(a) *If $\gamma \notin \mathbb{Z}$, then for almost every $x \in \mathbb{R}$ the set $A_x$ is thick, and hence contains $B + C$ for some infinite $B, C \subset \mathbb{N}$.*

(b) *If $\gamma = d \in \mathbb{N}$ and $x \notin \mathbb{Q}$, then $A_x$ contains no set $B + F$ with $B \subset \mathbb{N}$ infinite and $F \subset \mathbb{Z}$, $|F| = d + 1$; in particular $A_x$ contains no $B + C$ with $B, C$ infinite.*

*Consequently, $A_x$ contains an infinite sumset $B + C$ for almost every $x$ if and only if $\gamma \notin \mathbb{N}$.*

For integer $\gamma = d$ and rational $x = p/q$ one has $q\mathbb{N} \subset A_x$, so $A_x \supset B + C$; this does not affect the almost-everywhere statement. Part (a) is an application of the standard second-moment method of metric Diophantine approximation; part (b) is a short argument with divided differences supplying the proof omitted in [22].

**Lemma 10.2.** *If $A \subset \mathbb{N}$ is thick, then $B + C \subset A$ for some infinite $B, C \subset \mathbb{N}$.*

*Proof.* Choose $b_1 < c_1 < b_2 < c_2 < \cdots$ inductively (with $c_0 = 0$): $b_j > c_{j-1}$ is the left endpoint of an interval of $A$ of length $> c_{j-1}$, and $c_j > b_j$ is the left endpoint of an interval of $A$ of length $> b_j$. For $i < j$, $b_j + c_i \in [b_j, b_j + c_{j-1}]$ lies in the interval starting at $b_j$; for $i \le j$, $c_j + b_i \in [c_j, c_j + b_j]$ lies in the interval starting at $c_j$. Thus $\{b_j\} + \{c_j\} \subset A$. □

10.1. **Proof of Theorem 10.1(b).** Let $F = \{f_0, \dots, f_d\}$ be distinct integers and suppose $B + F \subset A_x$ with $B$ infinite. For fixed $b$, the polynomial $P_b(f) = (b + f)^d x$ has degree $d$ and leading coefficient $x$, so its $d$-th divided difference at the nodes $f_0, \dots, f_d$ equals $x$:

$$\sum_{i=0}^{d} \lambda_i (b + f_i)^d x = x, \qquad \lambda_i = \prod_{j \neq i} (f_i - f_j)^{-1} \in \mathbb{Q}.$$

Let $M \in \mathbb{N}$ be a common denominator of the $\lambda_i$ and $e_i = M\lambda_i \in \mathbb{Z}$; then $\sum_i e_i (b+f_i)^d x = Mx$, so $Mx \equiv \sum_i e_i \{(b + f_i)^d x\} \pmod 1$ and

$$\|Mx\| \le \sum_i |e_i| \, \{(b + f_i)^d x\} \le \frac{\sum_i |e_i|}{\log(b + \min F)}.$$

Letting $b \to \infty$ in $B$ gives $\|Mx\| = 0$, i.e. $x \in \mathbb{Q}$. If $B + C \subset A_x$ with $B, C$ infinite, apply this to a subset $F \subset C$ with $|F| = d + 1$. □

10.2. **Proof of Theorem 10.1(a).** Let $k = \lfloor \gamma \rfloor \ge 0$, $\varsigma = \gamma - k \in (0, 1)$ and $c_j(n) = \binom{\gamma}{j} n^{\gamma - j}$ for $0 \le j \le k$. Since $\gamma \notin \mathbb{Z}$, every $\binom{\gamma}{j}$ $(j \ge 0)$ and every $\gamma - j$ is nonzero. Constants below may depend on $\gamma$, $L$ and the cut-off $\psi$ introduced later, but never on $N$, $n$ or $x$.

*Taylor expansion.* For $n \ge 1$ and $0 \le h < L$, Taylor's formula with Lagrange remainder gives

$$(n + h)^\gamma = \sum_{j=0}^{k} c_j(n) h^j + r_n(h), \qquad |r_n(h)| = \left| \tbinom{\gamma}{k+1} (n + \xi)^{\varsigma - 1} h^{k+1} \right| \le C_1 n^{\varsigma - 1}, \tag{24}$$

for some $\xi \in [0, h]$, because $\varsigma - 1 < 0$.

Fix $L \in \mathbb{N}$ and $X_0 > 0$. For $N \in \mathbb{N}$ put

$$\eta = \eta_N = \frac{1}{\log 4N}, \qquad \rho = \frac{\eta}{16(k+1)L^k},$$

where $N$ ranges over powers of 2; for $N \le n < 2N$ let $E_n$ be the set of $x \in \mathbb{R}$ with $\{c_0(n)x\} \in [\eta/4, \eta/2]$ and $\|c_j(n)x\| \le \rho$ for $1 \le j \le k$.

**Lemma 10.3.** *There is $N_0$ such that for $N \ge N_0$, $N \le n < 2N$ and $x \in E_n$ with $|x| \le X_0$ we have $\{n, n+1, \dots, n+L-1\} \subset A_x$.*

*Proof.* Write $c_j(n)x = m_j + \theta_j$ with $m_j \in \mathbb{Z}$, $\theta_0 \in [\eta/4, \eta/2]$ and $|\theta_j| \le \rho$ for $j \ge 1$. By (24), $(n+h)^\gamma x = \sum_j m_j h^j + \theta_0 + \sum_{j\ge1} \theta_j h^j + x r_n(h)$. Here $|\sum_{j\ge1} \theta_j h^j| \le kL^k\rho \le \eta/16$, and $|x r_n(h)| \le X_0 C_1 N^{\varsigma-1} \le \eta/16$ once $N$ is large, since $\eta \asymp 1/\log N$. Hence $\{(n+h)^\gamma x\} \in [\eta/8, 5\eta/8]$, and $\eta \le 1/\log(n+h)$ because $2 \le n+h < 4N$ (for $N \ge L$). □

*Smooth windows.* Fix $\Phi_0 \in C_c^\infty(-\frac12, \frac12)$ with $0 \le \Phi_0 \le 1$ and $\Phi_0 = 1$ on $[-\frac14, \frac14]$, and fix an exponent $P \ge 4k+10$. For $\alpha \in \mathbb{R}$ and $0 < w < \frac12$ let $\varphi_{\alpha,w}(t) = \sum_{m\in\mathbb{Z}} \Phi_0\big((t-\alpha-m)/(2w)\big)$. Then $0 \le \varphi_{\alpha,w} \le \mathbf{1}_{[\alpha-w,\alpha+w]+\mathbb{Z}}$, $\int_0^1 \varphi_{\alpha,w} \ge w$, and $\hat\varphi_{\alpha,w}(\ell) = 2w\, e(-\ell\alpha)\hat\Phi_0(2w\ell)$, where $e(t) = e^{2\pi i t}$ and $|\hat\Phi_0(u)| \le C_P(1+|u|)^{-P}$. Comparing sums with integrals one gets, uniformly in $\alpha$ and $w$,

$$\sum_{\ell\in\mathbb{Z}} |\hat\varphi_{\alpha,w}(\ell)| \le C_2, \qquad \sum_{|\ell|>Q} |\hat\varphi_{\alpha,w}(\ell)| \le C_2 (wQ)^{1-P} \quad (wQ \ge 1). \tag{25}$$

Let $\varphi_0 = \varphi_{3\eta/8,\eta/8}$ and $\varphi_j = \varphi_{0,\rho}$ for $1 \le j \le k$, and

$$F_n(x) = \prod_{j=0}^{k} \varphi_j(c_j(n)x), \qquad Z_N = \sum_{N\le n<2N} F_n.$$

Then $0 \le F_n \le \mathbf{1}_{E_n}$. Expanding in Fourier series (absolutely convergent by (25)),

$$F_n(x) = \sum_{\ell\in\mathbb{Z}^{k+1}} a(\ell)\, e\big(x\,\Xi_n(\ell)\big), \qquad a(\ell) = \prod_j \hat\varphi_j(\ell_j), \qquad \Xi_n(\ell) = \sum_j \ell_j c_j(n),$$

with $\sum_\ell |a(\ell)| \le C_2^{k+1}$ and $\mu := a(0) = \prod_j \int_0^1 \varphi_j \ge (\eta/8)\rho^k \ge c_3(\log N)^{-(k+1)}$. Put $Q = T = (\log N)^3$. Since both window widths are $\gg 1/\log N$, (25) gives

$$\sum_{\|\ell\|_\infty > Q} |a(\ell)| \le C_4 (\log N)^{-2(P-1)} = o(\mu^2). \tag{26}$$

Fix $\psi \in C_c^\infty(\mathbb{R})$ with $0 \le \psi \le 1$, $\operatorname{supp}\psi \subset [-X_0, X_0]$ and $m_\psi = \int \psi > 0$, and let $\check\psi(\xi) = \int \psi(x) e(\xi x)\, dx$, so that $|\check\psi(\xi)| \le C_P(1+|\xi|)^{-P}$ and $|\check\psi| \le m_\psi$. Write $\mathbf{E}_\psi[f] = m_\psi^{-1} \int f\psi\, dx$. In what follows $N \to \infty$ and $o(1)$ is uniform in $n, n' \in [N, 2N)$.

**Lemma 10.4.** *If $N$ is large, $n \in [N, 2N)$ and $0 \ne \ell \in \mathbb{Z}^{k+1}$ with $\|\ell\|_\infty \le Q$, then $|\Xi_n(\ell)| \ge c_5 N^\varsigma$.*

*Proof.* Let $j_0 = \min\{j : \ell_j \ne 0\}$. Then $|\Xi_n(\ell)| \ge |\binom{\gamma}{j_0}| n^{\gamma-j_0} - Qk \max_j |\binom{\gamma}{j}|\, n^{\gamma-j_0-1} \ge \frac12 |\binom{\gamma}{j_0}| n^{\gamma-j_0}$ for $N$ large, and $\gamma - j_0 \ge \varsigma$. □

*First moment.* $\mathbf{E}_\psi F_n = m_\psi^{-1} \sum_\ell a(\ell) \check\psi(\Xi_n(\ell))$. The term $\ell = 0$ is $\mu$; the terms with $0 < \|\ell\|_\infty \le Q$ contribute $O(C_2^{k+1} N^{-\varsigma P})$ by Lemma 10.4; the others contribute $o(\mu^2)$ by (26). Hence $\mathbf{E}_\psi F_n = \mu(1+o(1))$ and $\mathbf{E}_\psi Z_N = N\mu(1+o(1))$.

*Second moment.* For $n \ne n'$, $\mathbf{E}_\psi F_n F_{n'} = m_\psi^{-1} \sum_{\ell,\ell'} a(\ell)a(\ell') \check\psi\big(\Xi_n(\ell) + \Xi_{n'}(\ell')\big)$. The pair $(0,0)$ gives $\mu^2$. Pairs with $\ell$ or $\ell'$ outside the box $\|\cdot\|_\infty \le Q$ contribute $o(\mu^2)$ by (26). Pairs with exactly one of $\ell, \ell'$ equal to 0 and the other in the box contribute $O(\mu N^{-\varsigma P}) = o(\mu^2)$ by Lemma 10.4. Call $(n, n')$ *good* if $|\Xi_n(\ell) + \Xi_{n'}(\ell')| \ge T$ for all nonzero $\ell, \ell'$ in the box, and

*bad* otherwise. For good pairs the remaining terms contribute $O(T^{-P}) = o(\mu^2)$, so $\mathbf{E}_\psi F_n F_{n'} \le \mu^2(1+o(1))$.

**Lemma 10.5.** *The number of bad pairs is at most* $C_6(\log N)^{6k+9}N^{2-\varsigma}$.

*Proof.* Fix $n$ and nonzero $\ell, \ell'$ in the box, let $j_1 = \min\{j : \ell'_j \neq 0\}$ and $g(s) = \Xi_s(\ell')$ for real $s \in [N, 2N]$. Then

$$|g'(s)| \ge \left|\tbinom{\gamma}{j_1}(\gamma - j_1)\right| s^{\gamma - j_1 - 1} - C_7 Q s^{\gamma - j_1 - 2} \ge c_8 N^{\varsigma - 1}$$

for $N$ large, using $\gamma - j_1 \neq 0$ and $\gamma - j_1 - 1 \ge \varsigma - 1$ (if the exponent is negative, evaluate at $s = 2N$). So $g$ is strictly monotone on $[N, 2N]$, and the set of $s$ with $|g(s) + \Xi_n(\ell)| < T$ is an interval of length at most $2T/(c_8 N^{\varsigma-1})$, containing at most $2TN^{1-\varsigma}/c_8 + 1$ integers $n'$. Summing over the $N$ values of $n$ and the at most $(2Q+1)^{2k+2}$ pairs $(\ell, \ell')$ gives the claim. □

For bad pairs we use $F_n F_{n'} \le F_n$, and $F_n^2 \le F_n$ on the diagonal. Thus

$$\mathbf{E}_\psi Z_N^2 \le N\mu(1+o(1)) + N^2\mu^2(1+o(1)) + C_6(\log N)^{6k+9}N^{2-\varsigma}\mu(1+o(1)).$$

Since $N\mu \to \infty$ and $(\log N)^{6k+9}N^{-\varsigma}/\mu \to 0$, we get $\mathbf{E}_\psi Z_N^2 \le (\mathbf{E}_\psi Z_N)^2(1+o(1))$, and Chebyshev's inequality yields $\mathbf{P}_\psi(Z_N = 0) \le \mathrm{Var}_\psi(Z_N)/(\mathbf{E}_\psi Z_N)^2 \to 0$, where $\mathbf{P}_\psi$ is the probability measure $m_\psi^{-1}\psi\,dx$.

Let $G_r = \{x : Z_{2^r}(x) > 0\}$. Then $\mathbf{P}_\psi(\limsup_r G_r) \ge \limsup_r \mathbf{P}_\psi(G_r) = 1$. If $x \in G_r$ then $F_n(x) > 0$, hence $x \in E_n$, for some $n \in [2^r, 2^{r+1})$. By Lemma 10.3, for $\mathbf{P}_\psi$-almost every $x$ there are infinitely many $n$ with $\{n, \dots, n+L-1\} \subset A_x$. Taking $\psi = 1$ on $[-X_0+1, X_0-1]$, and then intersecting over $X_0 \in \mathbb{N}$ and $L \in \mathbb{N}$, we conclude that $A_x$ is thick for almost every $x \in \mathbb{R}$. Lemma 10.2 completes the proof. □

*Formalization.* Part (b) of Theorem 10.1 and Lemma 10.2 are formalized in `Kmrr25/Q413.lean` (`thm413b`, `thm413b_sumset`, `lem_thick`). Part (a) and the final equivalence are formalized unconditionally: `Kmrr25/Q413a.lean` contains Lemma 10.3 and the deduction of part (a) from the second-moment estimate, and `Kmrr25/Q413SecondMoment.lean` proves that estimate (`secondMomentBound`, `thm413a`, `thm413_iff`). The formal proof follows the second-moment method above, with Fejér kernels in place of the smooth windows and the indicator of $[-X_0, X_0]$ in place of $\psi$.

## 11. Sumsets in random sets

In this section we answer the question raised in §4.4 of [22], which we call Question §4.4(a). Let $p = (p_n)_{n\in\mathbb{N}} \in [0,1]^{\mathbb{N}}$ and let $A \subset \mathbb{N}$ be the random set in which the events $\{n \in A\}$ are independent and $\mathbf{P}(n \in A) = p_n$.

**Question §4.4(a)** ([22])**.** For which sequences $(p_n)_{n\in\mathbb{N}}$ does the set $A$ contain almost surely sumsets?

The survey shows: if $p$ is decreasing and $p_n n^\varepsilon \to \infty$ for every $\varepsilon > 0$, then $A$ is almost surely thick and hence contains $B + C$; if $p_n n^\varepsilon \to 0$ for some $\varepsilon > 0$, then for some $k$ almost surely $A$ contains no $B + F$ with $B$ infinite and $|F| = k$. Our answer has two parts.

- For monotone $p$ (indeed for $p$ comparable to a monotone sequence) the answer is completely explicit (Theorem 11.1): $A$ almost surely contains $B + C$ if and only if $A$ is almost surely thick, if and only if $\sum_n p_n^k = \infty$ for all $k$, if and only if $p_n n^\varepsilon \not\to 0$ for every $\varepsilon > 0$. In particular the gap left between the two conditions above is entirely on the positive side.

- For general $p$ there is no explicit answer: the set of $p$ for which $A$ almost surely contains $B+C$ (or $B\oplus B$, or $B+B$) is $\Sigma^1_1$-complete, hence not Borel, even within $(0,1)^{\mathbb{N}}$; for $B+C$ this persists within $\{p\in(0,1)^{\mathbb{N}} : p_n\to 0\}$ (Theorem 11.7). On the positive side we give the exact criterion for finite configurations $B+F$ (Theorem 11.3), an exact "strategy" criterion with a 0–1 law for $B+C$, $B\oplus B$ and $B+B$ (Theorems 11.5 and 11.6), which are close to a reformulation of the event, and simple sufficient and necessary conditions, with examples showing that all implications between them are strict (Theorem 11.8).

Theorems 11.5 and 11.6 are close to a reformulation of the event in question; it is Theorem 11.7 which shows that their existential form cannot be avoided, and Theorem 11.1 which settles the regime discussed in [22]. Theorem 11.3 is standard. The non-Borel part of Theorem 11.7 for $B\oplus B$ in the deterministic setting is due to Mazurkiewicz and Żeberski [25, Theorem 2.6], who proved that the ideal of sets containing no $B\oplus B$ (their $\mathcal{H}_2$) is $\Pi^1_1$-complete (their method, reducing ill-founded trees to additive families, is the one we use); for $B+C$ and $B+B$, and for the random versions, we did not find these statements in the literature.

11.1. **Notation and statements.** We set $p_m=0$ for integers $m\le 0$. For finite $U\subset\mathbb{Z}$ and $x\in\mathbb{Z}$ put $w_U(x)=\prod_{u\in U}p_{x+u}$ (so $w_\emptyset\equiv 1$), and for finite nonempty $F\subset\mathbb{Z}$ put $S_F=\sum_{n\in\mathbb{Z}}w_F(n)\in[0,\infty]$; clearly $S_{F+r}=S_F$ for $r\in\mathbb{Z}$, and $S_{F'}\ge S_F$ if $F'\subset F$. Write $B\oplus B=\{b+b' : b,b'\in B,\ b\ne b'\}$. Consider the events

- $\mathcal{E}_{BC}$: there are infinite $B,C\subset\mathbb{N}$ with $B+C\subset A$;
- $\mathcal{E}_{\oplus}$, $\mathcal{E}_{+}$: there is an infinite $B\subset\mathbb{N}$ with $B\oplus B\subset A$, respectively $B+B\subset A$;
- $\mathcal{E}_F$ ($F\subset\mathbb{Z}$ finite nonempty): there is an infinite $B\subset\mathbb{N}$ with $B+F\subset A$;
- $\mathcal{E}_k$: $\mathcal{E}_F$ holds for some $F\subset\mathbb{N}$ with $|F|=k$; and $\mathcal{E}_{<\omega}=\bigcap_k\mathcal{E}_k$.

We identify subsets of $\mathbb{N}$ with points of $2^{\mathbb{N}}$. Each of these events is an analytic subset of $2^{\mathbb{N}}$ (a projection of a Borel subset of $2^{\mathbb{N}}\times 2^{\mathbb{N}}\times 2^{\mathbb{N}}$), hence universally measurable (Lusin's theorem, see [16, Theorem 21.10]); $\mathbf{P}=\mathbf{P}_p$ denotes the completed product measure. The events $\mathcal{E}_{BC},\mathcal{E}_{\oplus},\mathcal{E}_{+},\mathcal{E}_F$ are invariant under finite modifications of $A$: adding points is harmless, and if $E$ is finite and $B+C\subset A$, then $(B\cap(\max E,\infty))+C\subset A\setminus E$ (similarly in the other cases).

**Theorem 11.1** (Monotone sequences)**.** *Suppose there are a non-increasing sequence $(q_n)$ and constants $0<\alpha\le\beta$ with $\alpha q_n\le p_n\le\beta q_n$ for all $n$. Then the following are equivalent:*

(1) *$A$ almost surely contains $B+C$ for some infinite $B,C$;*
(2) *$A$ is almost surely thick;*
(3) *$\sum_n p_n^k=\infty$ for every $k\in\mathbb{N}$;*
(4) *for every $\varepsilon>0$, $p_n n^{\varepsilon}$ does not tend to $0$;*
(5) *almost surely, for every $k$ there are an infinite $B$ and $F\subset\mathbb{N}$ with $|F|=k$ and $B+F\subset A$.*

*If they fail, there is $k$ such that almost surely $A$ contains no $B+F$ with $B$ infinite and $|F|=k$. In each case the probability of the event in (1), (2), (5) is $0$ or $1$.*

**Example 11.2.** Let $N_1=10$, $N_{j+1}=N_j^{j+1}$, $p_n=1$ for $n<N_1$ and $p_n=N_j^{-1/2}$ for $N_j\le n<N_{j+1}$. Then $p$ is non-increasing. At $n=N_j$, $p_n n^{\varepsilon}=N_j^{\varepsilon-1/2}\to 0$ for $\varepsilon<\frac12$, so the first condition of the survey fails; at $n=N_{j+1}-1$, $p_n n^{\varepsilon}\ge N_j^{-1/2}(N_j^{j+1}/2)^{\varepsilon}\to\infty$, so $p_n n^{\varepsilon}\not\to 0$ for every $\varepsilon$ and the second condition fails as well. By Theorem 11.1, $A$ is almost surely thick.

**Theorem 11.3** (Finite configurations)**.** *For finite nonempty $F\subset\mathbb{Z}$, $\mathbf{P}(\mathcal{E}_F)=1$ if $S_F=\infty$ and $\mathbf{P}(\mathcal{E}_F)=0$ otherwise. Consequently $\mathbf{P}(\mathcal{E}_k)=1$ if $S_F=\infty$ for some $F\subset\mathbb{N}$ with $|F|=k$,*

*and* $\mathbf{P}(\mathcal{E}_k) = 0$ *otherwise;* $\mathbf{P}(\mathcal{E}_{<\omega}) \in \{0,1\}$ *with value* 1 *if and only if this holds for every* $k$*; and* $A$ *is almost surely thick if* $S_{\{0,\dots,L-1\}} = \infty$ *for all* $L$*, and almost surely not thick otherwise.*

**Definition 11.4.** A *BC-strategy* for $p$ is a set $\mathcal{S}$ of pairs $(U,V)$ of finite subsets of $\mathbb{N}$ such that $(\emptyset,\emptyset) \in \mathcal{S}$ and for every $(U,V) \in \mathcal{S}$

$$\text{(a)} \sum_{x \in X_{\mathcal{S}}(U,V)} w_U(x) = \infty, \qquad \text{(b)} \sum_{y \in Y_{\mathcal{S}}(U,V)} w_V(y) = \infty,$$

where $X_{\mathcal{S}}(U,V) = \{x \in \mathbb{N}\backslash V : (U, V\cup\{x\}) \in \mathcal{S}\}$ and $Y_{\mathcal{S}}(U,V) = \{y \in \mathbb{N}\backslash U : (U\cup\{y\}, V) \in \mathcal{S}\}$. An *⊕-strategy* (resp. *+-strategy*) is a set $\mathcal{S}$ of finite subsets of $\mathbb{N}$ with $\emptyset \in \mathcal{S}$ such that for every $U \in \mathcal{S}$, $\sum_{x \in X_{\mathcal{S}}(U)} w_U^*(x) = \infty$, where $X_{\mathcal{S}}(U) = \{x \in \mathbb{N} \setminus U : U \cup \{x\} \in \mathcal{S}\}$ and $w_U^* = w_U$ (resp. $w_U^*(x) = p_{2x} w_U(x)$).

**Theorem 11.5.** $\mathbf{P}(\mathcal{E}_{BC}) \in \{0,1\}$*, and* $\mathbf{P}(\mathcal{E}_{BC}) = 1$ *if and only if* $p$ *has a BC-strategy.*

**Theorem 11.6.** $\mathbf{P}(\mathcal{E}_{\oplus}) \in \{0,1\}$*, and* $\mathbf{P}(\mathcal{E}_{\oplus}) = 1$ *if and only if* $p$ *has an ⊕-strategy; likewise for* $\mathcal{E}_+$ *and +-strategies.*

For $p \in \{0,1\}^{\mathbb{N}}$, i.e. $A = D$ deterministic, a strategy exists if and only if $D$ contains the configuration in question (see the end of Section 11.2); the two criteria in Theorem 11.6 differ, as $D = \{2^i + 2^j : i < j\}$ shows (Example 11.10).

To state the non-Borel results, recall that a *tree* is a set $\mathcal{T} \subset \mathbb{N}_0^{<\mathbb{N}_0}$ of finite sequences of elements of $\mathbb{N}_0 = \{0,1,2,\dots\}$ closed under initial segments; the trees form a closed subset Tr of $2^{\mathbb{N}_0^{<\mathbb{N}_0}}$, and IF $\subset$ Tr denotes the set of trees with an infinite branch. A subset $Z$ of a Polish space $W$ is $\Sigma^1_1$*-complete* if it is analytic and for every analytic subset $Z'$ of a zero-dimensional Polish space $W'$ there is a continuous $f\colon W' \to W$ with $Z' = f^{-1}(Z)$. The set IF is $\Sigma^1_1$-complete (see [16, Theorem 27.1]); hence an analytic set $Z$ is $\Sigma^1_1$-complete as soon as IF $= g^{-1}(Z)$ for some continuous $g\colon \mathrm{Tr} \to W$, and $\Sigma^1_1$-complete sets are not Borel.

**Theorem 11.7.** (a) *The families* $\mathfrak{D}_{BC} = \{D \subset \mathbb{N} : \exists$ *infinite* $B, C,\ B + C \subset D\}$*,* $\mathfrak{D}_{\oplus} = \{D : \exists$ *infinite* $B,\ B \oplus B \subset D\}$ *and* $\mathfrak{D}_{+} = \{D : \exists$ *infinite* $B,\ B + B \subset D\}$ *are* $\Sigma^1_1$*-complete subsets of* $2^{\mathbb{N}}$*.*

(b) *For* $* \in \{BC, \oplus, +\}$ *let* $\mathfrak{A}_* = \{p \in [0,1]^{\mathbb{N}} : \mathbf{P}_p(\mathcal{E}_*) = 1\}$*. Then* $\mathfrak{A}_*$ *is a* $\Sigma^1_1$*-complete subset of* $[0,1]^{\mathbb{N}}$*, and* $\mathfrak{A}_* \cap (0,1)^{\mathbb{N}}$ *is a* $\Sigma^1_1$*-complete subset of* $(0,1)^{\mathbb{N}}$*.*

(c) *Let* $\mathfrak{C} = \{p \in (0,1)^{\mathbb{N}} : p_n \to 0\}$*. There is a continuous* $g\colon \mathrm{Tr} \to \mathfrak{C}$ *with* $g^{-1}(\mathfrak{A}_{BC}) = \mathrm{IF}$*. In particular there is no Borel set* $W \subset [0,1]^{\mathbb{N}}$ *with* $W \cap \mathfrak{C} = \mathfrak{A}_{BC} \cap \mathfrak{C}$*.*

We do not know whether the analogue of (c) holds for $\mathcal{E}_{\oplus}$ and $\mathcal{E}_+$.

**Theorem 11.8.** *Consider the conditions*

*(F) there are infinite* $B_0, C_0 \subset \mathbb{N}$ *such that* $\sum_{c \in C_0} w_U(c) = \infty$ *for every finite* $U \subset B_0$ *and* $\sum_{b \in B_0} w_V(b) = \infty$ *for every finite* $V \subset C_0$*;*

*(N) there are infinite* $B_0, C_0 \subset \mathbb{N}$ *such that* $S_G = \infty$ *for every finite nonempty* $G \subset B_0$ *and for every finite nonempty* $G \subset C_0$*.*

*Then* (F) $\Rightarrow \mathbf{P}(\mathcal{E}_{BC}) = 1 \Rightarrow$ (N) $\Rightarrow \mathbf{P}(\mathcal{E}_{<\omega}) = 1$*, and none of the three implications can be reversed (Examples 11.14 to 11.16).*

Condition (F) holds, for instance, with $B_0 = C_0 = \mathbb{N}$ if $S_G = \infty$ for every finite nonempty $G \subset \mathbb{N}$ (as in the situation of Theorem 11.1), or if some level set $\{n : p_n \geq \delta\}$, $\delta > 0$, contains $B_0 + C_0$; in particular, for $p = q\mathbf{1}_D$ with $0 < q \leq 1$ we have $\mathbf{P}(\mathcal{E}_{BC}) = 1$ if and only if $D \in \mathfrak{D}_{BC}$. In (N) the two families of finite sets are separate: we do not require $S_G = \infty$ for finite $G \subset B_0 \cup C_0$.

11.2. **Proofs of Theorems 11.3, 11.5 and 11.6.**

**Lemma 11.9.** *Let $(K_x)_{x\in I}$ be a countable family of finite subsets of $\mathbb{Z}$ (recall $p_m = 0$ for $m \le 0$, so $\mathbf{P}(K_x \subset A) = 0$ unless $K_x \subset \mathbb{N}$) such that each $K_x$ meets $K_{x'}$ for at most $M$ indices $x' \neq x$. If $\sum_x \mathbf{P}(K_x \subset A) = \infty$, then almost surely $K_x \subset A$ for infinitely many $x$.*

*Proof.* Enumerate $I$ and colour each $x$ with the least colour in $\{1, \dots, M+1\}$ not used by an earlier $x'$ with $K_{x'} \cap K_x \neq \emptyset$. Some colour class $I_0$ satisfies $\sum_{x\in I_0} \mathbf{P}(K_x \subset A) = \infty$. The sets $K_x$, $x \in I_0$, are pairwise disjoint, so the events $\{K_x \subset A\}$, $x \in I_0$, are independent, and the second Borel–Cantelli lemma applies. □

We use this with $K_x = x + U$, where $K_x \cap K_{x'} \neq \emptyset$ forces $x' - x \in U - U$ (so $M \le |U|^2$), and with $K_x = \{2x\} \cup (x + U)$, where it forces $x' \in x + (U - U)$, $2x' \in x + U$ or $2x \in x' + U$ (so $M \le |U|^2 + 2|U|$).

*Proof of Theorem 11.3.* $\mathcal{E}_F$ holds if and only if $n + F \subset A$ for infinitely many $n \in \mathbb{Z}$: if $B + F \subset A$ take $n \in B$; conversely such $n$ satisfy $n \ge 1 - \min F$, so infinitely many of them lie in $\mathbb{N}$. Since $\mathbf{P}(n + F \subset A) = w_F(n)$, the first Borel–Cantelli lemma handles $S_F < \infty$ and Lemma 11.9 (with $K_n = n + F$) handles $S_F = \infty$. Next, $\mathcal{E}_k$ is the countable union of the $\mathcal{E}_F$ with $F \subset \mathbb{N}$, $|F| = k$, each of probability 0 or 1, and $\mathcal{E}_{<\omega}$ is a countable intersection. Finally, $A$ is thick if and only if for every $L$ there are infinitely many $n$ with $\{n, \dots, n + L - 1\} \subset A$ (an interval of length $L' \ge L$ contains $L' - L + 1$ such blocks), i.e. if and only if $\mathcal{E}_{\{0,\dots,L-1\}}$ holds for all $L$. □

*Proof of Theorem 11.5. If $\mathcal{S}$ is a BC-strategy, then $\mathbf{P}(\mathcal{E}_{BC}) = 1$.* As $\mathcal{S}$ is countable, (a), (b) and Lemma 11.9 give an event $\Omega_0$ of probability 1 on which, for every $(U, V) \in \mathcal{S}$, there are infinitely many $x \in X_{\mathcal{S}}(U, V)$ with $x + U \subset A$ and infinitely many $y \in Y_{\mathcal{S}}(U, V)$ with $y + V \subset A$. (For $U = \emptyset$, (a) just says that $X_{\mathcal{S}}(\emptyset, V)$ is infinite.) Fix an outcome in $\Omega_0$. Starting from $(U_0, V_0) = (\emptyset, \emptyset)$, choose $x \in X_{\mathcal{S}}(U_r, V_r)$ with $x > \max(U_r \cup V_r)$ and $x + U_r \subset A$, put $V_{r+1} = V_r \cup \{x\}$; then choose $y \in Y_{\mathcal{S}}(U_r, V_{r+1})$ with $y > \max(U_r \cup V_{r+1})$ and $y + V_{r+1} \subset A$, and put $U_{r+1} = U_r \cup \{y\}$. Inductively $(U_r, V_r) \in \mathcal{S}$ and $U_r + V_r \subset A$, since the new sums are exactly $x + U_r$ and $y + V_{r+1}$. Then $B = \bigcup_r U_r$ and $C = \bigcup_r V_r$ are infinite and $B + C \subset A$.

*If $\mathbf{P}(\mathcal{E}_{BC}) > 0$, then $p$ has a BC-strategy.* For finite $U, V \subset \mathbb{N}$ let $\mathcal{E}(U, V)$ be the event that there are infinite $B', C' \subset \mathbb{N}$ with $U + C' \subset A$, $B' + V \subset A$ and $B' + C' \subset A$. It is analytic, and $\mathcal{E}(\emptyset, \emptyset) = \mathcal{E}_{BC}$. Let $\mathcal{S}^* = \{(U, V) : \mathbf{P}(\mathcal{E}(U, V)) > 0\}$, which contains $(\emptyset, \emptyset)$. We show that $\mathcal{S}^*$ satisfies (a); (b) is symmetric. Let $(U, V) \in \mathcal{S}^*$, write $X^* = X_{\mathcal{S}^*}(U, V)$, and suppose $\sum_{x\in X^*} w_U(x) < \infty$. On $\mathcal{E}(U, V)$, with witnesses $B', C'$, every $x \in C' \setminus V$ satisfies $x + U \subset A$, and $\mathcal{E}(U, V \cup \{x\})$ holds with witnesses $B', C' \setminus \{x\}$ (because $B' + x \subset B' + C' \subset A$). Hence $\mathcal{E}(U, V) \subset N_1 \cup N_2$, where $N_1$ is the union of the null events $\mathcal{E}(U, V \cup \{x\})$ over $x \in \mathbb{N} \setminus (V \cup X^*)$, and $N_2$ is the event that $x + U \subset A$ for infinitely many $x \in X^*$, which is null by the first Borel–Cantelli lemma. Thus $\mathbf{P}(\mathcal{E}(U, V)) = 0$, a contradiction. Combined with the first part this also proves the 0–1 law. □

*Proof of Theorem 11.6.* The proof is the same, with single sets. Given a strategy, choose new elements $x \in X_{\mathcal{S}}(U_r)$, $x > \max U_r$, with $x + U_r \subset A$ (and also $2x \in A$ in the + case); for $x \notin U$ we have $2x \notin x + U$, so the probability of this event is exactly $w^*_U(x)$, and Lemma 11.9 applies with $K_x = x + U$, resp. $K_x = \{2x\} \cup (x + U)$. Conversely, let $\mathcal{E}^*(U)$ be the event that there is an infinite $B'$ with $U + B' \subset A$ and $B' \oplus B' \subset A$ (and $2b \in A$ for $b \in B'$ in the + case). On $\mathcal{E}^*(U)$, every $x \in B' \setminus U$ satisfies $x + U \subset A$ (and $2x \in A$), and $\mathcal{E}^*(U \cup \{x\})$ holds with witness $B' \setminus \{x\}$, since $x + (B' \setminus \{x\}) \subset B' \oplus B'$. The rest is as before. □

*The deterministic case.* If $p = \mathbf{1}_D$, then $\sum_{x\in X} w_U(x) = \infty$ means that $x + U \subset D$ for infinitely many $x \in X$. If $D \supset B + C$ with $B, C$ infinite, the set of pairs of finite subsets of $B$ and of $C$ is a $BC$-strategy (in (a), $X_{\mathcal{S}}(U, V) \supset C \setminus V$ and all these terms equal 1); conversely a strategy yields $B + C \subset D$ by the greedy construction. The same holds for $\oplus$ and $+$ with the finite subsets of $B$.

**Example 11.10.** Let $D = \{2^i + 2^j : 0 \le i < j\}$. Then $B = \{2^i : i \ge 0\}$ satisfies $B \oplus B \subset D$. But no infinite $B$ satisfies $B + B \subset D$: from $2b \in D$, every $b \in B$ has exactly two binary digits 1. Fix $b = 2^a + 2^c \in B$ $(a < c)$. If $b' \in B$ has lowest binary digit above position $c + 1$, then $b + b'$ has four binary digits 1; so infinitely many $b' \in B$ have lowest digit at most $c + 1$, and two of them, $b'_1 = 2^\alpha + 2^{c_1}$ and $b'_2 = 2^\alpha + 2^{c_2}$, share the lowest digit $\alpha$ and have $\alpha+1 < c_1 < c_2$. Then $b'_1 + b'_2 = 2^{\alpha+1} + 2^{c_1} + 2^{c_2} \notin D$.

11.3. **Rigid sets.** Fix $\Lambda = \{\lambda_1 < \lambda_2 < \cdots\} \subset \mathbb{N}$ with $\lambda_{j+1} > 8(\lambda_1 + \cdots + \lambda_j)$ for all $j$ (for instance $\lambda_j = 9^j$).

**Lemma 11.11.** *If $\sum_j e_j\lambda_j = 0$ with $e_j \in \mathbb{Z}$ and $\sum_j |e_j| \le 8$, then all $e_j = 0$. Consequently, every $n \in \Lambda + \Lambda$ has a unique representation $n = \lambda + \lambda'$ as an unordered pair (multiset) $P(n) = \{\lambda, \lambda'\}$.*

*Proof.* If $J$ is the largest index with $e_J \ne 0$, then $|e_J\lambda_J| \ge \lambda_J > 8\sum_{j<J} \lambda_j \ge |\sum_{j<J} e_j\lambda_j|$. Unique representation is the case of a relation with $\sum |e_j| \le 4$. □

**Lemma 11.12.** *Let $X, \Gamma \subset \mathbb{Z}$ with $|X| \ge 2$ and $|\Gamma| \ge 2$, and suppose that $X + \Gamma \subset \Lambda + \Lambda$. Then there is a unique $t = t(X, \Gamma) \in \mathbb{Z}$ such that $x - t \in \Lambda$ for all $x \in X$, $c + t \in \Lambda$ for all $c \in \Gamma$, and $P(x + c) = \{x - t, c + t\}$ for all $x \in X$, $c \in \Gamma$. If $X' \subset X$ and $\Gamma' \subset \Gamma$ with $|X'|, |\Gamma'| \ge 2$, then $t(X', \Gamma') = t(X, \Gamma)$.*

*Proof. Uniqueness.* If $t \ne t'$ both work and $x \in X$, then for $c \in \Gamma$, $\{x-t, c+t\} = \{x-t', c+t'\}$ forces $x - t = c + t'$; this determines $c$, contradicting $|\Gamma| \ge 2$. The last assertion follows, since $t(X, \Gamma)$ works for $(X', \Gamma')$.

*Two rows.* Let $x_1 \ne x_2$ in $X$ and write $P_{ic} = P(x_i+c)$. First, some $c_0 \in \Gamma$ has $P_{1c_0} \cap P_{2c_0} \ne \emptyset$. Otherwise, take $c \ne c'$ in $\Gamma$. The identity $(x_1 + c) + (x_2 + c') = (x_1 + c') + (x_2 + c)$ is a relation among at most eight elements of $\Lambda$, so by Lemma 11.11 the multisets $P_{1c} \uplus P_{2c'}$ and $P_{1c'} \uplus P_{2c}$ coincide. As $P_{1c}$ is disjoint from $P_{2c}$, it is contained in $P_{1c'}$ as a multiset, so $P_{1c} = P_{1c'}$ and $c = c'$, a contradiction. Write $x_1 + c_0 = u + v$ and $x_2 + c_0 = u' + v$ with $u, u', v \in \Lambda$; then $u \ne u'$ because $x_1 \ne x_2$. Put $t = x_1 - u$, so that $x_1 - t = u$ and $x_2 - t = u'$ lie in $\Lambda$. Let $c \in \Gamma$. Since $x_1 - x_2 = u - u'$, the identity $(x_1 + c) + u' = (x_2 + c) + u$ is a relation among six elements of $\Lambda$, so $P_{1c} \uplus \{u'\} = P_{2c} \uplus \{u\}$. As $u \ne u'$, we get $u \in P_{1c}$, that is, $x_1 + c = u + w$ with $w \in \Lambda$. Then $c + t = w \in \Lambda$, and $P(x_i + c) = \{x_i - t, c + t\}$ for $i = 1, 2$ by unique representation.

*General $X$.* Fix $x_1 \ne x_2$ in $X$ and let $t = t(\{x_1, x_2\}, \Gamma)$. For $x \in X$, $x \ne x_1$, the two-row case gives $t' = t(\{x_1, x\}, \Gamma)$; both $t$ and $t'$ work for $(\{x_1\}, \Gamma)$ in the sense of the uniqueness argument, which only used one row, so $t' = t$. Hence $x - t \in \Lambda$, and $P(x + c) = \{x - t, c + t\}$ by unique representation. □

**Corollary 11.13.** *If $B, C \subset \mathbb{Z}$ are infinite and $B + C \subset \Lambda + \Lambda$, there is $t \in \mathbb{Z}$ with $B - t \subset \Lambda$, $C + t \subset \Lambda$ and $P(b + c) = \{b - t, c + t\}$ for all $b \in B$, $c \in C$.*

11.4. **Proof of Theorem 11.7.** Fix injective labellings $s \mapsto \beta_s$, $s \mapsto \gamma_s$ of $\mathbb{N}_0^{<\mathbb{N}_0}$ by elements of $\Lambda$ with disjoint images. Comparability of finite sequences means that one is an initial segment of the other.

*Upper bounds.* $\mathfrak{D}_*$ is the projection of a Borel subset of $2^{\mathbb{N}} \times 2^{\mathbb{N}} \times 2^{\mathbb{N}}$, hence analytic. For $\mathfrak{A}_{BC}$, by Theorem 11.5 it is the projection to $[0, 1]^{\mathbb{N}}$ of the set of pairs $(\mathcal{S}, p) \in 2^{\mathcal{P}_{\mathrm{fin}}(\mathbb{N})^2} \times [0, 1]^{\mathbb{N}}$

such that $\mathcal{S}$ is a $BC$-strategy for $p$; this set is Borel, since each partial sum of the series in (a), (b) is a continuous function of $(\mathcal{S}, p)$ and "$=\infty$" means "$\forall M\,\exists N$: partial sum $> M$". The same applies to $\mathfrak{A}_\oplus$, $\mathfrak{A}_+$, and to the intersections with the Polish space $(0,1)^{\mathbb{N}}$.

(a) *for $B+C$.* For a tree $\mathcal{T}$ let $D_\mathcal{T} = \{\beta_s + \gamma_{s'} : s, s' \in \mathcal{T} \text{ comparable}\}$. By Lemma 11.11, whether $n \in D_\mathcal{T}$ depends on at most two coordinates of $\mathcal{T}$, so $\mathcal{T} \mapsto D_\mathcal{T}$ is continuous. If $z$ is an infinite branch of $\mathcal{T}$, then $B = \{\beta_{z|n}\}$, $C = \{\gamma_{z|n}\}$ give $B + C \subset D_\mathcal{T}$. Conversely let $B + C \subset D_\mathcal{T}$ with $B, C$ infinite. By Corollary 11.13, $\{b - t, c + t\} = \{\beta_s, \gamma_{s'}\}$ with $s, s' \in \mathcal{T}$ comparable. Fixing $c$, if $c + t$ is a $\gamma$-label then every $b - t$ is a $\beta$-label, and then every $c + t$ is a $\gamma$-label; so either $B - t$ consists of $\beta$-labels and $C + t$ of $\gamma$-labels, or vice versa. In either case we get infinite sets $Q_1, Q_2 \subset \mathcal{T}$ such that every element of $Q_1$ is comparable with every element of $Q_2$. A node has finitely many initial segments, so for $q \in Q_1$ all but finitely many elements of $Q_2$ properly extend $q$, and vice versa. Choosing alternately we obtain $q_1 \subsetneq q_1' \subsetneq q_2 \subsetneq q_2' \subsetneq \cdots$ in $\mathcal{T}$, which defines an infinite branch. So $\mathcal{T} \mapsto D_\mathcal{T}$ reduces IF to $\mathfrak{D}_{BC}$.

(a) *for $B \oplus B$ and $B + B$.* Let $D^\oplus_\mathcal{T} = \{\gamma_s + \gamma_{s'} : s \neq s' \in \mathcal{T} \text{ comparable}\}$ and $D^+_\mathcal{T} = \{\gamma_s + \gamma_{s'} : s, s' \in \mathcal{T} \text{ comparable}\}$; both maps are continuous, and a branch $z$ gives $B = \{\gamma_{z|n}\}$. Conversely, let $B$ be infinite with $B \oplus B \subset D^\oplus_\mathcal{T}$ or $B + B \subset D^+_\mathcal{T}$. Split $B$ into disjoint infinite sets $B_1, B_2$; then $B_1 + B_2 \subset B \oplus B \subset \Lambda + \Lambda$, and Corollary 11.13 gives $t$ with $P(b + b') = \{b - t, b' + t\}$ for $b \in B_1$, $b' \in B_2$. Since $b + b' = \gamma_s + \gamma_{s'}$ with $s, s' \in \mathcal{T}$ comparable, $b - t$ and $b' + t$ are $\gamma$-labels of comparable nodes of $\mathcal{T}$. So the nodes labelling $B_1 - t$ and those labelling $B_2 + t$ form infinite sets $Q_1, Q_2 \subset \mathcal{T}$ such that every element of $Q_1$ is comparable with every element of $Q_2$, and the argument for $B + C$ gives an infinite branch.

(b). The embedding $D \mapsto \mathbf{1}_D$ of $2^{\mathbb{N}}$ into $[0,1]^{\mathbb{N}}$ is continuous, and $\mathbf{P}_{\mathbf{1}_D}(\mathcal{E}_*) = 1$ if and only if $D \in \mathfrak{D}_*$; so the first claim follows from (a). For $(0,1)^{\mathbb{N}}$, let $p^D_n = \frac{1}{2}$ for $n \in D$ and $p^D_n = 2^{-n-2}$ for $n \notin D$; $D \mapsto p^D$ is continuous. Under $\mathbf{P}_{p^D}$, $A \setminus D$ is almost surely finite (first Borel–Cantelli lemma), so by invariance under finite modifications $\mathbf{P}_{p^D}(\mathcal{E}_*) = \mathbf{P}(A \cap D \in \mathcal{E}_*)$, where $A \cap D$ is a random subset of $D$ containing each element independently with probability $\frac{1}{2}$. If $D \notin \mathfrak{D}_*$ this probability is 0. If $D \supset B_0 + C_0$, the pairs of finite subsets of $B_0$ and $C_0$ form a $BC$-strategy for $\frac{1}{2}\mathbf{1}_D$ (all weights are $2^{-|U|}$ or $2^{-|V|}$, with infinitely many candidates), so the probability is 1 by Theorem 11.5. Likewise, if $D \supset B_0 \oplus B_0$ (resp. $B_0 + B_0$), the finite subsets of $B_0$ form an $\oplus$- (resp. $+$-) strategy with weights $\geq 2^{-|U|-1}$. Hence $\mathbf{P}_{p^D}(\mathcal{E}_*) = 1$ if and only if $D \in \mathfrak{D}_*$, and composing with the reductions in (a) proves the second claim.

(c). For $s \in \mathbb{N}_0^{<\mathbb{N}_0}$ let $\ell(s) = |s| + \sum_i s_i$, and let $\mathcal{N}$ be the set of pairs $x = (s, \eta)$ with $s \in \mathbb{N}_0^{<\mathbb{N}_0}$ and $\eta \in \{0,1\}^{\ell(s)}$, partially ordered by $(s, \eta) \preceq (s', \eta')$ if $s \subset s'$ and $\eta$ is an initial segment of $\eta'$. Write $\ell(x) = \ell(s)$. Then: if $s \subsetneq s'$ then $\ell(s) < \ell(s')$; every $x \in \mathcal{N}$ has finitely many predecessors; for each $L$ there are finitely many $x$ with $\ell(x) = L$; and two distinct elements with the same first coordinate are incomparable. Fix injective labellings $x \mapsto \beta_x$, $x \mapsto \gamma_x$ of $\mathcal{N}$ by elements of $\Lambda$ with disjoint images, and let $\theta_L = 1/(L+2)$. For a tree $\mathcal{T}$ define $q^\mathcal{T} \in [0, \frac{1}{2}]^{\mathbb{N}}$ by $q^\mathcal{T}_n = \theta_{\max(\ell(x),\ell(y))}$ if $n = \beta_x + \gamma_y$ with $x, y$ comparable and with first coordinates in $\mathcal{T}$, and $q^\mathcal{T}_n = 0$ otherwise; and let $g(\mathcal{T}) = p^\mathcal{T}$, where $p^\mathcal{T}_n = q^\mathcal{T}_n$ if $q^\mathcal{T}_n > 0$ and $p^\mathcal{T}_n = 2^{-n-2}$ otherwise. Each coordinate of $p^\mathcal{T}$ depends on at most two coordinates of $\mathcal{T}$ (Lemma 11.11), so $g$ is continuous, and $p^\mathcal{T} \in (0,1)^{\mathbb{N}}$. Moreover $p^\mathcal{T}_n \to 0$: $q^\mathcal{T}_n \geq \varepsilon$ forces $\max(\ell(x), \ell(y)) \leq 1/\varepsilon$, which happens for finitely many pairs $(x, y)$ only.

Suppose $z$ is an infinite branch of $\mathcal{T}$ and let $\ell_m = \ell(z|m)$, a strictly increasing sequence. Let $\mathcal{N}_z = \{(z|m, \eta) \in \mathcal{N} : m \geq 0\}$ and let $\mathcal{S}$ consist of the pairs $(\{\beta_x\}_{x \in P}, \{\gamma_y\}_{y \in Q})$ with $P, Q \subset \mathcal{N}_z$ finite and $P \cup Q$ a chain. We check (a) for $q^\mathcal{T}$; (b) is symmetric. If $P \cup Q = \emptyset$, (a) holds since $U = \emptyset$ and there are infinitely many candidates. Otherwise let $\delta = (z|d, \eta_\delta)$ be the largest element of $P \cup Q$. For $m > d$ each of the $2^{\ell_m - \ell_d}$ elements $y = (z|m, \eta) \succ \delta$ extends the chain, and

$w_U(\gamma_y) = \prod_{x\in P} q^{\mathcal{T}}_{\beta_x+\gamma_y} = \theta_{\ell_m}^{|P|}$. Hence the sum in (a) is at least $\sum_{m>d} 2^{\ell_m-\ell_d}(\ell_m+2)^{-|P|} = \infty$. So $\mathcal{S}$ is a $BC$-strategy for $q^{\mathcal{T}}$, hence for $p^{\mathcal{T}} \geq q^{\mathcal{T}}$, and $\mathbf{P}_{p^{\mathcal{T}}}(\mathcal{E}_{BC}) = 1$.

Conversely suppose $\mathbf{P}_{p^{\mathcal{T}}}(\mathcal{E}_{BC}) > 0$. Almost surely $A \setminus \operatorname{supp} q^{\mathcal{T}}$ is finite, so with positive probability $A \cap \operatorname{supp} q^{\mathcal{T}}$ contains some $B + C$; in particular $\operatorname{supp} q^{\mathcal{T}} \supset B + C$ for some infinite $B, C$. As $\operatorname{supp} q^{\mathcal{T}} \subset \Lambda + \Lambda$, the argument in (a) (which only used that every element has finitely many predecessors) yields an infinite chain $x_1 \prec x_2 \prec \cdots$ in $\mathcal{N}$ whose first coordinates lie in $\mathcal{T}$. Distinct comparable elements have distinct first coordinates, so the first coordinates form a strictly increasing sequence in $\mathcal{T}$, i.e. an infinite branch. Thus $g^{-1}(\mathfrak{A}_{BC}) = \mathrm{IF}$. If $W$ were Borel with $W \cap \mathfrak{C} = \mathfrak{A}_{BC} \cap \mathfrak{C}$, then $\mathrm{IF} = g^{-1}(W)$ would be Borel, which it is not. □

11.5. **Proof of Theorem 11.8 and the examples.** *(F) implies* $\mathbf{P}(\mathcal{E}_{BC}) = 1$*:* the pairs of finite subsets of $B_0$ and $C_0$ form a $BC$-strategy, since removing the finitely many terms $x \in V$ from a divergent series of terms in $[0, 1]$ keeps it divergent. For the level-set condition, $w_U(c) \geq \delta^{|U|}$ for $c \in C_0$; for $p = q\mathbf{1}_D$ necessity is clear since $A \subset D$.

$\mathbf{P}(\mathcal{E}_{BC}) = 1$ *implies (N):* by the first Borel–Cantelli lemma, almost surely every finite $G$ with $S_G < \infty$ has only finitely many translates in $A$. Fix an outcome in this event and in $\mathcal{E}_{BC}$, with $B + C \subset A$. Every finite $G \subset C$ has the infinitely many translates $b + G \subset A$, $b \in B$, so $S_G = \infty$; similarly for $G \subset B$. Take $B_0 = B$, $C_0 = C$.

*(N) implies* $\mathbf{P}(\mathcal{E}_{<\omega}) = 1$*:* by Theorem 11.3 applied to $k$-subsets of $C_0$.

**Example 11.14** ($\mathbf{P}(\mathcal{E}_{BC}) = 1$ but (F) fails)**.** Label the binary tree $\{0,1\}^{<\mathbb{N}_0}$ injectively by $\beta_\sigma, \gamma_\sigma \in \Lambda$ (disjoint images), let $\theta_j = 1/(j+1)$, and let $p_{\beta_\sigma+\gamma_\tau} = \theta_{\max(|\sigma|,|\tau|)}$ for comparable $\sigma, \tau$, and $p_n = 0$ otherwise. Exactly as in the proof of Theorem 11.7(c), the pairs $(\{\beta_\sigma\}_{\sigma\in P}, \{\gamma_\tau\}_{\tau\in Q})$ with $P \cup Q$ a finite chain form a $BC$-strategy: below the deepest node, at depth $d$, there are $2^{j-d}$ candidates at depth $j$, each of weight $\theta_j^{|P|}$, and $\sum_j 2^{j-d}(j+1)^{-|P|} = \infty$. So $\mathbf{P}(\mathcal{E}_{BC}) = 1$.

Suppose $B_0, C_0$ satisfy (F). Fix $b_1 \neq b_2$ in $B_0$ and let $\Gamma = \{c \in C_0 : w_{\{b_1,b_2\}}(c) > 0\}$, which is infinite by (F). As $\operatorname{supp} p \subset \Lambda + \Lambda$, let $t = t(\{b_1, b_2\}, \Gamma)$ (Lemma 11.12). For $b \in B_0$, the set $\Gamma_b = \{c \in C_0 : w_{\{b_1,b_2,b\}}(c) > 0\} \subset \Gamma$ is infinite, and $t(\{b_1, b_2, b\}, \Gamma_b) = t(\{b_1, b_2\}, \Gamma_b) = t$; hence $B_0 - t \subset \Lambda$. For each $c \in \Gamma$, $P(b_i + c) = \{b_i - t, c + t\}$ contains one $\beta$- and one $\gamma$-label, so $b_1 - t$ and $b_2 - t$ (and then, by the same argument, all of $B_0 - t$) have the same type; as $p$ is symmetric under exchanging the roles of $\beta$ and $\gamma$, we may assume $B_0 - t = \{\beta_\sigma : \sigma \in \Sigma_B\}$. Let $c \in C_0$. By (F) with $V = \{c\}$ there are distinct $b = \beta_{\sigma_1} + t$, $b' = \beta_{\sigma_2} + t$ in $B_0$ with $b + c = \beta_{\sigma_1'} + \gamma_{\tau_1}$ and $b' + c = \beta_{\sigma_2'} + \gamma_{\tau_2}$. Eliminating $c$ gives a relation of six elements of $\Lambda$, so by Lemma 11.11 the multisets $\{\beta_{\sigma_1'}, \gamma_{\tau_1}, \beta_{\sigma_2}\}$ and $\{\beta_{\sigma_2'}, \gamma_{\tau_2}, \beta_{\sigma_1}\}$ coincide; hence $\tau_1 = \tau_2$, and since $\sigma_1 \neq \sigma_2$ we get $\beta_{\sigma_1'} = \beta_{\sigma_1}$, i.e. $c + t = \gamma_{\tau_1}$. So $C_0 + t = \{\gamma_\tau : \tau \in \Sigma_C\}$, and $b + c \in \operatorname{supp} p$ (for $b = \beta_\sigma + t$, $c = \gamma_\tau - t$) if and only if $\sigma, \tau$ are comparable. By (F) with $V = \{c, c'\}$, infinitely many $\sigma \in \Sigma_B$ are comparable with both $\tau$ and $\tau'$; if $\tau, \tau'$ were incomparable, such $\sigma$ would be common initial segments of $\tau$ and $\tau'$, of which there are finitely many. So $\Sigma_C$ is a chain, with at most one node at each depth. Then for $U = \{b_1, b_2\}$, $\sum_{c\in C_0} w_U(c) \leq \sum_{\tau\in\Sigma_C} \theta^2_{|\tau|} \leq \sum_j (j+1)^{-2} < \infty$, contradicting (F).

**Example 11.15** ((N) holds but $\mathbf{P}(\mathcal{E}_{BC}) = 0$)**.** Label $\{\beta_{k,m} : k, m \in \mathbb{N}\} \cup \{\gamma_i : i \in \mathbb{N}\}$ injectively by elements of $\Lambda$ and let $p = \mathbf{1}_D$ with $D = \{\beta_{k,m} + \gamma_i : i \leq k,\ m \in \mathbb{N}\}$. Condition (N) holds with $B_0 = C_0 = \{\gamma_i\}$: a finite $G \subset \{\gamma_1, \ldots, \gamma_k\}$ has the translates $\beta_{k,m} + G \subset D$, $m \in \mathbb{N}$. But $D \notin \mathfrak{D}_{BC}$: if $B + C \subset D$, then by Corollary 11.13 $\{b - t, c + t\} = \{\beta_{k,m}, \gamma_i\}$ with $i \leq k$. If the $b - t$ are $\beta$-labels, fix $b_1$ with $b_1 - t = \beta_{k_1,m_1}$; then every $c + t$ lies in $\{\gamma_1, \ldots, \gamma_{k_1}\}$, so $C$ is finite. Otherwise, symmetrically, $B$ is finite. In particular $\mathcal{E}_{<\omega}$ holds but $\mathcal{E}_{BC}$ fails, for a deterministic set.

**Example 11.16** ($\mathbf{P}(\mathcal{E}_{<\omega}) = 1$ but (N) fails)**.** Label $\{\beta_{k,m}, \gamma_{k,i} : k, m \in \mathbb{N},\ 1 \le i \le k\}$ injectively by elements of $\Lambda$ and let $p = \mathbf{1}_D$ with $D = \{\beta_{k,m} + \gamma_{k,i} : 1 \le i \le k,\ m \in \mathbb{N}\}$. Then $\{\beta_{k,m} : m \in \mathbb{N}\} + \{\gamma_{k,1}, \dots, \gamma_{k,k}\} \subset D$, so $\mathcal{E}_k$ holds for every $k$. Suppose $C_0$ is infinite and $S_G = \infty$ (i.e. $G$ has infinitely many translates in $D$) for all $G \subset C_0$ with $|G| \le 3$. Fix $c_1 \ne c_2$ in $C_0$, let $\Gamma$ be the (infinite) set of $n$ with $n + \{c_1, c_2\} \subset D$, and $t = t(\{c_1, c_2\}, \Gamma)$. For each $n \in \Gamma$, $n + t$ has one type and $c_1 - t$, $c_2 - t$ have the other. If $c_1 - t = \beta_{k_1,m_1}$ and $c_2 - t = \beta_{k_2,m_2}$, then $n + t \in \{\gamma_{k_1,i}\} \cap \{\gamma_{k_2,i}\}$, a finite set, for all $n \in \Gamma$, which is impossible. Hence $c_j - t = \gamma_{k,i_j}$ and $n + t = \beta_{k,m_n}$ with the same $k$. For any $c \in C_0$, applying the same reasoning to $\{c_1, c_2, c\}$ and its infinite set $\Gamma_c \subset \Gamma$ of translates, Lemma 11.12 gives the same $t$, and $c - t$ is a $\gamma$-label paired with $n + t = \beta_{k,m_n}$, so $c - t \in \{\gamma_{k,1}, \dots, \gamma_{k,k}\}$. Thus $C_0$ is finite, a contradiction; so (N) fails.

11.6. **Proof of Theorem 11.1.** (3)⇒(2): Fix $L$. The events $\{[jL, (j+1)L) \subset A\}$, $j \ge 1$, are independent, with probabilities $\prod_{i=jL}^{(j+1)L-1} p_i \ge \alpha^L q^L_{(j+1)L}$. Since $q_n \le q_{(j+1)L}$ for $(j+1)L \le n < (j+2)L$, we have $\sum_{j\ge1} q^L_{(j+1)L} \ge L^{-1} \sum_{n\ge 2L} q^L_n = \infty$, as $p^L_n \le \beta^L q^L_n$ and $\sum_n p^L_n = \infty$. By the second Borel–Cantelli lemma $A$ almost surely contains infinitely many intervals of length $L$; intersecting over $L$, $A$ is almost surely thick.

(2)⇒(1) is Lemma 10.2, and (1)⇒(5) is Theorem 11.8.

(5)⇒(3): Suppose $\sum_n p^k_n < \infty$ for some $k$, so $\sum_n q^k_n < \infty$. For $F \subset \mathbb{N}$ with $|F| = k$ and $\min F = f_0$, monotonicity gives $S_F = \sum_n w_F(n) \le \beta^k \sum_n q^k_{n+f_0} < \infty$, so $\mathbf{P}(\mathcal{E}_F) = 0$ by Theorem 11.3. Taking the union over the countably many $F$, almost surely $A$ contains no $B + F$ with $B$ infinite and $|F| = k$. This also proves the statement following the list.

(3)⇔(4): Both conditions are unchanged when $p$ is replaced by $q$. If $\sum_n q^k_n = S < \infty$, then $n q^k_n \le \sum_{m\le n} q^k_m \le S$, so $q_n \le (S/n)^{1/k}$ and $q_n n^\varepsilon \to 0$ for $\varepsilon < 1/k$. Conversely, if $q_n n^\varepsilon \to 0$, then $q_n \le n^{-\varepsilon}$ for large $n$ and $\sum_n q^k_n < \infty$ for $k > 1/\varepsilon$.

The 0–1 statements follow from Theorems 11.3 and 11.5. □

*Formalization.* Theorems 11.1 and 11.3 and Example 11.2 are formalized in `Kmrr25/S44a.lean` (`thm_monotone`, `thm_finite_int`, `ex_gap`), with the random set modelled by an infinite product of Bernoulli measures. Theorems 11.5, 11.6 and 11.8, Lemmas 11.9, 11.11 and 11.12 and Examples 11.10 and 11.14 to 11.16 are formalized in `Kmrr25/S44b.lean` (`thm_strategy`, `thm_strategy2_oplus`, `thm_strategy2_plus`, `thm_hierarchy`, `thm_hierarchy_strict`). For Theorem 11.7 the same file proves analyticity, the continuity of the reductions and the reduction property on trees (`thm_nonborel_a_BC`, `thm_nonborel_b_BC`, `thm_nonborel_c` and their variants); the classical facts that IF is $\Sigma^1_1$-complete and not Borel [16, Theorems 14.2 and 27.1] are stated as explicit hypotheses.

## 12. Ordered sumsets in the primes with summands in $\mathbb{P} - 1$

In this section we prove Conjecture 4.5 of [22]. It turns out to be a short corollary of the theorem of Tao and Ziegler on ascending chains of prime-producing tuples [30].

**Conjecture 4.5** ([22])**.** There exist infinite sets $B \subset \mathbb{N}$ and $C \subset \mathbb{P} - 1$ such that $\mathbb{P} \supset \{b + c : b \in B,\ c \in C;\ b < c\}$.

The survey remarks that the analogous statement with $B \subset \mathbb{P} - 1$ instead of $C \subset \mathbb{P} - 1$ follows from the results of [30] together with the existence of an infinite admissible subset of $\mathbb{P} - 1$. Following [30, Definition 1.1], a tuple $(h_1, \dots, h_k)$ of natural numbers is *prime-producing* if there are infinitely many $n$ such that $n + h_1, \dots, n + h_k$ are simultaneously prime.

**Theorem** (Tao–Ziegler [30, Theorem 1.5])**.** *There exists an infinite sequence $h_1 < h_2 < \dots$ of natural numbers, such that the $k$-tuple $(h_1, \dots, h_k)$ is prime-producing for every $k$.*

**Theorem 12.1.** *For all integers $s,t$ there are infinite sets $B,C\subset\mathbb{N}$ such that $C+t\subset\mathbb{P}$ and $b+c+s\in\mathbb{P}$ for all $b\in B$, $c\in C$ with $b<c$. In particular $(s=0,\ t=1)$ Conjecture 4.5 of [22] holds, and $(s=t=1)$ there are infinite $B,C$ with $C\subset\mathbb{P}-1$ and $\{b+c : b\in B,\ c\in C,\ b<c\}\subset\mathbb{P}-1$.*

*Proof.* Let $h_1<h_2<\cdots$ be as in the Tao–Ziegler theorem and put $\beta_i=h_{i+1}-h_1+t-s$; fix $i_0$ with $\beta_i\geq 1$ for $i\geq i_0$. For $k\geq i_0$ let

$$M_k=\{m\in\mathbb{N}:\ m+t\in\mathbb{P}\text{ and }m+s+\beta_i\in\mathbb{P}\text{ for }i_0\leq i\leq k\}.$$

Each $M_k$ is infinite: there are infinitely many $n$ with $n+h_1,\dots,n+h_{k+1}$ all prime; they satisfy $n\geq 2-h_1$, so infinitely many of them exceed $t-h_1$, and for those $m=n+h_1-t\in\mathbb{N}$ satisfies $m+t=n+h_1\in\mathbb{P}$ and $m+s+\beta_i=n+h_{i+1}\in\mathbb{P}$.

Now choose $i_1=i_0$ and inductively $c_j\in M_{i_j}$ with $c_j>\beta_{i_j}$, and then $i_{j+1}>i_j$ with $\beta_{i_{j+1}}>c_j$ (possible since $\beta_i\to\infty$). Then $\beta_{i_1}<c_1<\beta_{i_2}<c_2<\cdots$. Let $B=\{\beta_{i_j}:j\in\mathbb{N}\}$ and $C=\{c_j:j\in\mathbb{N}\}$. Then $C+t\subset\mathbb{P}$. If $b=\beta_{i_l}$ and $c=c_j$ with $b<c$, the interlacing gives $l\leq j$, so $i_0\leq i_l\leq i_j$, and $c_j\in M_{i_j}$ yields $b+c+s=c_j+s+\beta_{i_l}\in\mathbb{P}$. □

*Formalization.* Theorem 12.1 is formalized in `Kmrr25/C45.lean` (`c45_of_TZ`, `conjecture45_of_TZ`), with the Tao–Ziegler theorem stated as an explicit hypothesis.

*Remark* 12.2. Theorem 12.1 is a short consequence of [30, Theorem 1.5], and all the substance lies in the Tao–Ziegler theorem. The only observation is that being prime-producing is invariant under translation, so the chain can be translated to contain 1; then every witness $m$ satisfies $m+1\in\mathbb{P}$. The variant mentioned in the survey, with $B\subset\mathbb{P}-1$ in place of $C\subset\mathbb{P}-1$, is equally a consequence of the argument of [30]. There, Proposition 3.1 (the Maynard sieve input) is stated for an arbitrary admissible tuple, and in the proof of Theorem 1.5 the odd squares serve only to make every finite subtuple admissible. Running the same proof with the odd squares replaced by an infinite admissible subset of $\mathbb{P}-1$ gives a chain inside $\mathbb{P}-1$, as indicated in the introduction of [30]; Corollary 1.6 there and the interlacing above then give the variant. So Conjecture 4.5 and this variant are both short corollaries of [30]. Conjecture 4.5 uses only the statement of Theorem 1.5, whereas the variant re-runs its proof.

Neither argument gives the two conditions at once. To get $B,C\subset\mathbb{P}-1$ with $b+c\in\mathbb{P}$ for $b<c$ in the same way, one would need a prime-producing chain with entries in $\mathbb{P}-1$ whose translate by $1-h_1$ also has its entries in $\mathbb{P}-1$, and the argument of [30] does not obviously provide this. The statement that seems intended, the ordered analogue of [22, Conjecture 4.3] with $k=2$, also needs this: it asks for $B\cup C\cup\{b+c:b<c\}\subset\mathbb{P}-1$, which is the case $s=t=1$ of Theorem 12.1 with the additional requirement $B\subset\mathbb{P}-1$. We do not know whether these strengthenings hold.

## 13. Unrestricted sums $B+B$ and lower Banach density

In this section we answer, negatively, both parts of a question of Kousek and Radić [19, Question 5.2] about lower Banach density thresholds for unrestricted sumsets $B+B=\{b+b':b,b'\in B\}$; this question is not from [22]. We show that the lower Banach density thresholds are exactly $\frac{1}{2}$ (with a shift) and $\frac{3}{4}$ (without a shift), the same as the lower density thresholds found in [19].

For $A\subset\mathbb{N}$ let

$$\mathsf{d}_*(A)=\liminf_{N-M\to\infty}\frac{|A\cap[M,N]|}{N-M}$$

denote the lower Banach density, with the normalisation of [19]; $\mathbb{N}_0=\mathbb{N}\cup\{0\}$. Kousek and Radić proved that $\underline{\mathsf{d}}(A)>\frac{3}{4}$ implies $B+B\subset A$ for some infinite $B\subset\mathbb{N}$ [19, Theorem 1.2], and

that $\underline{\mathsf{d}}(A) > \frac{1}{2}$ implies $B+B+t \subset A$ for some infinite $B$ and some $t \in \{0,1\}$ [19, Theorem 1.3]. They also constructed a syndetic set of lower Banach density $\frac{1}{3}$ containing no $B+B+t$, and, by adding the odd numbers, a set of lower Banach density $\frac{2}{3}$ containing no $B+B$ [19, Proposition 5.1 and the discussion after it], and asked:

**Question** (Kousek–Radić [19, Question 5.2]). *Let $A \subset \mathbb{N}$.*

(1) *If $\mathsf{d}_*(A) > 1/3$, does there exist an infinite set $B \subset \mathbb{N}$ and a number $t \in \mathbb{N}$ so that $B+B+t \subset A$?*
(2) *If $\mathsf{d}_*(A) > 2/3$, does there exist an infinite set $B \subset \mathbb{N}$ so that $B+B \subset A$?*

**Theorem 13.1.** *Let $\alpha \in \mathbb{R} \setminus \mathbb{Q}$, let $\kappa > 0$ satisfy $\kappa \log 2 \equiv \frac{1}{2} \pmod 1$ (for instance $\kappa = 1/(2\log 2)$), let $0 < \delta < \frac{1}{4}$ and $c = \frac{1}{2} - 2\delta$. Put $\varphi(n) = n\alpha - \kappa \log n$ and*

$$E = \{n \in \mathbb{N} : \ \{\varphi(n)\} \in [0,c)\}, \qquad A' = (2\mathbb{N}-1) \cup 2E.$$

(a) *$E$ has uniform density $c$: for every $\varepsilon > 0$ there is $H_0$ such that $\big||E \cap [M, M+H)| - cH\big| \le \varepsilon H$ for all $M \ge 1$ and $H \ge H_0$. In particular $\mathsf{d}_*(E) = \mathsf{d}^*(E) = \frac{1}{2} - 2\delta$, and similarly $\mathsf{d}_*(A') = \mathsf{d}^*(A') = \frac{3}{4} - \delta$.*
(b) *For every $t \in \mathbb{Z}$ and every infinite $B \subset \mathbb{N}_0$, $B+B+t \not\subset E$.*
(c) *For every infinite $B \subset \mathbb{N}$, $B+B \not\subset A'$.*

**Corollary 13.2.** *Both parts of Question 5.2 of* [19] *have negative answers (take $0 < \delta < \frac{1}{12}$). Moreover*

$$\sup\{\mathsf{d}_*(A) : \ B+B+t \not\subset A \text{ for all } t \ge 0 \text{ and all infinite } B\} = \tfrac{1}{2},$$
$$\sup\{\mathsf{d}_*(A) : \ B+B \not\subset A \text{ for all infinite } B\} = \tfrac{3}{4}.$$

*Proof.* The lower bounds come from Theorem 13.1. For the upper bounds note that $\underline{\mathsf{d}}(A) \ge \mathsf{d}_*(A)$ (take $M = 1$ in the definition). Hence $\mathsf{d}_*(A) > \frac{1}{2}$ gives $B+B+t \subset A$ for some infinite $B$ and $t \in \{0,1\}$ by [19, Theorem 1.3], and $\mathsf{d}_*(A) > \frac{3}{4}$ gives $B+B \subset A$ by [19, Theorem 1.2]. □

*Remark* 13.3. The obstruction is the logarithmic-scale mechanism used in [22, Example 3.6] and in [19, Proposition 5.1]: for $b'$ fixed and $b \to \infty$, the numbers $2b+t$ and $b+b'+t$ differ by a factor close to 2, so the phase $\kappa \log n$ of the two differs by about $\frac{1}{2}$. We combine it with an irrational rotation $n\alpha$, which makes the set uniformly dense (the logarithmic obstruction alone has lower Banach density 0), while the rotation phases of $2b+t$ and $b+b'+t$ still nearly agree along a subsequence of $B$ on which $b\alpha$ converges, because both have total coefficient 2 on $B$. The proof uses the diagonal element $2b+t$. The sharpness examples of [19, Proposition 4.2] for lower density contain arbitrarily long gaps, so they have lower Banach density 0; Corollary 13.2 shows that lower Banach density gives no improvement over lower density. We do not know whether the boundary values are attained, that is, whether some $A$ with $\mathsf{d}_*(A) = \frac{1}{2}$ contains no $B+B+t$, or some $A$ with $\mathsf{d}_*(A) = \frac{3}{4}$ contains no $B+B$.

**Lemma 13.4.** *For $L \in \mathbb{N}$ let*

$$D_L = \sup_{\gamma \in \mathbb{R}} \sup_{0 \le a \le b \le 1} \frac{1}{L}\Big|\#\{0 \le j < L : \ \{j\alpha + \gamma\} \in [a,b)\} - (b-a)L\Big|.$$

*Then $D_L \to 0$ as $L \to \infty$.*

*Proof.* Let $K \in \mathbb{N}$. By Weyl's equidistribution theorem, for each of the $K$ arcs $I_i = [\frac{i}{K}, \frac{i+1}{K})$ of $\mathbb{T}$ we have $\#\{0 \le j < L : j\alpha \bmod 1 \in I_i\} = L/K + o(L)$, so there is $L_0$ such that for $L \ge L_0$ each of these counts lies in $[(1-\frac{1}{K})\frac{L}{K}, (1+\frac{1}{K})\frac{L}{K}]$. Now $\{j\alpha+\gamma\} \in [a,b)$ if and only if $j\alpha \bmod 1$ lies in the arc $J = [a-\gamma, b-\gamma)$ of length $b-a$. The arc $J$ contains at least $K(b-a)-2$ and

is covered by at most $K(b-a)+2$ of the arcs $I_i$. Hence for $L \ge L_0$ the count lies between $(K(b-a)-2)(1-\frac{1}{K})\frac{L}{K}$ and $(K(b-a)+2)(1+\frac{1}{K})\frac{L}{K}$, so $D_L \le 4/K$. □

*Proof of Theorem 13.1*(a). Let $0 < \varepsilon < \min(c, 1-c)$, choose $L$ with $D_L \le \varepsilon/4$, put $K = 4\kappa/\varepsilon$, and let $H \ge H_0 = 8(K+1)L/\varepsilon$. Let $W = [M, M+H)$ with $M \ge 1$. At most $KL \le \varepsilon H/8$ integers of $W$ are smaller than $KL$. Split the rest of $W$ into consecutive blocks $\{m, \dots, m+L-1\}$, $m \ge KL$, and a remainder of fewer than $L \le \varepsilon H/8$ integers. For $n = m+j$, $0 \le j < L$, we have

$$\varphi(n) = j\alpha + \gamma_m - \eta_n, \qquad \gamma_m = m\alpha - \kappa \log m, \qquad \eta_n = \kappa \log \frac{m+j}{m} \in \Big[0, \frac{\kappa L}{m}\Big) \subset \Big[0, \frac{\varepsilon}{4}\Big).$$

For $x \in \mathbb{R}$ and $0 \le \eta < \varepsilon/4$ we have, since $c + \varepsilon/4 < 1$,

$$\{x\} \in [\tfrac{\varepsilon}{4}, c) \implies \{x-\eta\} \in [0, c) \implies \{x\} \in [0, c+\tfrac{\varepsilon}{4}).$$

Applying this with $x = j\alpha + \gamma_m$ and using Lemma 13.4, each block contains between $L(c-\frac{\varepsilon}{2})$ and $L(c+\frac{\varepsilon}{2})$ elements of $E$. If $P$ denotes the number of integers of $W$ lying in blocks, then $(1-\frac{\varepsilon}{4})H \le P \le H$, and therefore

$$(c-\tfrac{\varepsilon}{2})(1-\tfrac{\varepsilon}{4})H \;\le\; |E \cap W| \;\le\; (c+\tfrac{\varepsilon}{2})H + \tfrac{\varepsilon}{4}H,$$

so $\big||E \cap W| - cH\big| \le \varepsilon H$. For $A'$: a window $[M, M+H)$ contains $\frac{H}{2} + O(1)$ odd numbers, and its even elements of $A'$ are the numbers $2n$ with $n$ in $E \cap [\frac{M}{2}, \frac{M+H}{2})$, a window of length $\frac{H}{2} + O(1)$; so by the above they number $\frac{cH}{2} + o(H)$, uniformly in $M$. Hence $A'$ has uniform density $\frac{1+c}{2} = \frac{3}{4} - \delta$. □

*Proof of Theorem 13.1*(b). Let $\eta = \frac{1}{2} - c = 2\delta$, and suppose that $t \in \mathbb{Z}$ and $B \subset \mathbb{N}_0$ is infinite with $B + B + t \subset E$. By compactness of $\mathbb{T}$ there are an infinite $B_1 \subset B$ and $\beta \in \mathbb{T}$ with $b\alpha \to \beta$ in $\mathbb{T}$ as $b \to \infty$ in $B_1$. Choose $b' \in B_1$ with $\|b'\alpha - \beta\| < \eta/4$, where $\|\cdot\|$ is the distance to the nearest integer. Since $(2b+t)/(b+b'+t) \to 2$ as $b \to \infty$, we can choose $b \in B_1$, $b > b'$, with

$$\|b\alpha - \beta\| < \frac{\eta}{4} \qquad \text{and} \qquad |r| < \frac{\eta}{2}, \quad \text{where } r = \kappa\Big(\log \frac{2b+t}{b+b'+t} - \log 2\Big).$$

The numbers $m_1 = 2b + t = b + b + t$ and $m_2 = b + b' + t$ belong to $B + B + t \subset E$ (here the diagonal element $b + b$ is used). Let $x = \{\varphi(m_1)\}$ and $y = \{\varphi(m_2)\}$. Then $x, y \in [0, c)$, so $x - y \in (-c, c)$ and

$$\operatorname{dist}\big(x - y, \tfrac{1}{2} + \mathbb{Z}\big) \;\ge\; \tfrac{1}{2} - c \;=\; \eta.$$

On the other hand,

$$\varphi(m_1) - \varphi(m_2) = (b-b')\alpha - \kappa \log \frac{2b+t}{b+b'+t} = (b-b')\alpha - \kappa \log 2 - r,$$

and $\kappa \log 2 \in \frac{1}{2} + \mathbb{Z}$. Since $x - y \equiv \varphi(m_1) - \varphi(m_2) \pmod 1$, we get

$$\operatorname{dist}\big(x - y, \tfrac{1}{2} + \mathbb{Z}\big) \le \|(b-b')\alpha\| + |r| \le \|b\alpha - \beta\| + \|\beta - b'\alpha\| + |r| < \eta,$$

a contradiction. □

*Proof of Theorem 13.1*(c). Suppose $B \subset \mathbb{N}$ is infinite and $B + B \subset A'$. Choose $\sigma \in \{0, 1\}$ such that $B_\sigma = B \cap (2\mathbb{Z} + \sigma)$ is infinite, and let $C = \{(b-\sigma)/2 : b \in B_\sigma\} \subset \mathbb{N}_0$. For $c_1, c_2 \in C$ the number $(2c_1 + \sigma) + (2c_2 + \sigma) = 2(c_1 + c_2 + \sigma)$ is an even element of $A'$, so $c_1 + c_2 + \sigma \in E$. Thus $C + C + \sigma \subset E$, contradicting part (b). □

*Formalization.* Theorem 13.1 and the lower bounds in Corollary 13.2 are formalized in `Kmrr25/KR52.lean` (`main`, `q52_negative`, `sup_lower_bounds`). The upper bounds in Corollary 13.2 are Theorems 1.2 and 1.3 of [19] and are not formalized.

## Part 3. Sums and products

### 14. Sums and products in $\mathbb{N}$ and $\mathbb{Q}$

This section is about three questions from [22, §5.1] that ask for sums and products of infinite sets at the same time. In Section 14.1 we answer Question §5.1(a) positively: the two explicit sets named in [22, §5.1], namely Beatty sequences and the set of integers with an even number of prime factors, both contain $B+C$ and $BC$ for some infinite $B, C \subset \mathbb{N}$. In Section 14.2 we answer Question 5.4 of [22] negatively. The counterexample is a set built at logarithmic scale; it has logarithmic density $3/10$ and it is aperiodic. The same set answers Question 5.2 of [22] negatively if "positive density" is read as positive upper density (Remark 14.11). In Section 14.3 we answer Question 5.6 of [22] negatively, as it is literally stated. The counterexample comes from the sign obstruction at the real place, which survives the removal of a set of density zero.

*Notation.* For $x \in \mathbb{R}$ we write $\{x\} \in [0,1)$ for the fractional part of $x$ and $\|x\|$ for the distance from $x$ to the nearest integer. We identify $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ with $[0,1)$, and we write $\|x-y\|$ for the distance in $\mathbb{T}$. For $X, Y \subset \mathbb{N}$ (or $\mathbb{Q}$) we write $X+Y = \{x+y : x \in X,\, y \in Y\}$, $XY = \{xy : x \in X,\, y \in Y\}$, $X+t = \{x+t : x \in X\}$ and $dX = \{dx : x \in X\}$. Recall from [22, eq. (19)] that $A \subset \mathbb{N}$ is *aperiodic* if

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} \mathbf{1}_A(n)\, e(\alpha n) = 0 \qquad \text{for all } \alpha \in \mathbb{Q} \setminus \mathbb{Z}, \tag{27}$$

where $e(x) = e^{2\pi i x}$.

14.1. **The explicit examples.** Question 5.2 of [22] asks whether every aperiodic $A \subset \mathbb{N}$ of positive density contains $B+C$ and $BC$ for some infinite sets $B, C \subset \mathbb{N}$. Just before that question the survey names two typical aperiodic sets: the set of all numbers with an even number of prime factors, and the Beatty sequences $\{\lfloor n\alpha + \gamma \rfloor : n \in \mathbb{N}\}$, where $\alpha > 0$ is irrational and $\gamma \in \mathbb{R}$. It then remarks: "Even for the explicit examples we have mentioned, we consider the answer to be interesting."

**Question §5.1(a)** ([22])**.** Let $A$ be either a Beatty sequence $\{\lfloor n\alpha + \gamma \rfloor : n \in \mathbb{N}\}$, with $\alpha > 0$ irrational and $\gamma \in \mathbb{R}$, or the set of all $n \in \mathbb{N}$ with an even number of prime factors. Are there infinite sets $B, C \subset \mathbb{N}$ with $B + C \subset A$ and $BC \subset A$?

The survey does not say whether prime factors are counted with multiplicity. We answer the question for both readings. The answer is yes in all cases (Corollary 14.2 and Theorem 14.4).

14.1.1. *Bohr sets and Beatty sequences.*

**Theorem 14.1.** *Let $\vartheta \in \mathbb{R} \setminus \mathbb{Q}$, $\delta \in \mathbb{R}$, let $J \subset \mathbb{T}$ be a nonempty open set and let $M_0 \in \mathbb{N}$. Suppose that*

$$A \supseteq \{m \in \mathbb{N} :\ m \geq M_0,\ m\vartheta + \delta \bmod 1 \in J\}.$$

*Then there is an infinite set $Z \subset \mathbb{N}$ such that $z + z' \in A$ and $zz' \in A$ for all distinct $z, z' \in Z$. In particular, there are disjoint infinite sets $B, C \subset \mathbb{N}$ with $B + C \subset A$ and $BC \subset A$.*

*Proof.* Since $J$ is open and nonempty, there are $\xi_0 \in \mathbb{T}$ and $\ell \in (0,1)$ such that every $x \in \mathbb{T}$ with $\|x-\xi_0\| < \ell/2$ lies in $J$. Fix $q \in \mathbb{N}$ with $q \geq 4/\ell$. The points $2r/q$ ($r \in \mathbb{Z}$) form a subgroup of $\mathbb{T}$ in which consecutive points are at distance at most $2/q$. Hence we can choose $r \in \mathbb{Z}$ with $\|2r/q + \delta - \xi_0\| \leq 1/q \leq \ell/4$. Put $\xi = 2r/q + \delta$. Then

$$\|x - \xi\| < \ell/4 \implies \|x - \xi_0\| < \ell/2 \implies x \in J. \tag{28}$$

We choose $z_1 < z_2 < \cdots$ recursively. Put $z_0 = 1$, and suppose $z_0, \ldots, z_{n-1}$ have been chosen. Let $\varepsilon_n = \ell/(8z_{n-1})$. We look for $z_n$ of the form $z_n = qk + 2$ with $k \in \mathbb{N}$. Then $z_n\vartheta - r/q = k(q\vartheta) + (2\vartheta - r/q)$. Since $q\vartheta$ is irrational, the sequence $(kq\vartheta \bmod 1)_{k>K}$ is dense in $\mathbb{T}$ for every $K$. So we can choose $k$ such that $z_n = qk+2 > \max(z_{n-1}, M_0)$ and $\|z_n\vartheta - r/q\| < \varepsilon_n$. Thus every $n \geq 1$ satisfies

$$z_n \equiv 2 \pmod q, \qquad z_n\vartheta = K_n + \frac{r}{q} + e_n, \quad K_n \in \mathbb{Z},\ |e_n| < \varepsilon_n = \frac{\ell}{8z_{n-1}} \leq \frac{\ell}{8}.$$

Let $1 \leq i < j$. Write $z_i = qk_i + 2$ with $k_i \in \mathbb{Z}$. We compute the two phases modulo 1.

- Sums: $(z_i + z_j)\vartheta \equiv 2r/q + e_i + e_j$, and $|e_i + e_j| < \ell/4$.
- Products: $z_iz_j\vartheta = z_iK_j + z_ir/q + z_ie_j$. Here $z_iK_j \in \mathbb{Z}$ and $z_ir/q = k_ir + 2r/q \equiv 2r/q$. Moreover $|z_ie_j| < z_i\ell/(8z_{j-1}) \leq \ell/8$, because $z_i \leq z_{j-1}$. So $z_iz_j\vartheta \equiv 2r/q + z_ie_j$.

Hence both $(z_i+z_j)\vartheta+\delta$ and $z_iz_j\vartheta+\delta$ lie within distance $\ell/4$ of $\xi$ in $\mathbb{T}$, so they lie in $J$ by (28). Since $z_i+z_j$ and $z_iz_j$ are at least $M_0$, both lie in $A$. Take $Z = \{z_n : n \geq 1\}$, $B = \{z_1, z_3, z_5, \ldots\}$ and $C = \{z_2, z_4, z_6, \ldots\}$. □

**Corollary 14.2.** *Let $\alpha > 0$ be irrational, let $\gamma \in \mathbb{R}$, and let $A = \{\lfloor n\alpha + \gamma \rfloor : n \in \mathbb{N}\}$. Then there are infinite sets $B, C \subset \mathbb{N}$ with $B + C \subset A$ and $BC \subset A$.*

*Proof.* Suppose first that $\alpha < 1$. Then consecutive terms of the nondecreasing sequence $\lfloor n\alpha+\gamma \rfloor$ differ by 0 or 1, and the sequence tends to infinity. So $A$ contains every integer $m \geq \lfloor \alpha + \gamma \rfloor$, and any infinite sets $B, C$ of sufficiently large integers work.

Now suppose $\alpha > 1$ and put $\beta = 1/\alpha \in (0,1)$. An integer $m$ lies in $A$ if and only if some $n \in \mathbb{N}$ satisfies $m \leq n\alpha + \gamma < m + 1$, that is, $y_m \leq n < y_m + \beta$, where $y_m = (m - \gamma)\beta$. The least integer that is at least $y$ is $y + \{-y\}$. So the interval $[y_m, y_m + \beta)$ contains an integer if and only if $\{-y_m\} < \beta$. When $m > \gamma$, this integer is at least $y_m > 0$, so it lies in $\mathbb{N}$. Therefore, for $m > \gamma$,

$$m \in A \iff \{m\vartheta + \delta\} \in [0, \beta), \qquad \text{where } \vartheta = -\beta,\ \delta = \gamma\beta.$$

Here $\vartheta$ is irrational. Apply Theorem 14.1 with $J = (0, \beta)$ and any $M_0 > \gamma$. □

*Remark* 14.3. Theorem 14.1 does not use any density assumption. It applies to every set that contains an inhomogeneous Bohr set with an irrational frequency, up to finitely many elements; this includes all Beatty sequences. The argument is elementary, and we expect the statement to be folklore. We did not find it in the literature.

14.1.2. *The Liouville function.* Let $\Omega(n)$ and $\omega(n)$ be the numbers of prime factors of $n$ counted with and without multiplicity, respectively. Let $\lambda(n) = (-1)^{\Omega(n)}$ be the Liouville function. It is completely multiplicative: $\lambda(mn) = \lambda(m)\lambda(n)$ for all $m, n \in \mathbb{N}$. Put

$$L_+ = \{n \in \mathbb{N} : \Omega(n) \text{ is even}\} = \{n : \lambda(n) = 1\}, \qquad L'_+ = \{n \in \mathbb{N} : \omega(n) \text{ is even}\}.$$

**Theorem 14.4.** *For each $E \in \{L_+, L'_+\}$ there are infinite sets $B, C \subset \mathbb{N}$ with $B + C \subset E$ and $BC \subset E$.*

*Proof.* Let $S$ be the set of primes $p \equiv 5 \pmod 6$. There are infinitely many such primes: if $p_1, \dots, p_r$ were all of them, then $6p_1 \cdots p_r - 1 \equiv 5 \pmod 6$ would be coprime to $6p_1 \cdots p_r$; so all its prime factors would be $\equiv 1 \pmod 6$, which is impossible. Let $E \in \{L_+, L'_+\}$, and colour each two-element subset $\{s, s'\}$ of $S$ according to whether $s + s' \in E$. By the infinite Ramsey theorem for pairs [28] there is an infinite set $S' \subset S$ such that either $s + s' \in E$ for all distinct $s, s' \in S'$, or $s + s' \notin E$ for all distinct $s, s' \in S'$. Split $S'$ into two disjoint infinite sets $B_0$ and $C_0$, and let $b_0 \in B_0$, $c_0 \in C_0$. Then $b_0 \neq c_0$ are primes different from 2 and 3, and $b_0 + c_0$ is even and $\equiv 1 \pmod 3$. Also $\Omega(b_0c_0) = \omega(b_0c_0) = 2$, so $b_0c_0 \in L_+ \cap L'_+$.

If $s + s' \in E$ for all distinct $s, s' \in S'$, then $B = B_0$ and $C = C_0$ work. Otherwise we put $B = dB_0$ and $C = dC_0$, where $d = 2$ if $E = L_+$ and $d = 6$ if $E = L'_+$.

- $E = L_+$: $\Omega(2(b_0 + c_0)) = \Omega(b_0 + c_0) + 1$ is even, and $\Omega(4b_0c_0) = 4$.
- $E = L'_+$: since $2 \mid b_0 + c_0$ and $3 \nmid b_0 + c_0$, we get $\omega(6(b_0 + c_0)) = \omega(b_0 + c_0) + 1$, which is even. Since $b_0, c_0 \notin \{2, 3\}$, $\omega(36b_0c_0) = 4$.

In both cases $b + c \in E$ and $bc \in E$ for all $b = db_0 \in B$ and $c = dc_0 \in C$. □

For $L_+$, the same idea with Hindman's theorem in place of Ramsey's theorem gives the following generalization.

**Proposition 14.5.** *Let $G$ be a finite abelian group, and let $f\colon \mathbb{N} \to G$ be completely multiplicative, i.e. $f(mn) = f(m)f(n)$ for all $m, n \in \mathbb{N}$. Then there are infinite sets $B, C \subset \mathbb{N}$ with $f(b + c) = f(bc) = 1_G$ for all $b \in B$ and $c \in C$.*

*Proof.* Hindman's theorem [12, Theorem 3.1], applied to the finite colouring $n \mapsto f(n)$, gives $h \in G$ and a sequence $(x_i)_{i \in \mathbb{N}}$ in $\mathbb{N}$ such that $f$ takes the value $h$ on $\mathrm{FS}((x_i)) = \{x_{i_1} + \cdots + x_{i_k} : k \in \mathbb{N},\ i_1 < \cdots < i_k\}$. We may assume that $x_1 < x_2 < \cdots$. Indeed, if $(x_i)$ is unbounded we pass to a strictly increasing subsequence, whose finite sums form a subset of $\mathrm{FS}((x_i))$. Otherwise some value $x$ occurs infinitely often, so $\mathrm{FS}((x_i))$ contains every multiple of $x$, and we replace $(x_i)$ by $(2^i x)$. Let $B_0 = \{x_1, x_3, \dots\}$ and $C_0 = \{x_2, x_4, \dots\}$. Then $f$ equals $h$ on $B_0$, on $C_0$ and on $B_0 + C_0$. The image $f(\mathbb{N})$ is a nonempty subset of the finite group $G$ that is closed under multiplication, hence a subgroup. So there is $d \in \mathbb{N}$ with $f(d) = h^{-1}$. Put $B = dB_0$ and $C = dC_0$. For $b_0 \in B_0$ and $c_0 \in C_0$ we get $f(db_0 + dc_0) = f(d)f(b_0 + c_0) = h^{-1}h = 1_G$ and $f(db_0 \cdot dc_0) = f(d)^2 f(b_0)f(c_0) = h^{-2}h^2 = 1_G$. □

For instance, for every $k \geq 2$ the set $\{n : \Omega(n) \equiv 0 \pmod k\}$ contains $B + C$ and $BC$ for some infinite $B, C \subset \mathbb{N}$.

*Formalization.* Theorems 14.1 and 14.4 and Corollary 14.2 are formalized in `Kmrr25/S51a.lean` (`bohr`, `beatty_sum_prod`, `liouville`). Proposition 14.5 is not formalized.

*Remark* 14.6. Corollary 14.2 and Theorem 14.4 answer Question §5.1(a) for both examples named in the survey. Question 5.2 of [22] itself remains open when "positive density" means that the natural density exists and is positive (see also Remark 14.11).

14.2. **Sums and shifted products at logarithmic scale.** Logarithmic density is defined in [22, Definition 3.22]: for $A \subset \mathbb{N}$,

$$\overline{\mathrm{d}}_{\log}(A) = \limsup_{N \to \infty} \frac{1}{\log N} \sum_{n=1}^{N} \frac{\mathbf{1}_A(n)}{n},$$

with the lower version defined by $\liminf$. When the two agree, their common value is the logarithmic density $\mathrm{d}_{\log}(A)$. Question 5.4 of [22] reads as follows.

**Question 5.4** ([22])**.** Suppose $A \subset \mathbb{N}$ has positive upper logarithmic density. Are there infinite sets $B, C \subset \mathbb{N}$ with $B + C \subset A$ and $BC + 1 \subset A$?

The survey introduces the shift $BC + 1$ because the unshifted question fails for congruence (local) reasons: the odd numbers contain no $B+C$ together with $BC$, since $b+c$ and $bc$ cannot both be odd. The survey also notes that Question 5.4 generalizes an earlier question of Moreira about singletons $B = \{x\}$, $C = \{y\}$. Taken literally, the singleton version is trivial ($x = 1$, $y = a - 1$ with $a \in A$); with $x, y \geq 2$ it is Moreira's conjecture, which Richter proved for every set of positive upper logarithmic density [29, Theorem 1.5 with $Q(y) = y - 1$]. Indeed, that theorem gives $x, y \geq 2$ with $x + y - 1 \in A - 1$ and $xy \in A - 1$, i.e. $x + y \in A$ and $xy + 1 \in A$, and $A - 1$ has the same upper logarithmic density as $A$. Richter also relates his results to Questions 5.2 and 5.4 of [22]. We show that the infinite version fails, because of an obstruction of a different kind: an archimedean one. Our set is aperiodic in the sense of (27), so it has no congruence obstruction.

Throughout this subsection let $I = [\frac{7}{20}, \frac{13}{20})$, let $\psi(n) = \log_2 n \bmod 1 \in \mathbb{T}$ for $n \in \mathbb{N}$, and let

$$A = \{n \in \mathbb{N} : \{\log_2 n\} \in I\} = \bigcup_{k \geq 0} \left[2^{k+7/20},\, 2^{k+13/20}\right) \cap \mathbb{N}.$$

Membership is decided in exact arithmetic: $n \in A$ if and only if $2^{20k+7} \leq n^{20} < 2^{20k+13}$, where $k = \lfloor \log_2 n \rfloor$.

**Proposition 14.7.** *The set $A$ has the following properties.*

(i) *Its logarithmic density exists and $\mathsf{d}_{\log}(A) = 3/10$.*
(ii) *It is aperiodic, i.e. it satisfies (27).*
(iii) $\overline{\mathsf{d}}(A) = 2 - 2^{7/10} \approx 0.3755$ *and* $\underline{\mathsf{d}}(A) = 2^{3/10} - 1 \approx 0.2311$. *In particular, the natural density of $A$ does not exist.*

*Proof.* Write $a = 7/20$ and $b = 13/20$. For real $1 \leq u < v$, comparing with $\int_u^v dx/x$ gives $\sum_{u \leq n < v} 1/n = \log(v/u) + O(1/u)$. So the $k$-th block contributes $(b - a)\log 2 + O(2^{-k})$ to $\sum \mathbf{1}_A(n)/n$.

(i) Let $2^K \leq N < 2^{K+1}$. Summing over the blocks with $k < K$, and bounding the contribution of the block $k = K$ by $(b-a)\log 2 + O(2^{-K})$, we get $\sum_{n \leq N} \mathbf{1}_A(n)/n = \frac{3}{10} K \log 2 + O(1)$. Since $\log N = K \log 2 + O(1)$, the claim follows.

(ii) Let $\alpha \in \mathbb{Q} \setminus \mathbb{Z}$. For integers $m \leq m'$ we have $|\sum_{n=m}^{m'} e(\alpha n)| \leq 1/|\sin \pi\alpha|$. The set $A \cap [1, N]$ is a union of at most $\log_2 N + 1$ intervals of consecutive integers. Hence $|\frac{1}{N} \sum_{n \leq N} \mathbf{1}_A(n) e(\alpha n)| \leq (\log_2 N + 1)/(N|\sin \pi\alpha|) \to 0$.

(iii) Let $N = 2^{K+s}$ with $K \in \mathbb{N}$ and $s \in [0, 1)$. Counting the blocks with $k < K$ and the part of block $K$ below $N$ gives

$$|A \cap [1, N]| = 2^K\big(2^b - 2^a + g(s)\big) + O(K), \qquad g(s) = \max\big(0, \min(2^s, 2^b) - 2^a\big).$$

So $|A \cap [1, N]|/N = \rho(s) + o(1)$, uniformly in $s$, where $\rho(s) = (2^b - 2^a + g(s))/2^s$. On $[0, a]$ the function $\rho(s) = (2^b - 2^a)2^{-s}$ decreases. On $[a, b]$ it equals $1 - (2^{a+1} - 2^b)2^{-s}$, which increases because $2^{a+1} > 2^b$. On $[b, 1)$ it equals $2(2^b - 2^a)2^{-s}$, which decreases. Therefore $\min \rho = \rho(a) = 2^{b-a} - 1 = 2^{3/10} - 1$ and $\sup \rho = \rho(b) = 2 - 2^{a+1-b} = 2 - 2^{7/10}$. Taking $N = 2^{K+a}$ and $N = 2^{K+b}$ (rounded down) as $K \to \infty$ shows that these values are $\underline{\mathsf{d}}(A)$ and $\overline{\mathsf{d}}(A)$. □

Let $\varepsilon = \log_2(1 + 1/58)$. Since $\log(1 + x) \leq x$ and $\log 2 > 0.693$, we have $\varepsilon \leq 1/(58 \log 2) < 0.0249$. The constant 58 is the least $M \in \mathbb{N}$ with $2\log_2(1 + 1/M) < 1/20$; this is the margin needed in Theorem 14.9(a).

**Lemma 14.8.** *If $x, y \in \mathbb{N}$ satisfy $y \geq 58x$ and $x + y \in A$, then $\{\log_2 y\} \in [\frac{7}{20} - \varepsilon, \frac{13}{20})$.*

*Proof.* We have $\log_2(x + y) = \log_2 y + \varepsilon'$ with $0 < \varepsilon' = \log_2(1 + x/y) \leq \varepsilon < \frac{7}{20}$. Write $\log_2(x + y) = k + \theta$ with $k \in \mathbb{Z}$ and $\theta = \{\log_2(x + y)\} \in I$. Then $\log_2 y = k + (\theta - \varepsilon')$, and $0 \leq \theta - \varepsilon' < 1$ because $\theta \geq \frac{7}{20} > \varepsilon'$. Hence $\{\log_2 y\} = \theta - \varepsilon' \in [\frac{7}{20} - \varepsilon, \frac{13}{20})$. □

**Theorem 14.9.**

(a) *There are no $b_0, c_0, b, c \in \mathbb{N}$ and $\tau \in \{0, 1\}$ with $b \geq 58c_0$ and $c \geq 58b_0$ such that*

$$b_0 + c \in A, \qquad c_0 + b \in A, \qquad bc + \tau \in A.$$

(b) *Let $t_1, t_2 \in \mathbb{Z}$. There are no infinite sets $B, C \subset \mathbb{N}$ with $b + c + t_1 \in A$ and $bc + t_2 \in A$ for all $b \in B$ and $c \in C$.*

*In particular, Question 5.4 has a negative answer: $\mathsf{d}_{\log}(A) = 3/10$, but there are no infinite sets $B, C \subset \mathbb{N}$ with $B + C \subset A$ and $BC + 1 \subset A$.*

*Proof.* (a) By Lemma 14.8, applied to $(x, y) = (b_0, c)$ and to $(x, y) = (c_0, b)$,

$$\{\log_2 c\},\ \{\log_2 b\} \in \left[\tfrac{7}{20} - \varepsilon,\ \tfrac{13}{20}\right).$$

Write $\log_2(bc + \tau) = \log_2 b + \log_2 c + \eta$, where $\eta = \log_2(1 + \tau/(bc))$. Since $bc \geq 58^2$, we have $0 \leq \eta \leq 1/(58^2 \log 2) < 0.001$. Let $u = \{\log_2 b\} + \{\log_2 c\} + \eta$. Then $\log_2(bc + \tau) - u \in \mathbb{Z}$, so $\{\log_2(bc + \tau)\} = \{u\}$, and $u \in [\frac{7}{10} - 2\varepsilon,\ \frac{13}{10} + \eta)$. If $u < 1$, then $\{u\} = u \geq \frac{7}{10} - 2\varepsilon > \frac{13}{20}$, because $2\varepsilon < 0.05$. If $u \geq 1$, then $\{u\} = u - 1 < \frac{3}{10} + \eta < \frac{7}{20}$. In both cases $\{\log_2(bc + \tau)\} \notin I$, so $bc + \tau \notin A$.

(b) Suppose such $B$ and $C$ exist. Fix $b_0 \in B$. For $c \in C$, $c \to \infty$, the distance in $\mathbb{T}$ between $\psi(c + b_0 + t_1)$ and $\psi(c)$ is at most $|\log_2(1 + (b_0 + t_1)/c)|$, which tends to 0. Since $\psi(c + b_0 + t_1) \in I$, every limit point of $\psi(c)$ as $c \to \infty$ in $C$ lies in the closure $\bar{I} = [\frac{7}{20}, \frac{13}{20}]$. By symmetry the same holds for $\psi(b)$ as $b \to \infty$ in $B$. Choose $b_n \in B$ and $c_n \in C$ with $b_n, c_n \to \infty$. Since $\mathbb{T}$ is compact, after passing to a subsequence we may assume that $\psi(b_n) \to \beta$ and $\psi(c_n) \to \gamma$, where $\beta, \gamma \in \bar{I}$. Now $\psi(b_n c_n + t_2) = \psi(b_n) + \psi(c_n) + \log_2(1 + t_2/(b_n c_n)) \to \beta + \gamma$ in $\mathbb{T}$. But $\beta + \gamma \in [\frac{7}{10}, \frac{13}{10}]$, and modulo 1 this set is $[\frac{7}{10}, 1) \cup [0, \frac{3}{10}]$, which is at distance $\frac{1}{20}$ from $\bar{I}$. On the other hand $\psi(b_n c_n + t_2) \in I$ for all $n$, so $\beta + \gamma \in \bar{I}$. This is a contradiction.

For the final statement, recall $\mathsf{d}_{\log}(A) = 3/10$ from Proposition 14.7(i). If $B, C \subset \mathbb{N}$ were infinite with $B + C \subset A$ and $BC + 1 \subset A$, we could fix $b_0 \in B$ and $c_0 \in C$ and then pick $b \in B$ with $b \geq 58c_0$ and $c \in C$ with $c \geq 58b_0$, contradicting (a) with $\tau = 1$. (It also follows from (b) with $t_1 = 0$, $t_2 = 1$.) □

*Formalization.* Lemma 14.8, both parts of Theorem 14.9 and Proposition 14.7(i) are formalized in `Kmrr25/Q54.lean` (`no_ratio58_config`, `no_shifted_config`, `q54_negative`); parts (ii) and (iii) of Proposition 14.7 are not.

*Remark* 14.10. (i) The obstruction is asymptotic, and it needs both $B$ and $C$ to be infinite. If $B = \{1\}$, then $B + C = BC + 1 = C + 1$, so $C = (A - 1) \cap \mathbb{N}$ works. By Richter's theorem quoted above, every set of positive upper logarithmic density, in particular $A$, contains $\{b + c, bc + 1\}$ with $b, c \geq 2$. Larger finite configurations also occur in $A$: for $B = \{1, 2, 4, 8, 16\}$ and $C = \{1306, 1307, \ldots, 1310\}$, all 25 sums $b + c$ and all 25 numbers $bc + 1$ lie in $A$. This is a direct check of 50 memberships with the exact test above. Part (a) of Theorem 14.9 is a quantitative form of the obstruction: it rules out a four-element configuration in which $b, c$ are large compared with $c_0, b_0$.

(ii) By Theorem 14.9(b), shifting the sums or the products by any fixed integers does not remove the obstruction. Unlike the congruence obstruction to the unshifted question exhibited

by the odd numbers, it is not local: $A$ is aperiodic (Proposition 14.7(ii)). It is also thick (the blocks have unbounded length), so $\mathsf{d}^*(A) = 1$. Similar constructions at logarithmic scale appear throughout §§2–3 of the survey [22], where they serve as counterexamples for sumsets $B+B+t$; Richter's Example 1.6 in [29] is another block-type construction of this kind, a set of positive upper density without $\{x+y-1, xy\}$, $x, y \geq 2$. Here the same idea is applied to infinite sums and shifted products. To our knowledge this counterexample is new.

(iii) The set $A$ is not syndetic. The survey also asks Question 5.4 for syndetic $A$, and Question 5.5 of [22] (whether every syndetic set contains $BC$ with $B, C$ infinite); both remain open.

Recall Question 5.2 of [22].

**Question 5.2** ([22])**.** Fix $A \subset \mathbb{N}$ that has positive density and is aperiodic. Are there infinite sets $B, C \subset \mathbb{N}$ with $B + C \subset A$ and $BC \subset A$?

*Remark* 14.11. Taking $\tau = 0$ in Theorem 14.9(a), or $t_1 = t_2 = 0$ in (b), shows that $A$ contains no $B + C$ together with $BC$ for infinite $B, C \subset \mathbb{N}$. By Proposition 14.7, $A$ is aperiodic and has positive upper density (and positive logarithmic and upper Banach density), but its natural density does not exist. So $A$ gives a negative answer to Question 5.2 if "positive density" there is read as positive upper density (or positive upper Banach or logarithmic density). This reading is consistent with the survey's text, which says that (19) is equivalent to equidistribution in residue classes "if the density of $A$ exists". It does not settle Question 5.2 when the natural density is required to exist; that case remains open.

14.3. **Sums and products in $\mathbb{Q}$.** Following [22, §5.1], where the notion is attributed to [6, Definition 1.3], we use the formulation of [18, Definition 1.7]: a *double Følner sequence* in $\mathbb{Q}$ is a sequence $\Phi = (F_N)_{N\in\mathbb{N}}$ of nonempty finite subsets of $\mathbb{Q}$ such that

$$\lim_{N\to\infty} \frac{|F_N \cap (x + F_N)|}{|F_N|} = 1 \quad (x \in \mathbb{Q}) \qquad \text{and} \qquad \lim_{N\to\infty} \frac{|F_N \cap xF_N|}{|F_N|} = 1 \quad (x \in \mathbb{Q}^*).$$

The survey describes $\Phi$ as simultaneously a Følner sequence in $(\mathbb{Q}, +)$ and in $(\mathbb{Q}^*, \times)$; strictly speaking the latter requires $F_N \subset \mathbb{Q}^*$. This makes no difference below: 0 is a single point and $|F_N| \to \infty$ (Lemma 14.12(i)), so all densities are unchanged if 0 is removed from every $F_N$. For $A \subset \mathbb{Q}$ we write $\overline{\mathsf{d}}_\Phi(A) = \limsup_N |A \cap F_N|/|F_N|$. When the limit exists we denote it by $\mathsf{d}_\Phi(A)$, as in [22, §1].

**Question 5.6** ([22])**.** Let $A \subset \mathbb{Q}$ have $\mathsf{d}_\Phi(A) > 0$ for some double Følner sequence $\Phi$. Are there infinite sets $B, C \subset \mathbb{Q}$ such that $B + C \subset A$ and $BC \subset A$?

We show that the answer is negative, for every double Følner sequence.

**Lemma 14.12.** *Let $\Phi = (F_N)$ be a double Følner sequence in $\mathbb{Q}$.*

(i) $|F_N| \to \infty$.
(ii) *For every bounded set $X \subset \mathbb{Q}$ we have $|F_N \cap X|/|F_N| \to 0$.*
(iii) $|F_N \cap \mathbb{Q}_{<0}|/|F_N| \to 1/2$.

*Proof.* (i) Suppose $|F_N| \leq M$ for infinitely many $N$. For those $N$ that are large enough, $|F_N \cap (1 + F_N)| > (1 - \frac{1}{M+1})|F_N| \geq |F_N| - 1$, hence $F_N \subset 1 + F_N$. Both sets have the same cardinality, so $F_N = 1 + F_N$. This is impossible for a finite nonempty $F_N \subset \mathbb{Q}$, since $\max(1 + F_N) = 1 + \max F_N > \max F_N$.

(ii) Choose $R \in \mathbb{N}$ with $X \subset [-R, R]$, fix $k \in \mathbb{N}$, and let $t_j = j(2R + 1)$ for $0 \leq j < k$. The sets $X + t_j$ are pairwise disjoint. For each $t \in \mathbb{Q}$,

$$|F_N \cap (X + t)| = |(F_N - t) \cap X| \geq |F_N \cap X| - |F_N \setminus (F_N - t)| = |F_N \cap X| - o(|F_N|),$$

since $|F_N \setminus (F_N - t)| = |F_N| - |F_N \cap (-t + F_N)|$. Summing over $j$ gives $k|F_N \cap X| \le |F_N| + o(|F_N|)$, so $\limsup_N |F_N \cap X|/|F_N| \le 1/k$. Now let $k \to \infty$.

(iii) Let $p_N = |F_N \cap \mathbb{Q}_{>0}|$ and $n_N = |F_N \cap \mathbb{Q}_{<0}|$. The map $y \mapsto -y$ gives $n_N = |(-F_N) \cap \mathbb{Q}_{>0}|$, and this differs from $p_N$ by at most $|(-F_N) \setminus F_N| = |F_N| - |F_N \cap (-1)F_N| = o(|F_N|)$. Moreover $p_N + n_N \ge |F_N| - 1$. By (i), $n_N/|F_N| \to 1/2$. □

**Lemma 14.13.** *Let $A \subset \mathbb{Q}_{<0}$ be such that $A \cap [-R, R]$ is finite for every $R > 0$. Then there are no infinite sets $B, C \subset \mathbb{Q}$ with $B + C \subset A$ and $BC \subset A$.*

*Proof.* Suppose such $B, C$ exist. Every product $bc$ is negative, so $0 \notin B \cup C$, and every $b \in B$ and $c \in C$ have opposite signs. If $B$ contained a positive and a negative element, no $c \in C$ could have the sign opposite to both. So $B$ has constant sign, and $C$ has the opposite constant sign. By symmetry of the hypotheses in $B$ and $C$ we may assume that $B \subset \mathbb{Q}_{<0}$ and $C \subset \mathbb{Q}_{>0}$. Fix $b^* \in B$. For $c \in C$ we have $b^* + c < 0$, so $0 < c < |b^*|$ and $b^* c \in (-(b^*)^2, 0)$. Since $c \mapsto b^* c$ is injective and $C$ is infinite, $b^* C$ is an infinite subset of $A \cap [-(b^*)^2, (b^*)^2]$, which is a contradiction. □

**Theorem 14.14.** *Let $\Phi = (F_N)$ be any double Følner sequence in $\mathbb{Q}$. There is a set $A \subset \mathbb{Q}_{<0}$ such that $\mathtt{d}_\Phi(A) = \lim_N |A \cap F_N|/|F_N| = 1/2$, and such that there are no infinite sets $B, C \subset \mathbb{Q}$ with $B + C \subset A$ and $BC \subset A$. In particular, Question 5.6 has a negative answer.*

*Proof.* For $R \in \mathbb{N}$ write $[-R, R]$ for $\mathbb{Q} \cap [-R, R]$, and let $\varepsilon_R(N) = |F_N \cap [-R, R]|/|F_N|$. By Lemma 14.12(ii), $\varepsilon_R(N) \to 0$ as $N \to \infty$ for each $R$. So we can choose $N_1 < N_2 < \cdots$ such that $\varepsilon_R(N) < 1/R$ for all $N \ge N_R$. Put

$$K_R = \bigcup_{N < N_{R+1}} F_N \ \text{(a finite set)}, \qquad Z = \bigcup_{R \in \mathbb{N}} \big([-R, R] \setminus K_R\big), \qquad A = \mathbb{Q}_{<0} \setminus Z.$$

*Bounded parts of $A$ are finite.* Let $R \in \mathbb{N}$. If $x \in A$ and $|x| \le R$, then $x \notin [-R, R] \setminus K_R$, so $x \in K_R$. Hence $A \cap [-R, R] \subset K_R$ is finite. By Lemma 14.13, $A$ contains no $B + C$ together with $BC$ for infinite $B, C$.

*$Z$ has density zero.* Let $N \ge N_1$, and let $R$ be such that $N_R \le N < N_{R+1}$. If $R' \ge R$, then $N < N_{R'+1}$, so $F_N \subset K_{R'}$ and $F_N$ does not meet $[-R', R'] \setminus K_{R'}$. If $R' < R$, then $[-R', R'] \subset [-R, R]$. Hence $F_N \cap Z \subset F_N \cap [-R, R]$, and so $|F_N \cap Z| \le \varepsilon_R(N)|F_N| < |F_N|/R$. As $N \to \infty$, also $R \to \infty$, so $|F_N \cap Z|/|F_N| \to 0$.

*Density of $A$.* By Lemma 14.12(iii), $|A \cap F_N|/|F_N| = |F_N \cap \mathbb{Q}_{<0}|/|F_N| - O(|F_N \cap Z|/|F_N|) \to 1/2$. □

*Formalization.* Lemmas 14.12 and 14.13 and Theorem 14.14 are formalized in `Kmrr25/Q56.lean` (`theoremN`), with double Følner sequences as in `Kmrr25/DoubleFolnerQ.lean`.

*Remark* 14.15. We record some easy facts that put Theorem 14.14 in context. Let $\Phi$ be a double Følner sequence and let $A \subset \mathbb{Q}$ satisfy $\overline{\mathtt{d}}_\Phi(A) > 0$.

(i) Each of the two conditions can be met separately. Moreira, Richter and Robertson proved the following [26, Theorem 1.3]: if $G$ is a countable group, $\Psi$ is a two-sided Følner sequence on $G$, and $E \subset G$ has positive upper density along $\Psi$, then $E \supset BC$ for some infinite $B, C \subset G$. Applied to $G = (\mathbb{Q}, +)$ and $\Psi = \Phi$, this gives infinite $B, C$ with $B + C \subset A$. Next apply it to $G = (\mathbb{Q}^*, \cdot)$ and $\Psi = (F_N \setminus \{0\})_{N \ge N_0}$, where $N_0$ is such that these sets are nonempty; this is a Følner sequence in $\mathbb{Q}^*$ by Lemma 14.12(i), and the upper density of $A \setminus \{0\}$ along it equals $\overline{\mathtt{d}}_\Phi(A)$. This gives infinite $B, C \subset \mathbb{Q}^*$ with $BC \subset A$. (Both groups are abelian, so one-sided and two-sided Følner sequences coincide.)

(ii) The set $\mathbb{Q}_{<0}$ itself contains the configuration: take $B = \{-n : n \geq 2\}$ and $C = \{1/m : m \in \mathbb{N}\}$. So the deletion of the null set $Z$ in Theorem 14.14 is essential.

(iii) The question requires two sets $B, C$. The version with $B = C$ fails even for restricted products: if $B \subset \mathbb{Q}$ is infinite, then $B \setminus \{0\}$ has two distinct elements of the same sign. So $\{b_1 b_2 : b_1, b_2 \in B,\ b_1 \neq b_2\}$ always contains a nonnegative number, and it is not contained in $\mathbb{Q}_{<0}$. On the other hand, $\mathsf{d}_\Phi(\mathbb{Q}_{<0}) = 1/2$ by Lemma 14.12(iii).

*Remark* 14.16. Theorem 14.14 answers Question 5.6 as stated. The question assumes only that $A$ has positive density along some double Følner sequence, and our set has density $1/2$ along any prescribed one. The obstruction is the sign of $x$, i.e. the real place of $\mathbb{Q}$. It plays the same role as congruence obstructions in $\mathbb{N}$, which is why the survey adds the aperiodicity hypothesis to Question 5.2. The point of Theorem 14.14 is that, once a set of density zero has been removed, this obstruction cannot be detected by density along a double Følner sequence. The $\Phi$-dependent deletion of finite pieces is similar to the device used in [22, Example 5.9] (compare also the proof of [22, Proposition 3.2]). The question that is presumably intended remains open: does every $A \subset \mathbb{Q}$ that is large and has no such local obstruction contain $B + C$ and $BC$? Possible formalizations are to require $\underline{\mathsf{d}}_\Phi(A) > 0$ for *every* double Følner sequence $\Phi$, which our construction (it depends on $\Phi$) does not address, or to impose a hypothesis on $A$ analogous to aperiodicity. We do not know the answer in either case.

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